Class 7

MATHEMATICS

Volume – 2

Solutions

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7. PLANE GEOMETRY

WORKSHEET – 1

LEVEL – I

1. A	2. C	3. B	4. D	5. A	6. B	7. D
8. C	9. C	10. A	11. D	12. D	13. A	14. B

LEVEL – II

1. C	2. A	3. C	4. B	5. B	6. D	7. A
8. B	9. B	10. A	11. B	12. C	13. B	14. A
15. A	16. A	17. C	18. C	19. C		

LEVEL – III

20. A, B, C,

21. A	22. A	23. B	24. C	25. A-R, B-S, C-P, D-Q
26. 900	27. C	28. C	29. D	30. A, C, D
31. A-P, B-R, C-Q, D-S		32. C	33. D	34. B

35. A-S, B-P, C-Q,S D-R

LEVEL – I

1. Supplement of x is also x.

   $x + x = 180° \Rightarrow 2x = 180° \Rightarrow x = 90°$

   $\therefore \frac{1}{4}$ complete angle $= \frac{1}{4} \times 360° = 90° \Rightarrow$ x $= 90°$

2. Given $(a + 20°) = (b + 30°) = 90° \Rightarrow a + 20° = 90° \Rightarrow a = 70°$

   $b + 30° = 90° \Rightarrow b = 60°$

   $\therefore$ Complement of a $= 90° - 70° = 20° \Rightarrow$ Complement of b $= 90° - 60° = 30°$

3. Given $\angle AOC = \angle ACO$ and $\angle BOD = \angle OBD$

   But $\angle AOC = \angle BOD$ (vertically opposite angles) $\Rightarrow \therefore \angle ACO = \angle OBD$

   These are alternate angles $\Rightarrow AC \parallel BD$

4. Given GED $= 126° \Rightarrow y^0 + 90° = 126° \Rightarrow \therefore y^0 = 36°$

   $\underline{GEC} = 180° - \angle GED = 180° - 126° = 54° \Rightarrow x = 180° - \angle GEC = 126°$ and $z° = 90°$

   ($\because$ EF $\perp$ CD and AB $\parallel$ CD)

   $\therefore x^0 + y^0 + z^0 = 126° + 36° + 90° = 252°$

5. Complementary angle of 33.5° is $90° - 33.5° = 56.5°$

6. Since $AB \parallel CD$, we have $\triangle ABC = BCD \Rightarrow y^0 + 25° = 75° \Rightarrow y^0 = 50°$

   Since $CD \parallel EF$ we have $x^0 + 25° = 180°$

   $\therefore x^0 + y^0 = 205° \rightarrow$ opt (1)

7. $\frac{a}{b} = \frac{5}{6} \Rightarrow 6a = 5b \Rightarrow b = \frac{6a}{5}$

we have $a + b + 70° = 180° \Rightarrow a + \frac{6a}{5} = 110° \Rightarrow \frac{11a}{5} = 110° \Rightarrow a = 50°$

and $b = \frac{6a}{5} = 60°$

Since b and 1 NOY are vertically opposite angles

we have $b + c = 180° \Rightarrow c = 120°$

8. Since I, m, n are parallel and $p \parallel q$, we have

and $4a + 120° = 180° \Rightarrow a = 15°, b = 20°; c = 12°$

$\therefore a + b - c = 15° + 20° - 12° = 23°$

9. $z = 180° - 108° = 72°$

But $4x + 5x = z^0$($\because$ corresponding angles are equal) $\Rightarrow 9x = z^0 \Rightarrow x = \frac{72^0}{9} = 8^0$

$\therefore q^0 = 5x^0 = 40^0; p^0 = 4x^0 = 32°$($\because$ vertically opposite angles are equal)

$\therefore p + q + z = 32° + 40° + 72° = 144°$

10. $ECD + 152° = 180°$

$\therefore ECD = 28°$

$\Rightarrow \angle BCD = \angle ABC = 56°$

$\therefore x^0 = \angle BCD - \angle ECD = 56° - 28° = 28°$

11. Given angles forms a complete angle.

$\Rightarrow 6x + 20° + 9y + 30° + 3z + 40° = 360°$

$\Rightarrow 6x + 9y + 3z = 270° \Rightarrow 3(2x + 3y + z) = 270° \Rightarrow 2x + 3y + z = 90°$

12. Given, $BC||DE$ Draw $FD||BC$ through D.

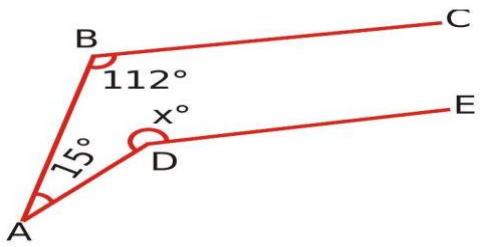
$\Rightarrow \angle AFD = \angle ABC = 112°$

($\because$ corresponding angles are equal)

$\therefore \angle FDA = 180° - (112° + 15°) = 53°$

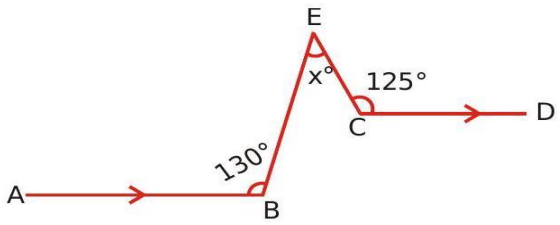
$\Rightarrow x = 180° + 53° = 233°$

$\therefore ADE = 360° - x = 127°$



13. Given, $AB||CD$. Draw FC parallel to AB through C.

$\Rightarrow \angle CFE = 180° - 130° = 50° \Rightarrow x = 125° - 50° = 75°$



(∵ Sum of opposite interior angles is equal to exterior angle)

14. Total ratio $= 3 + 4 + 2 = 9$

$\angle BOC = \frac{3}{9} \times 180° = 60°; \angle COD = \frac{4}{9} \times 180° = 80° \Rightarrow \angle AOD = \frac{2}{9} \times 180° = 40°$

OP is the bisector of $\angle BOC \Rightarrow \angle POC = \frac{1}{2}\angle BOC \therefore \angle POC = \frac{1}{2} \times 60° = 30°$

OR is the bisector of $\angle AOD \Rightarrow \angle DOR = \frac{1}{2}AOD \therefore DOR = \frac{1}{2} \times 40° = 20°$

Now, $\angle POR = \angle POC + \angle COD + \angle DOR = 30° + 80° + 20° = 130°$

15. Given the angles are in the ratio 2: 7 and they are supplement to each other.

Let the angles be $2x$ and $7x$.

$2x + 7x = 180° \Rightarrow 9x = 180° \Rightarrow x = 20°$

$\therefore$ Angles are $40°, 140°$

16. Let angles be $x, \frac{1x}{5} \Rightarrow x + \frac{x}{5} = 90° \Rightarrow \frac{6x}{5} = 90° \Rightarrow x = 75°$

$\therefore$ The angles are $75°, 90° - 15° = 75°, 15°$

17. $2x + 210° = 260° \Rightarrow 2x = 50° \Rightarrow x = 25°$

$\therefore$ Complement of $25° = 90° - 25° = 65°$

18. $c + p + q = 180°$ (straight angle) $\Rightarrow 28° + p + q = 180° \Rightarrow p + q = 152°$

but $p = a, b = q$ (vertically opposite angles)

$\therefore p + q + a + b = p + q + p + q = 2(p + q) = 2 \times 152° = 304^0$

19. $\underline{X} + \underline{A} = 90°$ (Complementary angles) $\Rightarrow \underline{A} = 90° - 30° = 60°$

$\underline{Y} + \underline{B} = 180°$(∵ Supplementary angles) $\Rightarrow \underline{B} = 180° - \lfloor \underline{y}$

$\Rightarrow B = 180° - 115° = 65° \Rightarrow \therefore \underline{A} + \underline{B} = 60° + 65° = 125°$

LEVEL – III

20. A, B, C

21. A

22. I ‖ m ‖ n&PR, SU are transversals

PR = 18Cm, UT = 6Cm, TS = 4Cm

From the fig. $\frac{PQ}{QR} = \frac{ST}{TU} \Rightarrow \frac{x}{18-x} = \frac{4}{6} \Rightarrow 6x = 72 - 4x \Rightarrow 10x = 72$

$X = 7.2$ cm $\Rightarrow QR = 18 - X = 18 - 7.2 \Rightarrow QR = 10.8$ cm

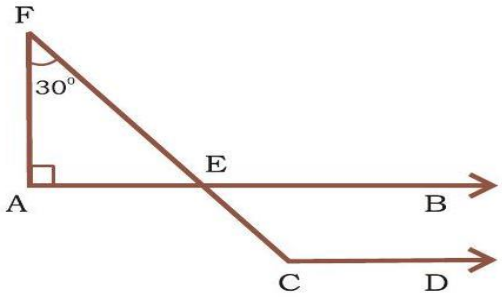
23. $\therefore$ PQ = 7.2 cm

24. $PR + UT = (PQ + QR) + UT = \not{x} + 18 - \not{x} + 6 = 18 + 6 = 24$ cm.
25. a-r, b-s, c-p, d-q
26. 90^0
27. C
28. C
29. D
30. A, C, D
31. a-p, b-r, c-q, d-s
32. $6x° + 9° = 3x + 30°$(∵ Alternate exterior angles are equal)

    $\Rightarrow 3x = 39° \Rightarrow x = 13^0$
33. $AB \parallel CD$ and transversal $DE, \angle AED = \lfloor CDP$

    $\Rightarrow \angle AED = 34°$ Now,$\Rightarrow \angle AEP + \angle PEF + \angle BEF = 180°$

    $\Rightarrow \angle PEF = 180° - 112° = 68° \therefore \angle DEF = 68°$
34. Through c, draw $CG \parallel AF$ Now $AF \parallel CG$, and transversal FC cuts their at $F\&C$.



    $\therefore \angle AFE = \angle GCE \Rightarrow \angle GCE = 30°$. Now $\angle FAE = \angle GHB \Rightarrow \angle GHB = 90°$

    Now $AB \parallel CD$ and transversal CG intersects them at H and C respectively.

    $\therefore DCH = \angle BHG \Rightarrow \angle DCH = 90°$

    $\therefore \angle ECD = \angle ECH + \angle DCH \Rightarrow \angle ECD = \angle GCE + \angle DCH \Rightarrow \underline{ECD} = 30° + 90 = 120°$
35. a-s, b-p, c-q,s d-r

WORKSHEET – 2

LEVEL – I

1. C	2. A	3. A	4. C	5. B	6. B	7. D
8. A	9. B	10. C	11. B	12. B	13. D	

LEVEL – II

1. B	2. C	3. D	4. B	5. D	6. C	7. C
8. D	9. C	10. D	11. C	12. D	13. A	14. C
15. D	16. A	17. A	18. B	19. A	20. A	21. B

LEVEL – III

22. A, B, C 23. D 24. B 25. C 26. A 27. a-q; b-p; c-s; d-r
28. 15050 29. A, B, C 30 a-q; b-r; c-s; d-p 31. D 32. B 33. A
34. 10

LEVEL – I

1. C
2. In a pentagon ⇒ number of sides(n) = 5
   Sum interior angles = $(2n-4)90^\circ = 540^\circ$
3. A
4. Number of sides $(n) = 6$ ⇒ sum of interior angles = $(2n-4)90^\circ = 720^\circ$
5. number of sides $(n) = 8$ ⇒ sum of interior angles = $(2n-4)90^\circ = 1080^\circ$
6. B
7. Number of sides $(n) = 5$
   Each interior angles of a polygon = $\frac{(2n-4)90^\circ}{n} = 108^\circ$
8. W.K.T each exterior angle ina polygon = $\frac{360^\circ}{n}$
   Given each exterior angle = $30^\circ \Rightarrow \frac{360^\circ}{n} = 30^\circ \Rightarrow n = 12$
9. B
10. Given sum of exterior angles of hexagon = 360°
11. Given number of sides in a Nanogon$(n) = 9$
    sum of interior angles = $(2n-4)90^\circ = 1260^\circ$
12. Given number of sides of hexagon $(n) = 6$
    the number of diagonals of n sided polygon = $\frac{n(n-3)}{2}$
    The number of diagonals of 6 sided polygon = $\frac{6(6-3)}{2} = 9$
13. D

LEVEL – II

1. Given that number of sides(n) = 4
   number of diagonals of n sided polygon = $\frac{n(n-3)}{2}$
   number of diagonals of 4 sided polygon = $\frac{4(4-3)}{2} = 2$
2. No. of triangles formed with common vertex = $(n-2) = 6-2 = 4$
3. given number of sides $(n) = 3$
   sum of interior angles of a polygon = $(2n-4)90 = [2(3)-4]90$
   sum of interior angles of a polygon = 180°

4. Given number of sides(n) = 5

   each exterior angle of a polygon $= \frac{360^\circ}{n} = \frac{360^\circ}{5} = 72^\circ$

5. Given interior of a regular polygon $= 60^\circ$

   $\frac{(2n-4)90^\circ}{n} = 60^\circ \Rightarrow (2n-4)90 = 60^\circ n, n = 3$

   Exterior angle of the polygon $= \frac{360}{n} = \frac{360}{3} = 120^\circ$

6. No. of diagonals of a polygon $= 54$

   $\frac{n(n-3)}{2} = 54 \Rightarrow n(n-3) = 108 \Rightarrow n^2 - 3n - 108 = 0 \Rightarrow n = 12, -9$

   $n = 12\ (n \neq -9)$

7. Given each interior angles of a regular polygon $= 144^\circ$

   $\Rightarrow \frac{(2n-4) \times 90^\circ}{n} = 144^\circ \Rightarrow (2n-4) \times 90^\circ = 144n \Rightarrow n = 10$

8. Given $n = 6 \Rightarrow \therefore$ sum of interior angles of the regular hexagon

   $= (2n-4) \times 90^\circ = [(2 \times 6) - 4] \times 90^\circ = 8 \times 90^\circ = 720^\circ$

9. $n = 5 \Rightarrow$ sum of interior angles of regular pentagon $= (2n-4) \times 90^\circ$

   $[2 \times (5) - 4] \times 90^\circ = 6 \times 90^\circ = 540^\circ$

10. Given $(2n-4) \times 90^\circ = 1080^\circ \Rightarrow (2n-4) = \frac{1080^\circ}{90^\circ}$

    $(2n-4) = 12 \Rightarrow 2n = 16 \Rightarrow n = 8$

11. Interior angles of a polygon ratio $= 2:3:4:5:6$

    let $\underline{A} = 2k\ \underline{B} = 3k$

    $C = 4k$

    $\underline{D} = 5k$

    $E = 6k$

    sum of the interior angles of a pentagon $= (2n-4) \times 90^\circ$

    $\Rightarrow 2k + 3k + 4k + 5k + 6k = [2(5) - 4] \times 90^\circ$

    $\Rightarrow 20k = 540^\circ \Rightarrow k = 27^\circ$

    $\therefore$ sum of 1st and 2st angle $= 2k + 3k = 5k = 5(27^\circ) = 135^\circ$

12. $10\alpha \cdot n = 360^\circ, n\alpha = 36^\circ,$ n represents the number of sides of a polygon, a closed figure, it cannot be 1 or 2, n is also a factor of 36 .

    $\therefore$ n = 3,4,6,9,12,18,36

13. Given interior angle of a regular polygon of n sides = 1 ts exterior angle of k sides

    i.e) $\frac{(2n-4)\times 90^\circ}{n} = \frac{360^\circ}{k} \Rightarrow \frac{2n-4}{n} = \frac{4}{k} \Rightarrow \frac{n-2}{n} = \frac{2}{k} \Rightarrow k = \frac{2n}{n-2}, \in, z^+$

for $n = 4, k = 4$ $n = 6, k = 3$

$\therefore$ (4,4) and (6,3) are only two pairs

14. One of the interior angle of a polygon = 150°

$\Rightarrow \frac{(2n-4)\times 90^\circ}{n} = 150^\circ \Rightarrow (2n-4)\times 9 = 15n \Rightarrow 3n = 36 \Rightarrow n = 12$

15. $\angle AER + \underline{REN} + \underline{NEP} + \underline{PEA} = 360^\circ$

$\Rightarrow \angle AER + \frac{1}{5}(540^\circ) + 90^\circ + 60^\circ = 360^\circ \Rightarrow \angle AER = 102^\circ$

Also $AE = EP = EN = ER \Rightarrow \angle EAR = \frac{180^\circ - 102^\circ}{2} = 39^\circ$

16. Interior angle $= \frac{(n-2)180^\circ}{n}$

$160^\circ = \frac{(n-2)\times 180^\circ}{n} \Rightarrow 8n = 9n - 18 \Rightarrow n = 18$

Number of diagonals of the polygon $= \frac{n(n-3)}{2} = \frac{18(18-3)}{2} = 9 \times 15 = 135$

17. Given $\frac{(2n-4)\times 90^\circ}{n} : \frac{360^\circ}{n} = 3:1 \Rightarrow \frac{2n-4}{4} = \frac{3}{1} \Rightarrow 2n - 4 = 12$

$\Rightarrow n = 8$ $\therefore$ The polygon is octagon.

18. n = 5

sum of interior angle of a pentagon $= (2n-4)\times 90^\circ$

$\Rightarrow a^\circ + (a+20)^\circ + (a+40) + (a+60) + (a+80^\circ) = [2(5)-4]\times 90^\circ$

$\Rightarrow 5a + 200^\circ = 540^\circ \Rightarrow 5a = 340^\circ \Rightarrow a = 68^\circ$

smallest angle $= a^\circ = 68^\circ$

19. $\frac{(2n-4)\times 90^\circ}{n} : \frac{360^\circ}{n} = 7:2 \Rightarrow (2n-4):4 = 7:2 \Rightarrow n-2:2 = 7:2$

$\Rightarrow n - 2 = 7 \Rightarrow n = 9$

20. Given $\frac{(2n-4)\times 90^\circ}{n} = 120^\circ \Rightarrow (2n-4)\times 3 = 4n \Rightarrow 6n - 12 = 4n \Rightarrow n = 6$

For new regular polygon; N = no. of sides = 2n = 12

$\therefore$ Interior angle of new regular polygon

$= \frac{(2N-4)\times 90^\circ}{N} = \frac{(2(12)-4)\times 90^\circ}{12} = \frac{20\times 90^\circ}{12} = 150^\circ$

21. Let No. of sides = n

By given data; $3(90^\circ) + (n-3)\times 150^\circ = (2n-4)\times 90^\circ$

$\Rightarrow 270^\circ + 150n - 450 = 180n - 360^\circ \Rightarrow 30n = 180^\circ \Rightarrow n = 6$

LEVEL – III

22. A, B, C

23. D

24. B

25. C
26. A
27. a-q; b-p; c-s; d-r
28. 15050
29. A, B, C
30. a-q; b-r; c-s; d-p
31. Given $\frac{(2n-4)\times 90^\circ}{n} = 2 \times \frac{360^\circ}{n} \Rightarrow (n-2) = 4 \Rightarrow n = 6$
32. $n = 9$, In Nonagon, each other five angles $= x^\circ$

    By given data; $50^\circ + 60^\circ + 70^\circ + 40^\circ + 5(x^0) = [2(9) - 4] \times 90^\circ$

    $220^\circ + 5x = 1260^\circ \Rightarrow 5x = 1040^\circ \Rightarrow x = 208^0$
33. Sum of the exterior angles in a polygon = 360

    ie $(2x+1)^0 + (3x+2)^0 + (4x-2)^0 + (5x+1)^0 + (6x+2)^0 + (6x-1)^0 + (x+1)^0$

    $+(x+6)^0 + (4x+3)^0 + (2x+7)^0 = 360^0$

    $\Rightarrow 34x + 20^\circ = 360^\circ \Rightarrow 34x = 340^\circ \Rightarrow x = 10^\circ$

    $\therefore$ Ratio of first and last exterior angle $= (2x+1)^\circ : (2x+7)^\circ$

    $= (2(10)+1) : (2(10)+7) = 21 : 27 = 7 : 9$
34. Given $\frac{(2n-4)\times 90^\circ}{n} = 4 \times \frac{360^\circ}{n} \Rightarrow (n-2) = 8 \Rightarrow n = 10$

WORKSHEET – 3

LEVEL – I

1.D	2. A	3. B	4. B	5. C	6. C	7.C
8. B	9. B	10.B	11. B	12. C	13.D	14.A
15. D	16. C	17. C	18. B	19.B	20. B	

LEVEL – II

1. A	2. A	3. B	4. C	5. A	6. D	7. D
8. C	9. C	10. B	11. D	12. D	13. B	14. B
15. B	16. B	17. B				

LEVEL – III

18. C	19. A, B	20. A, B, C	21. B, C	22. A-R; B-P; C-Q; D-S	
23. A	24. B	25. C	26. A	27. 6	28. A
29. A, B	30. A-2, B-3; C-4: D-1		31.B	32. B	33.A
34.A-Q; B-Q; C-R; D-Q					

LEVEL – I

1. Perimeter of $\triangle ABC = AB + BC + CA = 2 + 4 + 5 = 11$ cm
2. Sides of isosceles triangle $= 3$ cm, 3 cm, 4 cm

3. Given $\angle A + \angle B = \angle C$

   In a $\triangle ABC, \angle A + \angle B + \angle C = 180° \Rightarrow \angle C = \angle B = 180° \Rightarrow \angle C + \angle C = 180° \Rightarrow 2\angle C = 180°$

   $\Rightarrow \angle C = 90°$

7. $\angle A + \angle B + \angle C = 180° \Rightarrow 40° + 70° + \angle C = 180° \Rightarrow 110° + \angle C = 180° \Rightarrow \angle C = 70°$

8. Given $AB = AC\& \Rightarrow \angle A = 50° \Rightarrow \angle C = \lfloor B$

   But in $\triangle ABC, \angle A + \angle B + \angle C = 180° \Rightarrow 50° + \angle B + \angle B = 180°$

   $\Rightarrow 2\angle B = 130° \Rightarrow \angle B = 65° \Rightarrow \therefore \angle B = \angle C = 65°$

9. Each exterior angle of an Equilteral triangle is $180° - 60° = 120°$

10. W.K.T in a triangle sum of the two sides is greater than the third side and difference of two sides is less than the third side

    $AB + BC > CA \Rightarrow 5 + 8 > CA \Rightarrow CA < 13$

    $\therefore 3 < AC < 13$

    $\therefore$ Possible intergral values of $AC = 4,5,6,7,8,9,10,11,12$

    $\therefore$ No of possible integral values of $AC = 9$

11. In $\triangle ABC, \angle B = 90°$

    $\therefore AC^2 = AB^2 + BC^2 = 3^2 + 4^2 = 9 + 16 \Rightarrow \therefore AC = \sqrt{25} \Rightarrow AC = 5$ cm

12. C

13. D

14. A

15. D

16. In $\triangle ABC, AB = AC\& \angle A = 60° \Rightarrow \angle C = \angle B$

    In $\triangle ABC, \angle A + \angle B + \angle C = 180°$

    $60° + \angle B + \angle B = 180° \Rightarrow 2\angle B = 120° \Rightarrow \angle B = 60° \Rightarrow \therefore \angle C = 60°$

17. C

18. In $\triangle ABC, \therefore \angle B = 60° \& \angle C = 45°$

    $\therefore \angle A = 180°(\angle B + \angle C) = 180° - (60° + 45°) = 75°$

    W.K.T the side opposite to largest angle is largest side largest side=side oppo site is $\angle A = BC$

19. Verfiy $a + b > c \Rightarrow (8 + 9) > 16 \Rightarrow 17 > 16$

    $\therefore 8$ cm, 9 cm, 16 cm are the measuers of triangle

20. $\angle A: \angle B: \angle C = 7:3:5 \Rightarrow$ Total parts $= 7 + 3 + 5 = 15$

    $\angle A = \frac{7}{15} \times 180° = 84° \Rightarrow \angle B = \frac{3}{15} \times 180° = 36° \Rightarrow \angle C = \frac{5}{15} \times 180° = 60°$

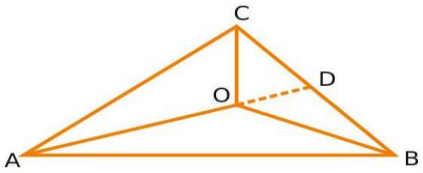
$\therefore$ Largest side = side opposite to largest angle = side opposite to $\angle A = BC$

LEVEL – II

1. A

4. We know that in a triangle the sum of two sides is greather than the third side.

   In the figure, extend segment AO to meet side BC at point D then by the tri angle inequality,



   $S_1 = OA + OB + OC$

   $S_2 = AB + BC + CA$

   In triangle $ADC, AD = AO + OD < AC + CD$

   In triangle $BOD, OB < OD + DB$

   by adding (1) &(2)

   $AO + OD + OB < AC + CD + OD + DB$

   $\Rightarrow AO + OB < AC + CD + DB = AC + CB \Rightarrow OA + OB < AC + CB$

   and in triangle $AOB, AB \angle AO + OB$

   $\therefore AB < OA + OB < AC + CB$

   similarly $BC < OB + OC < BA + AC$

   $CA < OC + OA < CB + BA$

   Adding (3),(4) and (5) we get $s_2 < 2\ s_1 < 2\ s_2 \Rightarrow s_1 > \frac{1}{2}\ s_2$& $s_1 < s_2$

5. In a $\triangle ABC$ let perimeter equal to $AB + BC + CA$ and sum of attitudes is equal to $AD + BE + CF$, now since $AD \perp BC$

   $\therefore AD < AB, AD < AC \therefore AD + AD < AB + AC$

   $2AD < AB + AC$

   $\therefore BE \perp CA$ similarly $2BE < BC + BA$

   $\therefore CF \perp AB$ similarly $2CF < CA + CB$

   Adding (1), (2) and(3) $2(AD + BE + CF) < 2(AB + BC + CA)$

   $\Rightarrow AD + BE + CF < AB + BC + CA$

6. D

7. D

8. In $\triangle PQR, PR > PQ$

   $\underline{PQR} > \underline{PRQ}$ (Angle opposite to greater side is greater)

   $\underline{1} = \underline{2}$ (PS is bisector of $\mathbb{P}$)

   $\underline{PQR} + \underline{1} > \underline{PRQ} + \underline{2} \Rightarrow 180° - y° > 180° - x°$

   $\therefore \angle PQS + 11 + y° = 180°$ and $\therefore \angle PQS + \angle 2 + x° = 180°$

   $\Rightarrow -y° > -x° \Rightarrow y° < x° \Rightarrow x° > y°$

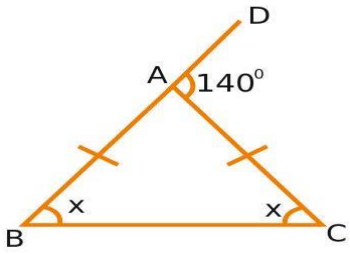
9. Given In $\triangle$ PQR which is equliateral Δ,

   given PO bisects $\underline{P} \Rightarrow \underline{QPO} = \underline{RPO}$

   given $\angle PQO = \angle PRO$

   given PQ = PR $\Rightarrow\therefore$ By ASA congruency condition, ΔQPO $\cong \Delta$RPO

10. In triangle ABC, given $\overline{AB} = \overline{AC}$



   $\Rightarrow \angle C = \angle B = x$

   by exterior angle property

   $X + X = 140° \Rightarrow 2x = 140° \Rightarrow x = 70° \Rightarrow\therefore ACB = 70°$

11. In triangle ABC, $\angle B + \angle C = 70° + 50° = 120° \Rightarrow\therefore A = 180° - 120° = 60$

   $\therefore$ ln$\triangle$ ABD; $\angle B + \angle x = 70° + 30° = 100° \Rightarrow$

   $\therefore \angle y = 180° - 100° = 80°$

   In triangle $ABD, \angle y > \angle B, AB > AD$

   In triangle $ACD, \angle C > \angle X, AD > CD$

   from (1) and (2) $AB > AD > CD$.

12. Verify the Inequalities of a triangle $\Rightarrow a + b > c|a - b| < c$.

13. Given $\triangle$ ABC $\cong\triangle$ PQR

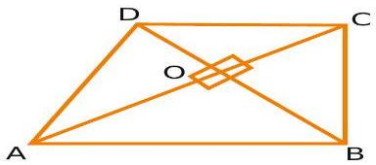
   By C.P.C.T $\angle A = \angle P$

   $\angle B = \angle Q$ and $\angle C = \angle R$

   $(x + 15)° = 70° \ (5y + 10)° = 35°$

   $\Rightarrow x = 55° \ y = 5°$

14. From the diagram



$\triangle ABC,\quad AB + BC > AC - - - - - (1)$ $\triangle ACD,\ CD + AD > AC$

$\triangle ABD,\quad AB + AD > BD - - - - - - (3)$ $\triangle BCD,\ BC + CD > BD$

$C - - - - - - (2)$

ADDING (1), (2), (3), (4) $\Rightarrow 2(AB + BC + CD + DA) > 2(AC + BD)$

$\Rightarrow AB + BC + CD + DA > AC + BD$

15. B

16. In triangle $ABE, AE = BE \Rightarrow \therefore BAE = \lfloor ABE = X°$

$y = \lfloor BAE + \lfloor ABE$ ($\because$ By exterior angle property) $= \lfloor x + \lfloor x \Rightarrow y = 2 \mid x$

17. In triangle $ABM, AM + BM > AB \Rightarrow AM + BM > AO + OB - - - (1)$

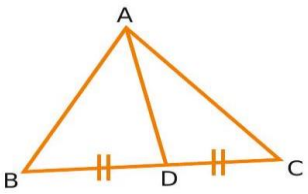
In triangle $\triangle CMD, CM + DM > CD \Rightarrow CM + DM > CO + OD$

Adding (1) and (2) $AM + BM + CM + DM > AO + OB + CO + OD$

$\Rightarrow AM + BM + CM + DM > AO + BO + CO + DO$

LEVEL – III

18. In triangle $ABC, AB = AC$



AD is median $\Rightarrow BD = DC$

In triangle ABD, triangle ACD

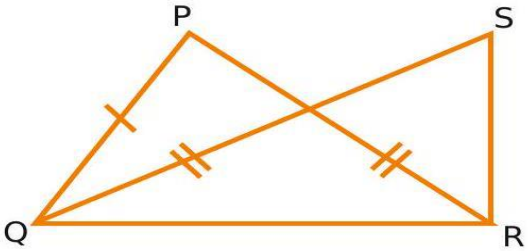
$AB = AC$ **GIVEN**

$AD = AD$ **common side**

$BD \Rightarrow DCAD$ is median

by sss congruency criterion, $\triangle ABD \cong \triangle ACD \Rightarrow$ by C.P.C.T $\angle ABD = \angle ACD$

19. In triangel PQR, triangle SQR



PQ = SR (given) $\Rightarrow$ PR = SQ (given) $\Rightarrow$ QR = QR (common side)

By sss congruency, criterion, $\triangle PQR = \Delta SRQ$

$PQR \cong \underline{SRQ}$

20. $3^2 + 4^2 = 5^2 \Rightarrow 8^2 + 15^2 = 17^2 \Rightarrow 7^2 + 24^2 = 25^2$

1. 3 cm. 4 cm, 5 cm
2. 8 cm, 15 cm, 17 cm
3. 7 cm, 24 cm, 25 cm

form a right angled triangle.

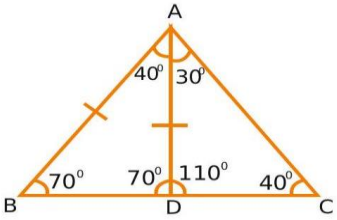
21. $GA^2 + GB^2 + GC^2 = \frac{1}{3}(AB^2 + BC^2 + AC^2)$
22. a-r; b-p; c-q; d-s
23. In triangle ABC, by exterior angle property $\Rightarrow \angle ACD = 30° + 45° = 75°$
24. Similarly in triangle CED,$\Rightarrow 75° + \angle CED + 46° = 180° \Rightarrow \underline{CED} = 180° - 121° = 59°$
25. $\angle AED = 75° + 46° = 121° \Rightarrow$ by exterior angle property
26. A
27. 6
28. In triangle $ABC, \angle A + \angle B + \angle C = 180°$

$55° + 57° + 2y = 180° (\therefore < B = \angle ABO + \angle CBO) \Rightarrow y = 34°$

In triangle BOC, $\underline{OBC} + \underline{BOC} + \underline{OCB} = 180°$

$\Rightarrow 37° + x + y = 180° \Rightarrow 37° + x + 34° = 180° \Rightarrow x + 71° = 180° \Rightarrow x = 109°$

30. a-2, b-3; c-4: d-1
31. Given $AB = AD$



$\Rightarrow \underline{ADB} = \underline{ABD} = 70°$

$\therefore \angle BAD = 180° - (70° + 70°) = 40° \Rightarrow \angle ADC = 180° - 70° = 110°$

In triangle $ADC, \angle CAD = 180° - (110° + 40°) = 30°$

$AB > CD$ ($\therefore ACD > | CAD$) $\Rightarrow AB > CD$ is true $\Rightarrow AB \neq AC$ is false

$AB > CD\big(|$ ABC $= 70° > \underline{ACB} = 40° \Rightarrow$ AC $=$ BC$(\therefore$ ABC $= \angle$BAC $= 70°$

32. $\triangle$ ADC $= 110°$
33. A $= 40° + 30° = 70°$
34. In Δ^{le}ABC, LET$\angle$C $=$ y

given $\angle B = 2\angle C \Rightarrow \angle B = 2y \Rightarrow$ given AD bisects $\angle BAC$

$\angle BAD = \angle CAD = x \Rightarrow$ given $AC = BC \Rightarrow \angle B = \angle A \Rightarrow 2y = 2x \Rightarrow x = y$

In $\triangle^{le}$ ABC, $\angle A + \angle B + \angle C = 180^\circ \Rightarrow 2x + 2y + y = 180^\circ \Rightarrow 2x + 3y = 180^\circ$

$\Rightarrow 2x + 3x = 180^\circ (\because x = y) \Rightarrow 5x = 180^\circ \Rightarrow x = 36^\circ$

$\angle A = 2x = 72^\circ \Rightarrow \angle B = 2y = 72^\circ$

$\angle ADC = x + 2y = 108^\circ \& \angle ADB = 72^\circ$

WORKSHEET – 4

LEVEL – I

1. A	2. C	3. B	4. A	5. C	6. B	7. A
8. B	9. B	10. B				

LEVEL – II

1. C	2. C	3. B	4. A	5. A	6. A	
7.A	8. A	9. A	10. A	11. D	12. B	13. C
14. C	15. A	16. C	17. A	18. A	19. B	20. C

LEVEL – III

21. A, B, C	22. A, B, C, D	23. A	24. B	25. C
26. A	27. a-s, b-q, c-t, d-p	28. B	29. A, B, C	30. A
31. B	32. A	33. C	34. C	35. a-q, b-r, c-p, d-t

LEVEL – II

1. C

2. $\triangle$ ABC $\sim \triangle$ PQR $AB = 6, BC = 4, AC = 8, PR = 6$

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} \Rightarrow \frac{6}{PQ} = \frac{4}{QR} = \frac{8}{6} \Rightarrow \frac{6}{PQ} = \frac{8}{6} \& \frac{4}{QR} = \frac{8}{6}$$

$$PQ = \frac{36}{8} = 4.5 \& QR = 3 \Rightarrow PQ + QR = 4.5 + 3 = 7.5$$

3. $\triangle$ ABC $\sim \triangle$ PQR; $AB = 3.6, PQ = 2.4 \& PR = 5.4$

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} \Rightarrow \frac{3.6}{2.4} = \frac{BC}{QR} = \frac{AC}{5.4} = \frac{3}{2} = \frac{AC}{5.4} \Rightarrow AC = 8.1$$

4. $\triangle$ ABC $\sim \triangle$ PQR, $m\angle A = 50^\circ$

$\angle A = \angle P$
$\angle B = \angle Q$
$\angle C = \angle R$

(Corresponding angles are equal)

Δ^{le}ABC;

$\angle A + \angle B + \angle C = 180^\circ \Rightarrow \angle 50 + \angle Q + \angle R = 180^\circ \Rightarrow \angle Q + \angle R = 130^\circ$

5. In $\Delta^{le} ABCDE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{AD}{AB} = \frac{AE}{AC} \Rightarrow \frac{3}{8} = \frac{AE}{5.6} \Rightarrow AE = 2.1$ cm

6. B

7. A

8. B

9. If two triangles are similar,

   w.k.t the ratio of areas of two similar triangles is equal to the square of their corresponding medians.

   $25:36$ = square of their corresponding medians

   Ratio of their corresponding Median $= \sqrt{25}:\sqrt{36} = 5:6$

10. $\triangle ABC \sim \triangle PQR$ $AB = 36, PQ = 2.4 \& PR = 5.4$

    $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} \Rightarrow \frac{3.6}{PQ} = \frac{BC}{QR} = \frac{8.1}{5.4} \Rightarrow PQ = 2.4 \text{ cm}$

11. D

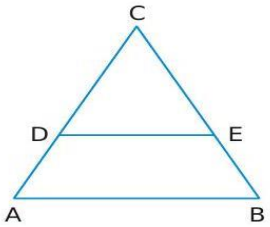
12. In $\triangle^{le}$ ABC bisector of $\angle A$ intersects BC at D.

    $BD:DC = 8:15 \& AC = 45$

    By the vertical angle bisector theorem;

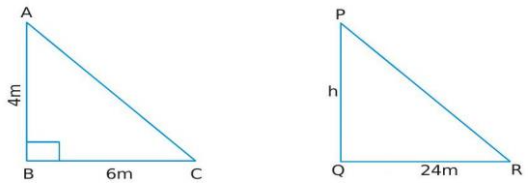
    $\frac{BD}{DC} = \frac{AB}{AC} \Rightarrow \frac{8}{15} = \frac{AB}{45} \Rightarrow AB = 24 \text{ cm}$

13. By B.P.T theorem



$$\frac{AD}{DC} = \frac{BE}{EC} \Rightarrow \frac{8x+9}{x+3} = \frac{3x+4}{x}$$
$$8x^2 + 9x = 3x^2 + 4x + 9x + 12$$
$$5x^2 - 4x - 12 = 0 \Rightarrow (5x+6)(x-2) = 0$$
$$5x = -6 \Rightarrow x = \frac{-6}{5}$$
$$x - 2 = 0$$
$$x = 2$$

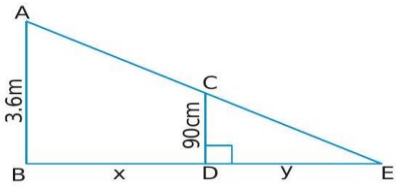
14.



Let PQ be the height of the building and QR be its shadow

$$\Delta ABC \sim \Delta PQR \Rightarrow \frac{AB}{PQ} = \frac{BC}{QR} \Rightarrow \frac{4}{h} = \frac{6}{24} = h = 16 \text{ m}$$

15. $AB = 3.6 \text{ cm} = 360 \text{ cm}, CD = 90 \text{ cm}$



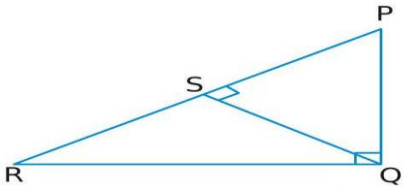
let length of shadow of the girl after 4 seconds be $y = 1.2 \times 4 = 4.8$ cm Distance travelled by the girl in 4 seconds be ' x ' Then $BD = x = 4.8$ m $= 480$ cm.

$\triangle$ ABE $\sim \triangle$ CDE (by AA Similarity)

$$\frac{AB}{CD} = \frac{BE}{DE} \Rightarrow \frac{360}{90} = \frac{x+y}{y} \Rightarrow 4 = \frac{x}{y} + 1 \Rightarrow \frac{x}{y} = 3 \Rightarrow y = \frac{x}{3}$$

$$y = \frac{480}{3} \Rightarrow y = 160 \text{ cm}$$

16. In Δ^{le}PQR, $\angle Q = 90°$, QS $\perp$ PR



Area of $\triangle$ PQR $= \frac{1}{2} \times$ PR $\times$ QS

also Area of $\triangle$ PQR $= \frac{1}{2} \times$ QR $\times$ PQ $- - -$ (2)

From (1) & (2), $\frac{1}{2} \times$ PR $\times$ QS $= \frac{1}{2} \times$ QR $\times$ PQ

PR $\times$ QS $=$ QR $\times$ PQ

Since $\triangle$ PQR is a right angled triangle at Q.

i.e) $PR^2 = PQ^2 + QR^2 \Rightarrow \frac{1}{PQ^2} + \frac{1}{QR^2} = \frac{QR^2+PQ^2}{PQ^2QR^2}$

$\frac{PR^2}{PQ^2QR^2} = \frac{PR^2}{PR^2 \times QS^2}$ (from (3)) $= \frac{1}{QS^2} \Rightarrow \frac{1}{PQ^2} + \frac{1}{QR^2} = \frac{1}{QS^2}$

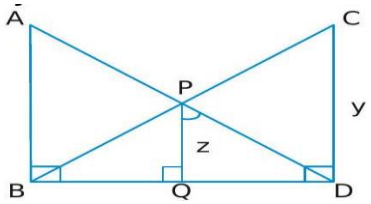
17. $\triangle$ ABC is an isosceles right angled at C.

i.e., $\Delta^{le\ ABC}, \angle C = 90° \& AC = BC$

$$\text{In } \Delta^{le}ABC, AC^2 + BC^2 = AB^2 \Rightarrow AC^2 + AC^2 = AB^2 \Rightarrow \boxed{AB^2 = 2AC^2}$$

18. Given AB, CD, PQ are perpendicular to BD.

$AB = x, CD = y, \& PQ = z$



By AA Similarity, $\triangle$ BPQ $\sim$ ΔBCD $\Rightarrow \frac{BQ}{BD} = \frac{z}{y} - - -$ (1)

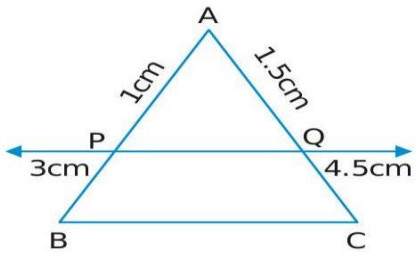
By AA Similarity, $\triangle DPQ \sim \triangle DAB \Rightarrow \frac{QQ}{BD} = \frac{z}{x} - - - (2)$

(1) +(2)

$\frac{BQ}{BD} + \frac{QD}{BD} = \frac{z}{y} + \frac{z}{x} \Rightarrow \frac{BQ + QD}{BD} = z\left(\frac{1}{y} + \frac{1}{x}\right) \Rightarrow \frac{BD}{BD} = z\left(\frac{1}{x} + \frac{1}{y}\right)$

$1 = z\left(\frac{1}{x} + \frac{1}{y}\right) \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

19. Given in $\triangle^{le} ABC, AP = 1$ cm, $BP = 3$ cm, $AQ = 1.5$ cm&$QC = 4.5$ cm



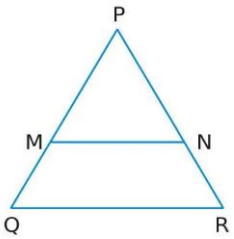
$\frac{AP}{PB} = \frac{AQ}{QC} \Rightarrow PQ \parallel BC$ (Converse of B.P.T theorem)

by AA Similarity, $\triangle APQ \sim \triangle ABC. \Rightarrow \frac{\text{area of } \triangle APQ}{\text{area of } \triangle ABC} = \frac{AP^2}{AB^2} = \frac{1^2}{(1+3)^2} = \frac{1}{16}$

Required ratio = 1: 16

20. In $\Delta^{le} PQR, MN \parallel QR$

$PM = x + 3, MQ = x + 5, PN = x, NR = x + 1$



$\frac{PM}{MQ} = \frac{PN}{NR} \Rightarrow \frac{x+3}{x+5} = \frac{x}{x+1} \Rightarrow (x+3)(x+1) = x(x+5)$

$x^2 + 4x + 3 = x^2 + 5x \Rightarrow x = 3$

LEVEL – III

21. A, B, C
22. A, B, C, D
23. A
24. $\triangle ABC \sim \triangle PQR \Rightarrow \frac{AB}{AC} = \frac{PQ}{PR} \Rightarrow \frac{15}{AC} = \frac{18}{12} \Rightarrow AC = 10$
25. $\triangle ABC \sim \triangle PQR;$

$m\angle Q + m\angle R = ?$

$\angle A + \angle B + \angle C = 180^\circ \Rightarrow 30^\circ + \angle Q + \angle R = 180^\circ \Rightarrow \angle Q + \angle R = 150^\circ$

26. $\triangle ABC \sim \triangle PQR;$

$\angle A = \angle P$

$\angle B = \angle Q$

$\angle C = \angle R$

$\angle A + \angle B + \angle C = 180° \Rightarrow m\angle A = 80°, m\angle B = 60° \Rightarrow \angle A + \angle B + \angle C = 180°$

$80° + 60° + \angle R = 180° \Rightarrow \angle R = 40°$

27. a-s, b-q, c-t, d-p

28. In $\triangle ABC$; bisector of $\angle A$ intersects BC in D.

$BD : BC = 4 : 7 \& AC = 3.5$

By vertical angle bisector theorem;

$\frac{BD}{BC} = \frac{AB}{AC} \Rightarrow \frac{4}{7} = \frac{AB}{3.5} \Rightarrow AB = 2$

29. A, B, C

30. A

31. $\triangle ABC \sim \triangle PQR$;

Ar. $\triangle ABC$: Ar. $\triangle PQR = AB^2 : PQ^2$

$1 : 4 = AB^2 : PQ^2 \Rightarrow AB : PQ = 1 : 2$

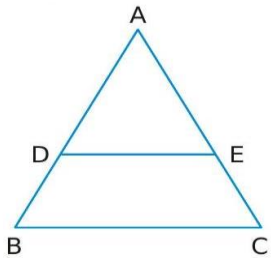
32. If two triangles are similar then

Ratio of their Areas = square of Ratio of their medians $= 6^2 : 7^2 = 36 : 49$

33. $\triangle ABC \sim \triangle PQR$

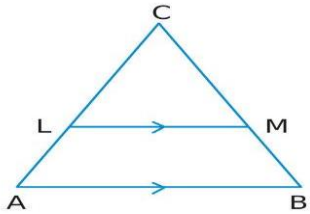
Ratio of perimeters = Ratio of their corresponding sides $= 2 : 3$

34. In $\triangle^{le} ABC, DE \parallel BC BD = 9$ cm, $AE = 1.8$ cm EC = 5.4 cm



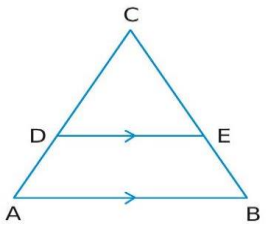
$\frac{AD}{BD} = \frac{AE}{EC} \Rightarrow \frac{AD}{9} = \frac{1.8}{5.4} \Rightarrow AD = 3$ cm

35. a) In $\Delta^{le} ABC, \frac{AL}{AC} = \frac{BM}{BC}$



$\frac{x-3}{2x} = \frac{x-2}{2x+3} \Rightarrow (x-3)(x+3) = 2x(x-2) \Rightarrow x = 9$

b) $\frac{AD}{CD} = \frac{BE}{CE}$

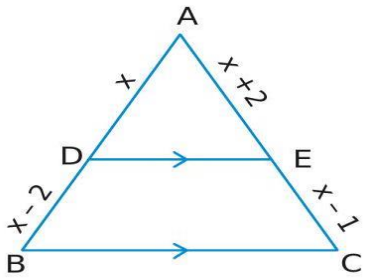


$\frac{8x+9}{x+3} = \frac{3x+4}{x} \Rightarrow (8x+9)x = 3x+4(x+3)$

$8x^2 + 9x = 3x^2 + 13x + 12 \Rightarrow 5x^2 - 4x - 12 = 0$

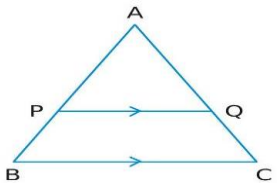
$(x-2)(5x+6) = 0 \Rightarrow x = 2$ (or) $x = -\frac{6}{5}$

c) DE ∥ BC



$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-2} = \frac{x+2}{x-1} \Rightarrow x^2 - x = x^2 - 4 \Rightarrow x = 4$

d) In $\Delta^{le} ABC, PQ \parallel BC$



$\frac{AP}{PB} = \frac{AQ}{QC} \Rightarrow \frac{1}{3} = \frac{1.5}{x} \Rightarrow x = 4.5$

WORKSHEET – 5
LEVEL – I

1. B	2. C	3. B	4. A	5. D	6. A	7. B
8. A	9. B	10. D	11. B	12. D	13. B	14. D
15. B	16. B					

LEVEL – II

1. A	2.B	3.D	4.C	5.A	6.B	7. C
8.B	9.B	10.D	11.A	12.A	13.D	14.A
15.B	16.B	17.B				

LEVEL – III

18. B, C 19. A, C, D 20. C 21. C 22. A 23. a-q; b-r; c-s; d-p

24. B 25. B 26. C 27. D

28. A, C 29. A 30. a-r, b-q, c-p, d-s 31.180 32.B 33.A

34.C 35 .a-r, b-s, c-p, d-q

LEVEL – I

1. $55 + 65 + 105 + x = 360 \Rightarrow 225 + x = 360 \Rightarrow x = 135$
2. $80 + 120 + 2x = 360 \Rightarrow 2x = 360 - 200 \Rightarrow x = 80$
3. $x + x - 10 + x + 30 + 2x = 360 \Rightarrow 5x + 20 = 360 \Rightarrow x = 68°$
4. $1x + 2x + 3x + 6x = 360 \Rightarrow x = 30; \angle 1 = 30; \angle 2 = 60; \angle 3 = 90; \angle 4 = 6(30)$ largest angle $= 180°$
5. $\angle D + \angle A = 180 \Rightarrow 50 + \angle A = 180 \Rightarrow \angle A = 130°$
6. $EF//AB$ then $AB//CD \Rightarrow AB = 10$ cm, $EF = 10$ cm

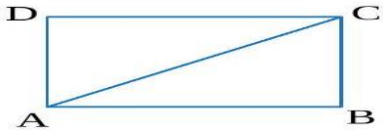
   In trapezium $ABCD, EF = \frac{1}{2}(AB + CD)$

   $10 = \frac{1}{2}(10 + CD) \Rightarrow 20 = 10 + CD \Rightarrow CD = 10$ cm
7. In a trapezium $OA = 5$ cm; $OC = 10$ cm $\Rightarrow \frac{DO}{OB} = \frac{OC}{OA} \Rightarrow \frac{10}{5} = \frac{2}{1} = 2:1$
8. The length of diagonals are equal in isosceles trapizium ($AC = BD$) so $BD = 10$ cm
9. The diagonals of a parallelogram bisect each other ($OA = OC, OB = OD$) $AC = OA + OC = 5$ cm $+ 5$ cm $= 10$ cm
10. D
11. In a rectangle the length of the diagonal are equal $BD = AC = 10$ cm
12. In a square, Diagonal AC bisects $\angle A$ ie $\angle BAC = \frac{90°}{2} = 45°$
13. In a parallelogram $\angle A = \angle C, \angle B = \angle D$

    $\therefore \angle B + \angle C = 180 \Rightarrow \angle B + 50 = 180 \angle B = 130°$
14. In a parallelogram; $OA = OC \Rightarrow 2x + 15 = 10x - 9$

    $8x = 24\ x = 3$

    $\therefore AC = 20A$ or $2OC = 2(2x + 15) = 2(6 + 15) = 42$ cm
15. In a rectangle $ABCD, AB = 4$ cm, $BC = 3$ cm



$AC^2 + AB^2 + BC^2 = 4^2 + 3^2 = 25 \Rightarrow AC = 5$ cm

16. In rectangle $ABCD, OA^2 + OC^2 = OB^2 + OD^2 = 3^2 + 4^2 = 25$ cm

LEVEL – II

1. Area of the trapezium $= \frac{1}{2} \times h \times (a + b) \Rightarrow 39 = \frac{1}{2} \times 6 \times (8 + b) \Rightarrow 39 = 3(8 + b)$

   $\Rightarrow 8 + \text{b} = 13$ b $= 5$ cm

2. Given area of square $= 16$ cm$^2 \Rightarrow a^2 = 16 = a = 4$ cm

   $\therefore$ Its perimeter $= 4\text{a} = 4 \times 4 = 16$ cm

3. In a parallelogram $PQRS$, let $PQ = x$ cm

   $\text{QR} = (\text{x} + 3)$cm

   Perimeter of parallelogram $PQRS = PQ + QR + SR + PS$

   In parallelogram opposite sides are equal $PQ = SR \& QR = PS$

   70 cm $= \text{PQ} + \text{QR} + \text{PQ} + \text{QR}$

   $70 = 2(\text{PQ} + \text{QR})$

   $70 = 2(x + x + 3) = 2x + 3 = 35 \Rightarrow x = 16$

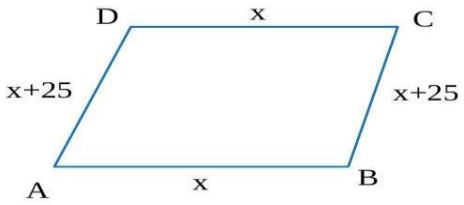
   $\text{QR} = \text{x} + 3 = 16 + 3 = 19$ cm

4. In parallelogram ABCD; $\angle\text{A}: \angle\text{B}: \angle\text{C}: \angle\text{D} = 2: 3: 7: 6$

   $\angle\text{C}$ = largest angle $= \frac{7}{18} \times 360^\circ = 140$

5. Let the adjacent sides of the parallelogram are x cm, $(x + 25)$cm

   Given perimeter of the parallelogram $= 150$ cm

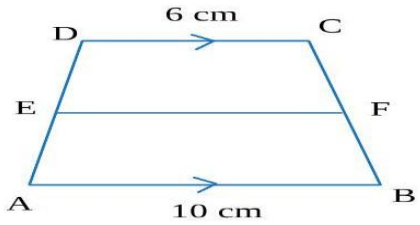


   i.e $2(x + x + 25) = 150 \Rightarrow 2x + 25 = 75 \Rightarrow x = 25$

   length of adjacent sides 25 cm, 50 cm

6. B

7. In a trapezium $ABCD$



   length of median, $EF = \frac{1}{2}(AB + CD) = \frac{1}{2}(10 + 6) = 8$ cm

8. In a parallelogram $ABCD$, opposite angles are equal

   ie $\angle A = \angle C \& \angle B = \angle D$

   $2x + 30° = x + 120° \Rightarrow x = 90°$

9. In a parallelogram $ABCD$, sum of adjacent angles is equal to 180°.

   $\angle A = (3x - 10)^0 \& \angle B = (5x + 30)^0$

   ie $\angle A + \angle B = 180° \Rightarrow (3x - 10)° + (5x + 30)° = 180° \Rightarrow 8x = 160° = x = 20°$

10. In rectangle $ABCD, AC^2 = AB^2 + BC^2 = 12^2 + 5^2 = 169 \Rightarrow AC = 13$ cm

11. Given area of a trapezium = 36 cm^2

$$\frac{1}{2} \times h \times (a + b) = 36 \Rightarrow \frac{1}{2} \times 4 \times (7 + b) = 36 \Rightarrow 7 + b = 18 \Rightarrow b = 11 \text{ cm}$$

12. In a parallelogram PQRS, let PQ = xcm; QR = (x + 3)cm Given its perimeter = 78 cm $[PQ + QR] = 78 \Rightarrow 2x + 3 = 39 \Rightarrow x = 18$ cm

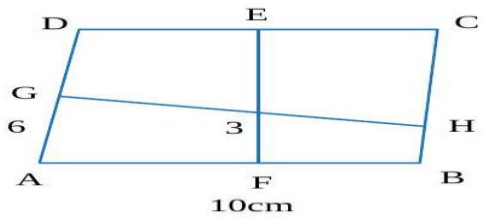
    $\therefore$ QR = x + 3 = 18 + 3 = 21 cm

13. In $\triangle ABCD$; given $\angle A = 72° \& \angle B = 56° \& \angle C : \angle D = 3 : 5 \Rightarrow \angle C = 3x \& \angle D = 5x$

$$\angle A + \angle B + \angle C + \angle D = 360°$$
$$\Rightarrow 72° + 56° + 3x + 5x = 360° \Rightarrow 8x + 128° = 360° \Rightarrow 8x = 232 \Rightarrow x = 29°$$
$$\therefore \angle C = 3x = 3(29°) = 87^0$$
$$\angle D = 5x = 5(29°) = 145°$$

    $\therefore$ largest angle, $\angle D = 145°$

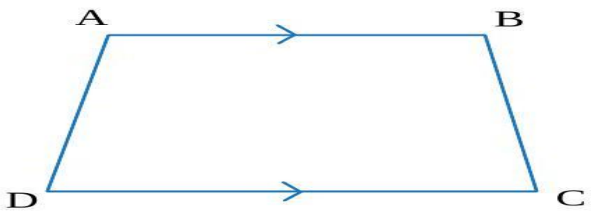
14. In a parallelogram $ABCD, AB = 10$ cm, $AD = 6$ cm Area of the parallelogram = base × height



    $\Rightarrow AB \times EF = AD \times GH \Rightarrow 10 \times 3 = 6 \times GH \Rightarrow GH = 5$ cm

    $\therefore$ Coresponding altitude of $\overline{AD} = 5$ cm

15. In $\triangle ABCD, AB // DC$

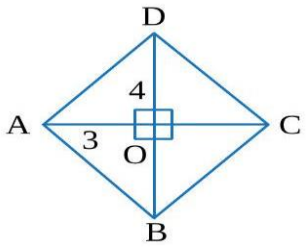


$$\angle A : \angle D = 2 : 3 \& \angle B : \angle C = 4 : 5$$

$\angle A = 2k, \angle D = 3k$
$\angle A + \angle D = 180°$ ($\therefore$ cointerior angles are supplementary)
$\Rightarrow 2k + 3k = 180° \Rightarrow 5k = 180° \Rightarrow k = 36°$
$\therefore \angle D = 3k = 3(36°) = 108°$

16. In a rhombus $ABCD, AB = 6$ cm$BD = 8$ cm

$AC\&BD$ are bisect each other



ie $OA = OC = \frac{AC}{2} = 3$ cm $\Rightarrow OB = OD = \frac{BD}{2} = 4$ cm

In $\triangle ACD, AD^2 = OA^2 + OD^2 = 3^2 + 4^2 = 25 \Rightarrow AD = 5$ cm

length side of rhombus = 5 cm

17. Ratio of areas of two squares = 4: 9

i.e $S_1^2 : S_2^2 = 4:9 \Rightarrow S_1 : S_2 = 2:3$

Ratio of their perimeters = 4 S_1: 4 S_2 = $S_1 : S_2 = 2:3$

LEVEL – III

18. sum of interior angles of a quadrilateral = 360°
19. A, B, D
20. C
21. C
22. A
23. a-q; b-r; c-s; d-p
24. In a trapezium $ABCD, EF$ is median $EF = \frac{1}{2}(AB + CD)$

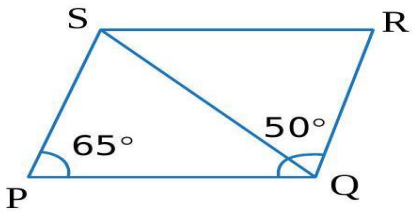
$\Rightarrow 4 = \frac{1}{2}(AB + 6) \Rightarrow AB + 6 = 8 \Rightarrow AB = 2$ cm

25. In $\square ABCD, \angle A + \angle B + \angle C + \angle D = 360°$

$\Rightarrow 5x + 5(x + 2) + 6x - 20 + 6(x + 3) = 360° \Rightarrow 5x + 5x + 10 + 6x - 20 + 6x + 18 = 360°$
$22x + 8 = 360° \Rightarrow 22x = 352 \Rightarrow x = 16 \Rightarrow \angle B = 5(x + 2) = 5(16 + 2) = 5(18°) = 90°$

26. $\angle C = 6x - 20 = 6(16) - 20 = 96 - 20 = 76°$
27. $\angle D = 6(x + 3) = 6(16 + 3) = 6(19) = 114^0$

28. In a parallelogram PQRS, $\angle SPQ = 65°, \& \angle SQR = 50°$



In $\triangle SQR, \angle RSQ + \angle SQR + \angle QRS = 180°$

$\Rightarrow \angle RSQ + 50° + 65° = 180°$ ($\therefore \angle SPQ = \angle QPS$)

$\Rightarrow \angle RSQ + 115° = 180° \Rightarrow \angle RSQ = 65°$

29. A

30. a-r, b-q, c-p, d-s

31. $\angle A = 60° \Rightarrow \angle C = 60°; \angle B = 120° \Rightarrow \angle D = 120° \Rightarrow \angle C + \angle D = 180°$

32. In a parallelogram $ABCD, \angle A : \angle B = 4:5$

sum of adjacent angles $= 180°$

$\Rightarrow \angle A + \angle B = 180° \Rightarrow 4k + 5k = 180° \Rightarrow 9k = 180° \Rightarrow k = 20°$

$\therefore \angle A = 4\ K = 4(20°) = 80°$

33. $\angle B = 5k = 5(20°) = 100°$

34. $\angle D - \angle C = \angle B - \angle A$ ($\therefore$ opposite angles are equal)

$= 100° - 80° = 20°$

35. $\angle SOR = \angle POQ$ (vertically opposite angles) $\therefore \angle POQ = 75°$

In $\triangle POQ, 45° + 75° + \angle PQO = 180°$ (sum of angles in a triangle $= 180°$)

$\Rightarrow \angle PQO = 180° - 120° = 60°$

$\angle PQO = \angle OSR = 60°$ (alternate angles)

$\angle SPO = \angle PRQ = 50°$ (alternate angles)

Now $\angle P + \angle Q = 180°$

$\therefore 50° + 45° + \angle PQR + \angle OQR = 180°$

$95° + 60° + \angle OQP = 180° \Rightarrow \angle OQR = 25°$

also $\angle OQR = \angle RQS = 25°$

also $\angle PQR = \angle PQO + \angle OQR = 60° + 25° = 85°$

WORKSHEET – 6

LEVEL – I

1. C	2. D	3. A	4. A	5. C	6. B	7. B
8. A	9. A	10. A				

LEVEL – II

1. B	2. C	3. B	4. B	5. B	6. C	7. A

8. B 9. D 10. D 11. B 12. C 13. B 14. B

15. A 16. A 17. A 18. B 19. B 20. A

LEVEL – III

21. A, B 22. A 23. D 24. B 25. D 26. a - s, b-t c -r d-q

27. A 28. A, B, C 29. C 30. C 31. B 32. a - r b-t c - t d-q 33. B

34. A 35. D

LEVEL – II

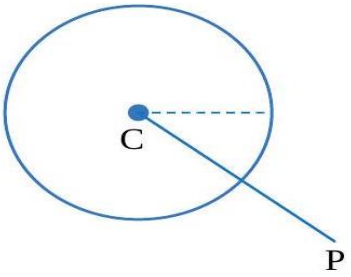
1. diameter, d = 14 cm ⇒ 2r = 14 ⇒ r = 7 cm

   ∴ circumference $= 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm

2. r = 21 cm

   length of the arc in semi circle $= \pi r = \frac{22}{7} \times 21 = 66$ cm

3. r = 10 cm CP = 12 cm

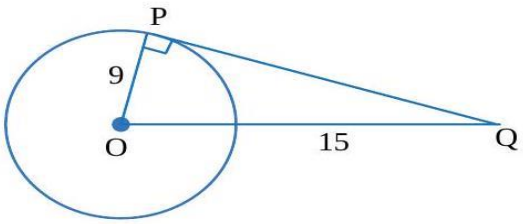


   here $CP > r$

   i.e P lies outside of the circle.

4. Infinite.

5. Given $r = OP = 9$ cm



   OQ = 15 cm OP ⊥ OQ

   ∴ Length of the tangent, $PQ = \sqrt{OQ^2 - OP^2}$

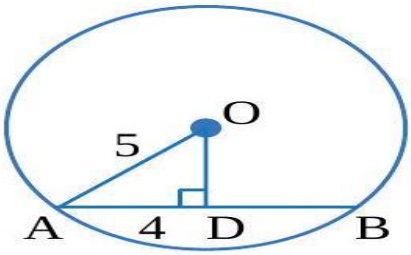
   $[\because PQ^2 + OP^2 = OQ^2] = \sqrt{15^2 - 9^2} = \sqrt{225 - 81} = \sqrt{144}$

6. Point of contact

   = 12 cm

7. $2\pi r : 2r \Rightarrow \pi : 1 = \frac{22}{7} : 1 = 22 : 7$

8. OA = r = 5 cm



Length of the chord, $AB = 8$ cm

$OD \perp AB$ and OD bisects AB

i.e $AD = DB = \frac{AB}{2} = 4$ cm

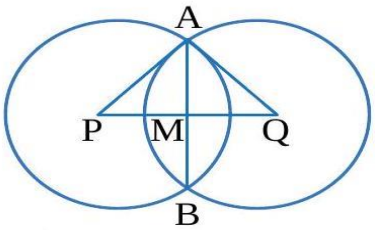
Distance of the chord, $OD = \sqrt{OA^2 - AD^2} = \sqrt{5^2 - 4^2} = \sqrt{9} = 3$ cm

9. OM ⊥ BC Then BM = MC also OM ⊥ AD Then AM = MD

$AM - BM = MD - MC$

i.e $AB = CD$

10.



$AB = 24$ cm, $AP = 15$ cm, $AQ = 13$ cm

$AM = \frac{1}{2}(AB) = \frac{1}{2}(24) = 12$ cm $\Rightarrow PM = 9$ cm

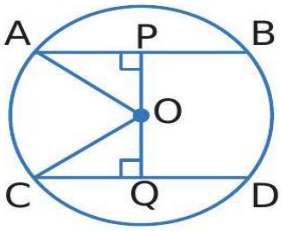
$MQ^2 = AQ^2 - A^2 = (13)^2 - (12)^2 = 169 - 144 = 25$

MQ = 5 cm

∴ PQ = PM + MQ = 9 + 5 = 14 cm

11. $AB = 6$ cm, $CD = 8$ cm, $PQ = 7$ cm

OP ⊥ AB, OQ ⊥ CD



$\therefore AP = \frac{1}{2}AB = \frac{1}{2}(6) = 3$ cm

$\& CQ = \frac{1}{2}(8) = 4$ cm

Let $OP = x$ Then $OQ = (7 - x)$

$OA^2 = OP^2 + AP^2 = x^2 + 3^2 = x^2 + 9$

$OC^2 = OQ^2 + CQ^2;\ = (7 - x)^2 + 4^2$

$\Rightarrow x^2 + 9 = (7 - x)^2 + 4^2$

$\Rightarrow x^2 + 9 = 49 + x^2 - 14x + 16 \Rightarrow x = 4$

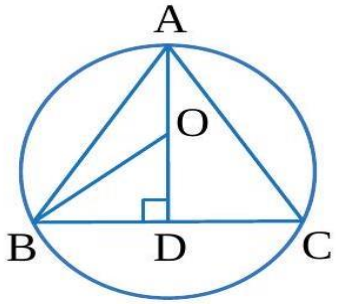
$\therefore OA^2 = 4^2 + 9 = 25 \Rightarrow OA = 5$ cm

12. $OD \perp AB \Rightarrow AD = DB$ Also $OB = OC$

In $\triangle ABC, O$ and D are mid points of $BC\&AB$

$\therefore ODIIAC$ and $OD = \frac{1}{2}AC \Rightarrow AC = 2.OD$

13. Side of Equilateral triangle, $a = 6$ cm



Height $(h) = \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{2} \times 6 = 3\sqrt{3}$ cm

Here O divides AD in the ratio2:1

$\therefore OD = \frac{1}{3}(AD) = \frac{1}{3}(h) = \frac{1}{3}(3\sqrt{3}) = \sqrt{3}$ cm

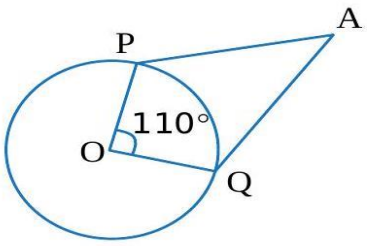
In $\Delta^{le}OBD, OB^2 = OD^2 + BD^2$

$= (\sqrt{3})^2 + \left(\frac{6}{2}\right)^2 = 3 + 9 = 12 \Rightarrow OB = \sqrt{12} \Rightarrow OB = 2\sqrt{3}$ cm

$\therefore$ Radius of the circle $= 2\sqrt{3}$ cm

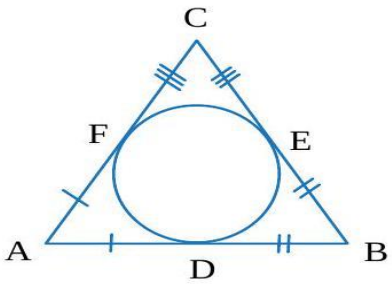
14. $POQ = 110°$

$\angle OPA = \angle OQA = 90°$



$\therefore PAQ = 360° - (110° + 90° + 90°) = 70°$

15. Given semiperimeter of $\triangle ABC = 28$ cm



i.e $\frac{AB+BC+CA}{2} = 28$ cm $\Rightarrow AB + BC + CA = 56$ cm

$\Rightarrow$ AD + DB + BE + EC + CF + FA = 56 $\Rightarrow$ AF + BD + BD + EC + CE + AF = 56

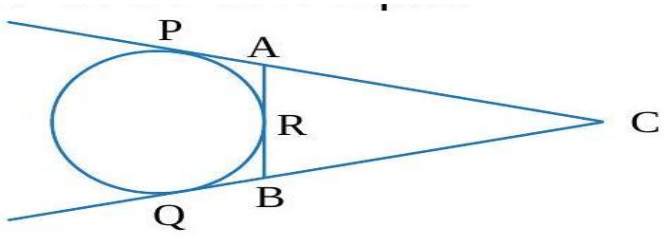
($\therefore$ length of the tangets fron on external point to a circle $AD = AF, BD = BE, CE = CF$

$\Rightarrow 2(AF + BD + CE) = 56 \Rightarrow AF + BD + CE = 28$ cm

16. W.K.T, Tangents drawn

from an external point to a circle are equal

i.e CP = CQ

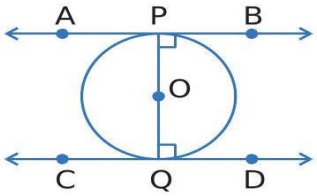


So $CQ = 11$ cm, $BC = 7$ cm

BQ and BR are tangents to the circle $BQ = BR$

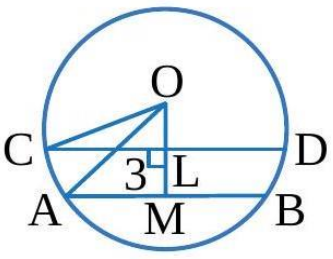
17.

$\Rightarrow$ BQ = CQ − BC = 4 cm



$AB||CD, r = 7$ cm $\Rightarrow PQ = 2r = 2(7) = 14$ cm

18.



$AB = 5$ cm, $CD = 11$ cm $\cdot AB||CD$

Let OL = x ∴ OM = OL + LM = (x + 3)

OL bisects CD ⇒ OL ⊥ CD

$CL = \frac{CD}{2} = \frac{11}{2} = 5.5$ cm ⇒ $AM = \frac{AB}{2} = \frac{5}{2} = 2.5$ cm

In Δ^{le}OAM; $OA^2 = OM^2 + AM^2 \Rightarrow OA^2 = (x+3)^2 + (2.5)^2 \rightarrow (1)$

In Δ^{le}OCL, $OC^2 = OL^2 + CL^2 \Rightarrow OC^2 = x^2 + (5.5)^2 \rightarrow (2)$

From (1) & (2) $OA^2 = OC^2 \Rightarrow (x+3)^2 + (2.5)^2 = x^2 + (5.5)^2$

$\Rightarrow 6x + 9 + 6.25 = 30.25 \Rightarrow 6x = 15 \Rightarrow x = \frac{15}{6} \Rightarrow OL = \frac{5}{2}$ cm

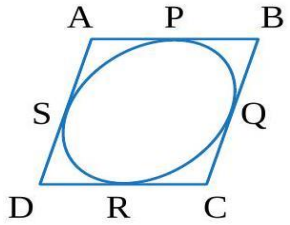
19. DC = BD = $\sqrt{13^2 - 8^2} = \sqrt{105}$

$\angle ACD = \angle ODB = 90°$

$$AD = \sqrt{AC^2 + DC^2} = \sqrt{16^2 + \left(\sqrt{105}\right)^2}$$

$= \sqrt{361} = 19$ cm

20.

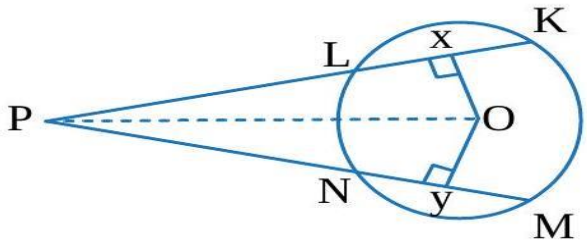


$AB + CD = AP + PB + CR + DR = AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ)$

$= AD + BC$

LEVEL – III

21.



OX ⊥ LK, OY ⊥ MN and join OP

LX = $\frac{1}{2}$LK and MY = $\frac{1}{2}$MN

In $\triangle OXP, \triangle OYP, \lfloor XP = \lfloor OYP = 90°$

$OP = OP(COMMON)$

OX = OY ∴ ΔOXP ≅ ΔOYP (RHS)

$\therefore$ PX = PY

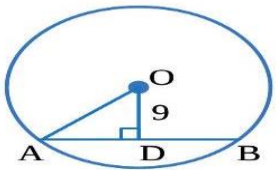
$\ddot{P}L$ + LX = PN + NY $\therefore$ PL = PN adding LK on both sides

$PL + LK = PN + LK$

PL + LK = PN + NM ($\therefore$ LK = NM)

$\Rightarrow$ PK = PM

22.



Statement I :

$AB = 12$ cm $\Rightarrow OD = 9$ cm $\Rightarrow OD \perp AB$

&OD bisects AB. i.e $AD = DB = \frac{AB}{2} = \frac{12}{2} = 6$ cm

$\therefore$ In Δ^{le}OAD, $OA^2 = OD^2 + AD^2 = 9^2 + 6^2 = 81 + 36 = 117$

$OA = \sqrt{117} \Rightarrow OA = 3\sqrt{13}$ cm

$\therefore$ Radius of the circle

$= 3\sqrt{13}$ cm

23. $I = \frac{\theta}{360^0} \times 2\pi r = \frac{70^0}{360} \times 2 \times \frac{22}{7} \times 6 = \frac{22}{3}$ cm

24. Area of the sector,

$$A = \frac{\theta}{360^0} \times \pi r^2$$
$$= \frac{45^0}{360^0} \times \frac{22}{7} \times 10^2 = \frac{275}{7} \text{ cm}^2$$

25. $r = 10$ cm, $I = \frac{110}{7}$

$$I = \frac{\theta}{360^\circ} \times 2\pi r$$
$$\Rightarrow \frac{110}{7} = \frac{\theta}{360^0} \times 2 \times \frac{22}{7} \times 10$$

26. a → s, b→t c →r d→q

27. A

28. Given diameter, d = 2r = 14 cm

r = 7 cm

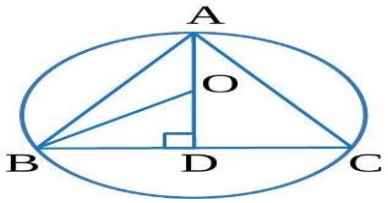
Area of circle $= \pi r^2 = \frac{22}{7} \times 7^2 = 154$ cm^2

Circumference of the circle $= 2\pi r$

$= 2 \times \frac{22}{7} \times 7$

$= 44$ cm

29.



Height (h) $= \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{2} \times 9 = \frac{9\sqrt{3}}{2}$ cm

30. Here ' O ' divides AD in th ratio $2:1$

$\therefore \text{OD} = \frac{1}{3}(\text{AD}) = \frac{1}{3}\left(\frac{9\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$ cm

$OB^2 = OD^2 + BD^2 = \left(\frac{3\sqrt{3}}{2}\right)^2 + \left(\frac{9}{2}\right)^2 = \frac{27}{4} + \frac{81}{4} = \frac{108}{4} = 27$

$\therefore \text{OB} = \sqrt{27} = 3\sqrt{3}$ cm

31. Area of the circle

$= \pi r^2 = \pi(3\sqrt{3})^2 = 27\pi \text{cm}^2$

32. $a \to r\ b \to t\ c \to t\ d \to q$

$AB = 6$ cm, $CD = 8$ cm, $OA = 5$ cm, $OC = 5$ cm

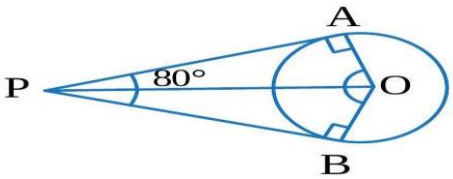
$\text{AP} = \frac{1}{2}\text{AB} = 3$ cm, $\text{CQ} = \frac{1}{2}\text{CD} = 4$ cm.

$OP^2 = OA^2 - AP^2 = 5^2 - 3^2 = 25 - 9 = 16 \Rightarrow OP = 4$

$\text{OQ}^2 = \text{OC}^2 - \text{CQ}^2 = 5^2 - 4^2 = 25 - 16 = 9 \Rightarrow \text{OQ} = 3$

$\therefore \text{PQ} = \text{OP} - \text{OQ} = 4 - 3 = 1$

33.



Given $\angle\text{APB} = 80°$ $\underline{\text{PAO}} = \underline{\text{PBO}} = 90°$

$\therefore \triangle$ OPA $\cong \triangle$ OPB (By RHS congruency criterion)

By C.P.C.T; $\underline{\text{OPA}} = \underline{\text{OPB}} = \frac{80^2}{2} = 40°$

34. $\text{POA} = 180° - (90° + 40°) = 180° - 130° = 50°$

35. $\therefore \angle\text{AOB} = \angle\text{POA} + \text{POB}$

$= 50° + 50° = 100°$

8. TRIGONOMETRY

WORKSHEET – 1

LEVEL – I

1. B	2.B	3.A	4.A	5.D	6.D	7.A
8. A	9.C	10.C	11.C	12.A	13.C	

LEVEL – II

1. D	2.A	3.A	4.A	5.C	6.A	7.B
8.D	9.D	10.D	11.D	12.D	13.B	14.C
15.0	16.B	17.B	18.A	19.A	20.D	21.6

LEVEL – III

22. A, B	23.A	24.C	25.B	26.a-t; b-p; c-s; d-q	27.B
28.D	29.A	30.A, B, C	31.A	32.D	33.D
34.a-q; b-s; c-p; d-r					

LEVEL – II

1. $\ell = r\theta \Rightarrow \theta = 60^\circ \Rightarrow r = 6$ cm

   $\ell = 6 \times \frac{\pi}{3} \Rightarrow \ell = 2\pi \Rightarrow \ell = 2x\frac{22}{7} = \frac{44}{7}$ cm

2. $\sin\theta = \frac{\text{oppposite side}}{\text{hypotenuse}} \Rightarrow$ oppposite side $= 3$ cm

   hypotenuse $= 5$ cm $\Rightarrow \sin\theta = \frac{3}{5}$

3. $\tan\theta = \frac{1}{2} \Rightarrow$ opposite side $= 1$ cm $\Rightarrow$ adjacent side $= 2$ cm

   hypotensue $= \sqrt{(1)^2 + (2)^2} = \sqrt{5} \Rightarrow \sec\theta = \frac{\text{hypotenuse}}{\text{oppposite side}} = \frac{\sqrt{5}}{2}$

4. XY $= 5$ cm $\Rightarrow$ XZ $= 13$ cm (hypotenuse) $\Rightarrow (XZ)^2 - (XY)^2 = (YZ)^2$

   $169 - 25 = (YZ)^2 \Rightarrow (YZ)^2 = 144 \Rightarrow YZ = \sqrt{144} \Rightarrow YZ = 12$ cm

5. $\cot A \cdot \sin A$

   $\frac{\text{adjacent side}}{\text{oppposite side}} \times \frac{\text{oppposite side}}{\text{hypotenuse}} \Rightarrow \frac{\text{adjacent side}}{\text{hypotenuse}} = \cos A$

6. $\sec^2\theta \times \cos^2\theta = 1$

7. $\sin\theta \times \text{cosec}\,\theta = 1 \Rightarrow \tan\theta \times \cot\theta = 1 \Rightarrow \sin\theta \times \text{cosec}\,\theta + \tan\theta \times \cot\theta = 1 + 1 = 2$

8. $\sin\theta = \frac{P}{Q} \Rightarrow \sin\theta = \frac{\text{oppposite side}}{\text{hypotenuse}}$

   opposite side ^{2} tadjacent side ^{2} = hypotenuse ^{2}

p^2 +adjacent side $^2 = Q^2$

adjacent side $^2 = Q^2 - P^2$

adjacent side $= \sqrt{Q^2 - P^2} \Rightarrow \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \Rightarrow \cos\theta = \frac{\sqrt{Q^2-P^2}}{Q}$

9. $\frac{\pi}{8} = \frac{180}{8} = \frac{45}{2} = 22\frac{1^0}{2}$

10. $\cot A = \frac{3}{4} \Rightarrow \frac{\text{adjacent side}}{\text{opposite side}} = \frac{3}{4} \Rightarrow$ hypotenuse $\Rightarrow\Rightarrow \sin A = \frac{4}{5} \Rightarrow \cos A = \frac{3}{5}$

$$= \frac{\sin A + \cos A}{\sin A - \cos A} = \frac{\frac{4}{5} + \frac{3}{5}}{\frac{4}{5} - \frac{3}{5}} = \frac{\frac{7}{5}}{\frac{1}{5}} = 7 = 7$$

11. $\sin^2\theta + \cos^2\theta = 1$

12. $2x + 3x + 4x = 180 \Rightarrow 9x = 180 \Rightarrow x = \frac{180}{9} = 20$

largest angle $= 4x = 4 \times 20 = 80$

13. $1 - \cos^2\theta = 1 - \frac{(5)^2}{(13)^2} = 1 - \frac{25}{169} = \frac{169-25}{169} = \frac{144}{169}$

14. $\sin^2\alpha + \cos^2\alpha = 1 \Rightarrow \sin^2\theta + \cos^2\theta = 1 \Rightarrow \sin^2\alpha + \cos\alpha^2 = \sin^2\theta + \cos^2\theta$

15. $\text{cosec}\,\theta = 1 \Rightarrow \frac{1}{\sin\theta} = 1 \Rightarrow \sin\theta = 1 \Rightarrow \theta = 90° \Rightarrow \cot\theta = \cot 90° = 0$

16. $\tan\theta = \frac{-4}{3} \Rightarrow \sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}} = \left(\frac{-4}{5} \text{ or } \frac{4}{5}\right) \Rightarrow \sin\theta = \frac{-4}{5} \text{ or } \frac{4}{5}$

17. $\cot\theta = x \Rightarrow \frac{1}{\tan\theta} = x \Rightarrow \tan\theta = \frac{1}{x} \Rightarrow \frac{1}{x} = \frac{\text{opposite side}}{\text{adjacent side}}$

hypotenuse $= \sqrt{x^2 + 1} \Rightarrow \text{cosec}\,\theta = \frac{\text{hypotenuse}}{\text{adjacent side}} = \frac{\sqrt{x^2+1}}{1} \Rightarrow \text{cosec}\,\theta = \sqrt{x^2 + 1}$

18. $\sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45° \Rightarrow \tan\theta = \tan 45° = 1$

19. $\text{cosec}\,\theta = \frac{17}{8} \Rightarrow \text{cosec}\,\theta = \frac{\text{hypotenuse}}{\text{opposite side}} \Rightarrow$ adjacent side $= \sqrt{289 - 64} = \sqrt{225} = 15$

$\cos\theta = \frac{\text{Adjacent side}}{\text{hypotenuse}} = \frac{15}{17}$

20. $\sin\theta = 0.5 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

$\cos\theta \times \sec\theta \times \tan\theta \times \cot\theta = 1$

21. $2\tan\theta \times 3\tan\theta \times \frac{1}{\tan^2\theta} = 6\tan^2\theta \times \frac{1}{\tan^2\theta} = 6$

LEVEL – III

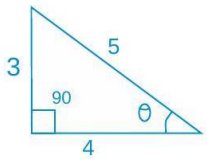
22. $\frac{g}{90} = \frac{G}{100} = \frac{2c}{\pi}$

23. $\ell = \frac{\theta}{360} \times 2\pi r = \frac{30}{360} \times 2 \times \frac{22}{7} \times 6 = \frac{22}{7}$ cm

24. $A = \frac{\theta}{360} \times \pi r^2 = \frac{30}{360} \times \frac{22}{7} \times 36 = \frac{66}{7}$ cm^2

25. $\ell = \frac{\theta}{360} \times 2\pi r \Rightarrow \frac{44}{7} = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 10 \Rightarrow \frac{44}{7} = \frac{11}{63} \times \theta \Rightarrow \theta = 36^0$

26. a) $\sin\theta + \cos\theta = \frac{3}{5} + \frac{4}{5} = \frac{7}{5}$



b) $\tan\theta - \cot\theta = \frac{3}{4} - \frac{4}{3} = \frac{9-16}{12} = \frac{-7}{12}$

c) $1 + \tan^2\theta = \sec^2\theta = \left(\frac{5}{4}\right)^2 = \frac{25}{16}$

d) $3\sin\theta - 4\cos\theta = 3 \cdot \frac{3}{5} - 4 \cdot \frac{4}{5} = \frac{9-16}{5} = \frac{-7}{5}$

27. $\sin^2 A + \cos^2 A = 1$

28. $AB^2 = AC^2 + BC^2$ (Pythogoras Theorem)

$AB^2 = 3 + 2 \Rightarrow AB = \sqrt{5} \Rightarrow \sin B = \frac{\text{oppositeside}}{\text{hypotenuse}} = \frac{AC}{AB} = \frac{\sqrt{3}}{\sqrt{5}}$

29. $\sec^2 A - \tan^2 A = 1$

30. $\angle A + \angle B + \angle C = 180^\circ \Rightarrow \angle A + \angle B + 90^\circ = 180^\circ \Rightarrow \angle A + \angle B = 180^\circ - 90^\circ = 90^\circ$

$\angle A + \frac{1}{2}\angle A = 90^\circ \Rightarrow \frac{3}{2}\angle A = 90^\circ \Rightarrow 3\angle A = 90 \times 2 = 180^\circ$

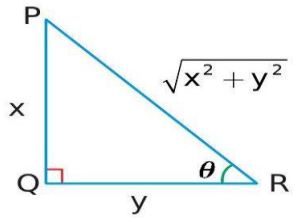
$\angle A = 60 \Rightarrow \angle B = 30$

$\Rightarrow 3\sin B - 4\sin^3 B$

$\Rightarrow 3x\sin 30 - 4\sin^3 30$

$\Rightarrow 3 \times \frac{1}{2} - 4 \times \left(\frac{1}{2}\right)^3 \Rightarrow \frac{3}{2} - 4 \times \left(\frac{1}{8}\right) \Rightarrow \frac{3}{2} - \frac{1}{2} \Rightarrow \frac{2}{2} = 1$

31. PQ = x



$QR = y \Rightarrow PR = \frac{\sqrt{x^2+y^2}}{x} \Rightarrow \sec P = \frac{\text{Hypotenuse}}{\text{adjacent side}} = \frac{\sqrt{x^2+y^2}}{x}$

32. $\sin^2 P + \sin^2 R = \frac{y^2}{\left[\sqrt{x^2+y^2}\right]^2} + \frac{x^2}{[x^2+y^2]^2} = \frac{x^2+y^2}{x^2+y^2} \Rightarrow \sin^2 P + \sin^2 R = 1$

33. $\text{cosec}^2 P - \cot^2 P = 1 \Rightarrow \text{cosec}\, p = \frac{\sqrt{x^2+y^2}}{y}; \cot p = \frac{x}{y}$

$\Rightarrow \text{cosec}^2 p - \cot^2 p \Rightarrow \frac{x^2+y^2}{y^2} - \frac{x^2}{y^2} = \frac{x^2+y^2-x^2}{y^2} = \frac{y^2}{y^2} = 1$

34. $25x^2 - 35x + 12 = 0 \Rightarrow 25x^2 - 15x - 20x + 12 = 0 \Rightarrow 5x(5x-3) - 4(5x-3) = 0$

$x = \frac{4}{5}$ or $\frac{3}{5} \Rightarrow \sin\alpha = \frac{4}{5}\ \sin\beta = \frac{3}{5} \Rightarrow \cos\alpha = \frac{3}{5}\ \cos\beta = \frac{4}{5}$

a) $\cos\alpha + \cos\beta = \frac{3}{5} + \frac{4}{5} = (a \to q)$

b) $\tan\alpha + \tan\beta = \frac{4}{3} + \frac{3}{4} = \frac{25}{12}(b \to s)$

c) $\sec\alpha + \text{cosec}\,\beta = \frac{5}{3} + \frac{5}{3} = \frac{10}{3}(c \to p)$

d) $\sec\beta + \cos\beta = \frac{5}{3} + \frac{5}{4} = \frac{35}{12}(d \to r)$

WORKSHEET – 2

LEVEL – I

1. B	2. C	3. C	4. A	5. B	6. D
7. C	8. C	9. B	10. B	11. D	

LEVEL – II

1. A	2. B	3. C	4. B	5. B	6. C
7. A	8. D	9. B	10. C	11. 0	12. A
13. C	14. A	15. A	16. C	17. C	18. C
19. C	20. B	21. B	22. A	23. A	24. C

LEVEL – III

25. A, B, C 26. B, C, D 27. B, D 28. a- p, r,s; b-p, r,s ; c- t,q; d -p, r, s
29. A, B 30. A, B, D 31. a -q, r; b-s; c-p; d -s 32. A 33. C 34. B

LEVEL – I

1. $\tan 45° = 1, 2\tan^2 60 = 2 \times \sqrt{3} \times \sqrt{3} = 6, \tan 45° + 2\tan^2 60 = 6 + 1 = 7$
2. $\sin 30° = \frac{1}{2}, \sin^2 30° = \frac{1}{4}, \cos 60° = \frac{1}{2}, \cos^2 60° = \frac{1}{4} \Rightarrow \sin^2 30 + \cos^2 60 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
3. $\cos 0° = 1, \sin 90° = 1, \sqrt{2}\sin 45° = \sqrt{2} \times \frac{1}{\sqrt{2}} \Rightarrow \cos 0° + \sin 90° + \sqrt{2}\sin 45° = 3$
4. $\sin^2\theta + \cos^2\theta = 1$
5. $\sin\theta \times \text{cosec}\,\theta = 1$
6. $\frac{1}{\tan\theta} = \cot\theta$
7. $\tan 90° = \alpha$
8. $\cot\frac{\pi}{4} = \cot 45° = 1 = \sqrt{2} \times 1 = \sqrt{2}$
9. $\sec^2 45° - \tan^2 45 = 1$
10. $\sin\theta + \cos\theta + \tan\theta = 0 + 1 + 0 = 1$
11. $\sec 2\theta = \sec 60° = 2$

LEVEL – II

1. $\frac{2\sin 45^\circ}{\operatorname{cosec} 30^\circ} = \frac{2\times\frac{1}{\sqrt{2}}}{2} = \frac{1}{\sqrt{2}}$

2. $\tan 45^\circ = 1, \cot 45^\circ = 1 \Rightarrow 1 + 1 = 2$

3. $\cos 45^\circ \cdot \sec 45^\circ + \sin 30^\circ \cdot \operatorname{cosec} 30^\circ \Rightarrow 1 + 1 = 2$

4. $\sin 30^\circ \cdot \cos 60^\circ + \sin 60^\circ \cdot \cos 30^\circ$

   $\sin 30^\circ \cdot \sin(90 - 60) + \cos(90 - 60) + \cos 30 = \sin^2 30 + \cos^2 30 = 1$

5. $\sin 0^\circ + \cos 30^\circ - \tan 45^\circ + \operatorname{cosec} 60^\circ + \cot 90^\circ = 0 + \frac{\sqrt{3}}{2} - 1 + \frac{2}{\sqrt{3}} + 0$

   $= \frac{3+4}{2\sqrt{3}} = \frac{7}{2\sqrt{3}} - \frac{1}{1} = \frac{(7-2\sqrt{3})\times 2\sqrt{3}}{2\sqrt{3}\times 2\sqrt{3}} = \frac{7\sqrt{3}-6}{6}$

6. $\sin 45^\circ = \frac{1}{\sqrt{2}}, \sin 90^\circ = 1, \tan 45^\circ = 1 = \sqrt{2}\times\frac{1}{\sqrt{2}} + 1 + 1 = 1 + 1 + 1 = 3$

7. $A = 45 \;\; B = 30$

   $\Rightarrow \frac{\sin A}{\cos A + \sin A \sin B} \Rightarrow \frac{\sin 45}{\cos 45 + \sin 45 \cdot \sin 30}$

   $\Rightarrow \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\times\frac{1}{2}} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}}} \Rightarrow \frac{\frac{1}{\sqrt{2}}}{\frac{2+1}{2\sqrt{2}}} \Rightarrow \frac{1}{\frac{3}{2}} = \frac{2}{3}$

8. $\frac{1-\cos 60^\circ}{1+\cos 60^\circ} = \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$

9. $\frac{2\tan 30}{1-\tan 30\cdot\tan 30} = \frac{2\times\frac{1}{\sqrt{3}}}{1-\frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \frac{2}{\sqrt{3}}\times\frac{3}{2} = \frac{3}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}$

10. $\cot^2 30 + \tan^2 60 - \frac{3}{2}\sin^2 45 + \operatorname{cosec}^2 30$

    $(\sqrt{3})^2 + (\sqrt{3})^2 - \frac{3}{2}\left(\frac{1}{\sqrt{2}}\right)^2 + 4 = 3 + 3 - \frac{3}{4} + 4 = 10 - \frac{3}{4} = \frac{40-3}{4} = \frac{37}{4}$

11. $4\times\frac{1}{2}\times\frac{1}{2} - 1 = 1 - 1 = 0$

    $\frac{\frac{1}{2}\times\frac{1}{2} + \frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{2}}{2\times\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}} = \frac{\frac{1}{4}+\frac{3}{4}}{1} = \frac{4}{4} = 1$

13. $\frac{1-\tan^2 30}{1+\tan^2 30} = \frac{1-\left(\frac{1}{\sqrt{3}}\right)^2}{1+\left(\frac{1}{\sqrt{3}}\right)^2} = \frac{\frac{2}{3}}{\frac{4}{3}} = \frac{1}{2} = \cos 60^\circ$

14. $\frac{\tan 60^\circ - \tan 30^\circ}{1+\tan 60^\circ\tan 30^\circ} = \frac{\sqrt{3}-\frac{1}{\sqrt{3}}}{1+\sqrt{3}\times\frac{1}{\sqrt{3}}} = \frac{\frac{3-1}{\sqrt{3}}}{2} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{1}} = \tan 30^\circ$

15. $\tan\theta = \frac{1}{\sqrt{3}}(\theta = 30^\circ) \Rightarrow \sin 3\theta = 3\sin\theta - 4\sin^3\theta = 3\times\frac{1}{2} - 4\times\frac{1}{8} = \frac{3}{2} - \frac{1}{2} = 1$

16. $4(\sin^4 30^\circ + \cos^4 60^\circ) - 3(\cos^2 45^\circ - \sin^2 90^\circ) + 5\cos^2 90^\circ$

$= 4\left(\frac{1}{16}+\frac{1}{16}\right)-3\left(\frac{1}{2}-1\right)+5(0) = 4\left(\frac{2}{16}\right)+\frac{3}{2}=\frac{4}{2}=2$

17. $\sqrt{3}\sin\theta-\cos\theta=0 \Rightarrow \sqrt{3}\sin\theta=\cos\theta=\sqrt{3}=\frac{\cos\theta}{\sin\theta}=\cot\theta \Rightarrow \cot\theta=\sqrt{3}$

$\theta = 30^\circ = \frac{1-\cot^2 30^\circ}{1+\cot^2 30^\circ}=\frac{1-3}{1+3}=-\frac{2}{4}=-\frac{1}{2}$

18. $\frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}}-\frac{1-\frac{\sqrt{3}}{2}}{\frac{1}{2}}=\frac{\frac{1}{2}}{\frac{2-\sqrt{3}}{2}}-\frac{\frac{2-\sqrt{3}}{2}}{\frac{1}{2}}=\frac{1}{2-\sqrt{3}}-(2-\sqrt{3})$

$\frac{1-(2-\sqrt{3})^2}{2-\sqrt{3}}=\frac{2\sqrt{3}}{1}=2\sqrt{3}=2\cot 30^\circ$

19. $\frac{\sqrt{3}}{1-\frac{1}{\sqrt{3}}}+\frac{\frac{1}{\sqrt{3}}}{1-\sqrt{3}}=\frac{-12-16\sqrt{3}}{-12}=\frac{12+16\sqrt{3}}{12}=1+\frac{4\sqrt{3}}{3}=1+\frac{4}{\sqrt{3}}$

$\Rightarrow 1+\operatorname{cosec}30^\circ\cdot\sec 30^\circ = 1+2\left(\frac{2}{\sqrt{3}}\right)$

$= 1+\frac{4}{\sqrt{3}}$

20. $4\left(\sin 30=\frac{1}{2},\cos 60=\frac{1}{2}\right)\Rightarrow 4\left(\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)+\left(\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\right)\right)+\frac{3}{2}=4\left(\frac{2}{16}\right)+\frac{3}{2}=\frac{4}{2}=2$

21. $\sqrt{3}\tan\theta_1=1,\tan\theta_1=\frac{1}{\sqrt{3}}(\theta_1=30^\circ)$

$\sin\theta_2=\frac{1}{2}(\theta_2=30^\circ)$ relation between $\theta_1 \& \theta_2=\theta_1=\theta_2$

22. $\frac{4}{3}\left[(\sqrt{3})^2+3\left(\frac{\sqrt{3}}{2}\right)^2-2\left(\frac{2}{\sqrt{3}}\right)^2-\frac{3}{4}\right]=\frac{4}{3}\left(3+\frac{9}{4}-\frac{8}{3}-\frac{1}{4}\right)=\frac{4}{3}\left(\frac{36+27-32-3}{12}\right)=\frac{4}{3}\left(\frac{28}{12}\right)=\frac{28}{9}$

23. $\frac{(\sqrt{3})^2+4\left(\frac{1}{\sqrt{2}}\right)^2+3\left(\frac{2}{\sqrt{3}}\right)^2+2(0)^2}{2(2)+3(2)-\frac{7}{3}(\sqrt{3})^2}=\frac{3+\frac{4}{2}+(2)^2}{4+6-7}=\frac{9}{3}=3$

24. $\frac{\tan 30^\circ}{\cot 60^\circ}+\frac{\sin 45^\circ}{\cos 45^\circ}+\frac{\sec 0^\circ}{\operatorname{cosec}90^\circ}=1+1+1=3$

LEVEL – III

25. (i) $\sin^2 A+\cos^2 B=\frac{5}{4},\sin^2 45^\circ+\cos^2 30^\circ=\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}+\frac{\sqrt{3}}{2}+\frac{\sqrt{3}}{2}=\frac{5}{4}$

(ii) $\cos^2 B=\sin^2 A,\cos^2 30-\sin^2 45^\circ=\frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{2}-\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}=\frac{1}{4}$

(iii) $\sin^2 A+\cos^2 A=1,\sin^2 45+\cos^2 45=1,\frac{1}{2}+\frac{1}{2}=\frac{2}{2}=1$

(iv) $2\sin^2 A+\cos^2 B=2\times\sin^2 45+\cos^2 30\neq 0$

26. (i) $\sin 2\theta=2\sin\theta\cdot\cos\theta=2\sin 30^\circ\cdot\cos 30^\circ=2\times\frac{1}{2}\times\frac{\sqrt{3}}{2}=\frac{\sqrt{3}}{2}$

(ii) $\cos 2\theta=\cos^2\theta-\sin^2\theta=\cos^2 30^\circ-\sin^2 30^\circ=\frac{\sqrt{3}}{2}\times\frac{\sqrt{3}}{2}-\left(\frac{1}{2}\times\frac{1}{2}\right)=\frac{3}{4}-\frac{1}{4}=\frac{2}{4}=\frac{1}{2}$

(iii) $\tan 2\theta=\frac{2\tan\theta}{1-\tan^2\theta}=\frac{2\times\frac{1}{\sqrt{3}}}{1-\frac{1}{3}}=\frac{\frac{2}{\sqrt{3}}}{1-\frac{1}{3}}=\frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}}=\frac{3}{\sqrt{3}}=\sqrt{3}$

(iv) $\cot 2\theta = \frac{1-\tan^2\theta}{2\tan\theta} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$

27. $a = b = 45°, \cot B = 1, \sin A = \cos B$

28. a –p, r,s; b–p, r,s ; c–t,q; d –p, r, s

29. If $\theta = 30°$

(i) $\tan\theta + \cot\theta = \tan 30° + \cot 30° = \frac{1}{\sqrt{3}} + \sqrt{3} = \frac{4\sqrt{3}}{3}$

(ii) If $\sin 3\theta = 3\sin\theta - 4\sin^3\theta = 3 \times \frac{1}{2} - 4 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{3}{2} - \frac{1}{2} = \frac{2}{2} = 1$

(iii) $\cot\theta = \frac{1}{\sqrt{3}}, \theta = 60°$ (iv) $\frac{\tan^2\theta}{3} = \frac{\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}}}{3} - \frac{\frac{1}{3}}{3} = 9$

30. $\theta_1 = 45°, \theta_2 = 90° - \theta_1$

(i) $\sin\theta_1 = \cos\theta_2$

(ii) $\sin 2\theta_1 = \sin 2(45) = \sin 90 = 1$

(iii) $\cos(\theta_1 + \theta_2) = \cos 90° = 0$

(iv) $\tan\theta_1 = \tan 45° = 1$

31. a) $\text{cosec}^2\ A + \cot^2\ B = 5, \text{cosec}^2 30 + \cot^2 45 = 5$

b) $\frac{1-\sin 60}{\cos 60} = \frac{1-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \frac{\frac{2-\sqrt{3}}{2}}{\frac{1}{2}} = 2 - \sqrt{3}$

c) $\sin^2 60 - 3\cos^2 60 + 2\tan 45° = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - 3 \times \frac{1}{2} \times \frac{1}{2} + 2 \times 1 = \frac{3}{4} - \frac{3}{4} + 2 = 2$

d) $\frac{1-\tan 30}{1+\tan 30} = \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{4-2\sqrt{3}}{2} = \frac{2(2-\sqrt{3})}{2} = 2 - \sqrt{3}$

32. Statement 1: $2\sin^2 45^0 + \cot^2 45^0 = 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + 1 \times 1 = \frac{2}{2} + 1 = 2$

Statement 2: $\text{cosec}^2 30 - \cot^2 30 = \text{cosec}^2 30 - \cot^2 30 = 4 - 3 = 1$ both statements are true

33. $A = 60°, B = 30°$

$$\frac{\sin 30}{1-\cos 30} + \frac{\sin 30}{1+\cos 30} = \frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}} + \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} = \frac{\frac{1}{2}}{\frac{2-\sqrt{3}}{2}} + \frac{\frac{1}{2}}{\frac{2+\sqrt{3}}{2}}$$

$$= \frac{1}{2-\sqrt{3}} + \frac{1}{2+\sqrt{3}} = \frac{2+\sqrt{3}+2-\sqrt{3}}{4-3} = \frac{4}{1} = 4$$

34. $$\frac{\cos 60}{1-\sin 60} + \frac{\cos 60}{1+\sin 60} = \frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}} + \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} = \frac{\frac{1}{2}}{\frac{2-\sqrt{3}}{2}} + \frac{\frac{1}{2}}{\frac{2+\sqrt{3}}{2}}$$

$$= \frac{1}{2-\sqrt{3}} + \frac{1}{2+\sqrt{3}} = \frac{2+\sqrt{3}+2-\sqrt{3}}{4-3} = \frac{4}{1} = 4$$

WORKSHEET – 3

LEVEL – I

1. B	2. C	3. D	4. C	5. A	6. B	7. A
8. A	9. B	10. D				

LEVEL – II

1.B	2.A	3.C	4.A	5.A	6.B	7.B
8.B	9.A	10.A	11.B	12.B	13.B	14.B
15.A	16.B	17.D	18.C	19.A	20.D	21.A
22.A	23.0	24.C	25.A	26.A		

LEVEL – III

27.A, C, D 28.A, B, C 29. a-r; b-p; c-q; d-r 30.B, D 31.B, C 32.a-s; b-r; c-p; d-q 33. A, C 34.B, D 35. a-r; b-p; c-q; d-t

LEVEL – I

1. $\sin(180+\theta) = -\sin\theta$
2. $\tan(90+\theta) = -\cot\theta$
3. $\operatorname{cosec}(270-\theta) = -\sec\theta$
4. $\cos(585) = \cos(360+225) = \cos(225) = \cos(180+45) = -\cos 45 = -\frac{1}{\sqrt{2}}$
5. $\cos 180^\circ = \cos(90+90) = -\sin(90) = -1$
6. $\sin\left(\frac{5\pi}{3}\right) = \sin(300) = \sin(360-60) = -\sin 60 = -\frac{\sqrt{3}}{2}$
7. $\cos\left(-\frac{\pi}{4}\right) = -\cos(-45) = \cos 45 = \frac{1}{\sqrt{2}}$
8. $\sin\theta = -\frac{1}{3}$, given θ lies in third quadrant then $\tan\theta$ is positive

   $AB^2 + BC^2 = AC^2, (-x)^2 + BC^2 = (3x)^2$

   $BC = \sqrt{9x^2 - x^2} = \sqrt{8x^2}$

   $BC = 2\sqrt{2}x$

   $\cos\theta = -\frac{2\sqrt{2}x}{3x} = -\frac{2\sqrt{2}}{3}$ [$\therefore \cos\theta$ isnegative in 3rd quadrant]

   $$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{1}{3}}{\frac{2\sqrt{2}}{3}} = \frac{1}{3}\times\frac{3}{2\sqrt{2}} = \frac{1}{2\sqrt{2}}$$
9. $\frac{\tan 35}{\cot 55} = \frac{\tan 35}{\cot(90-35)} = \frac{\tan 35}{\tan 35} = 1$
10. $\sin(90-\theta)\cdot\sec\theta = \cos\theta\cdot\sec\theta = 1$

LEVEL – II

1. $\cos^2 25 \cdot \operatorname{cosec} 25(\tan^{25} - \sin^2 25)$

   $\cos^2 25.\frac{1}{\sin^2 25}\left(\frac{\sin^2 25}{\cos^2 25} - \sin^2 25\right) = \cos^2 25 \cdot \frac{1}{\sin^2 25}\left(\frac{\sin^2 25 - \sin^2 25\cos^2 25}{\cos^2 25}\right)$

   $= \frac{\sin^2 25(1 - \cos^2 25)}{\sin^2 25} = (1 - \cos^2 25) = \sin^2 25$

2. $\sin^2 25 + \sin^2 65 = \sin^2 25 + \sin^2(90 - 25) = \sin^2 25 + \cos^2 25 = 1$

3. $\cos 1^0 \cdot \cos 2° \ldots\ldots\ldots\ldots \cos 179° = \cos 1° \cdot \cos 2° \ldots\ldots\ldots \cos 90° \ldots\ldots \cdot \cos 179°[\cos 90° = 0] = 0$

4. $\frac{\tan 35}{\cot 55} + \frac{\cot 78}{\tan 12} = \frac{\tan 35}{\cot(90-35)} + \frac{\cot 78}{\tan(90-78)} = \frac{\tan 35}{\tan 35} + \frac{\cot 78}{\cot 78} = 1 + 1 = 2$

5. Given $\sin A = \cos B$

   $\sin A = \sin(90 - B), A = 90 - B, A + B = 90$

6. $\frac{\sin 31°}{2\cos 59°} = \frac{\sin 31°}{2\cos(90-31)} = \frac{\sin 31°}{2\sin 31°} = \frac{1}{2}$

7. $\frac{\sin\left(\frac{\pi}{2}-\theta\right)\cdot\cot\left(\frac{\pi}{2}-\theta\right)}{\sin\theta} = \frac{\sin(90-\theta)\cdot\cot(90-\theta)}{\sin\theta} = \frac{\cos\theta\cdot\tan\theta}{\sin\theta} = \frac{\cos\theta}{\sin\theta}\cdot\tan\theta = \cot\theta\cdot\tan\theta = 1$

8. $\frac{\cos 18^0 - \cos 72^0}{\sin 72^0 - \sin 18^0} = \frac{\cos 18 - \cos(90-18)}{\sin(90-18) - \sin 18} = \frac{\cos 18 - \sin 18}{\cos 18 - \sin 18} = 1$

9. $\tan 9° \cdot \tan 27° \cdot \tan 45° \cdot \tan 63° \cdot \tan 81°$

   $= \tan 9° \cdot \tan 27° \cdot \tan 45° \cdot \cot 27° \cdot \cot 9° = \tan 45 = 1$

10. $\tan(225) = \tan(180 + 45) = \tan 45 = 1$

11. $\cot(90 - \theta) \cdot \sec(90 - \theta) = \tan\theta \cdot \operatorname{cosec}\theta = \frac{\sin\theta}{\cos\theta} \times \frac{1}{\sin\theta} = \frac{1}{\cos\theta}$

12. $\frac{\tan(90-\theta)\cdot\sec\theta\cdot\sin(90-\theta)}{\operatorname{cosec}(90-\theta)\cdot\cot\theta\cdot\cos\theta} = \frac{\cot\theta\cdot\sec\theta\cdot\cos\theta}{\sec\theta\cdot\cot\theta\cdot\cos\theta} = 1$

13. $\tan\theta = \frac{4}{3}$

    By this we can say that θ is in 2 nd (or) 4 th quadrant then $\sin\theta$ will be +ve and -ve

    $AB^2 + BC^2 = AC^2 = (4x)^2 + (3x)^2 = AC^2$

    $AC = \sqrt{25x^2} = 5x$

    $\sin\theta = \frac{4x}{5x}, -\frac{4x}{5x} = \frac{4}{5}, -\frac{4}{5}$

14. $\tan 10°\tan 20°\tan 30°\tan 40°\tan 50°\tan 60°\tan 70°\tan 80° =$

    $\tan 10°\tan 20°\tan 30°\tan 40°\cot 40°\cot 30°\cot 20°\cot 10° = 1$

15. $\left[\cos\left(-\frac{\pi}{4}\right)\right]^2 + \left[\cos\frac{3\pi}{4}\right]^2 = \left[\cos\left(\frac{\pi}{4}\right)\right]^2 + \left[\cos\frac{3\pi}{4}\right]^2 = (\cos 45°)^2 + (\cos 135°)^2$

    $= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1$

16. $\cot\left(\frac{\pi}{20}\right)\cdot\cot\left(\frac{3\pi}{20}\right)\cdot\cot\left(\frac{5\pi}{20}\right)\cdot\cot\left(\frac{7\pi}{20}\right)\cdot\cot\left(\frac{9\pi}{20}\right)\cdot\cot\left(\frac{15\pi}{20}\right)$

$= \cot 9^\circ \cdot \cot 27^\circ \cdot \cot 45^\circ \cdot \cot 63^\circ \cdot \cot 81^\circ \cdot \cot 135^\circ$

$= \cot 9^\circ \cdot \cot 27^0 \cdot \cot(90^\circ - 27^\circ) \cdot \cot(90^\circ - 9^\circ) \cdot \cot(90^\circ + 45^\circ)$

$= \cot 9^0 \cdot \cot 27^\circ \cdot 1 \cdot \tan 27^0 \cdot \tan 9^\circ(-1) = -1$

17. $\sin\theta = \mathrm{k}, \sin\theta = \frac{\mathrm{kx}}{\mathrm{x}}$

$\mathrm{AB}^2 + \mathrm{BC}^2 = \mathrm{AC}^2$

$(\mathrm{kx})^2 + \mathrm{BC}^2 = (\mathrm{x})^2$

$\mathrm{BC} = \sqrt{\mathrm{x}^2 - \mathrm{k}^2\mathrm{x}^2} = \mathrm{x}\left(\sqrt{1-\mathrm{k}^2}\right)$

$\cos\theta = \frac{\mathrm{x}\left(\sqrt{1-\mathrm{k}^2}\right)}{\mathrm{x}} = \sqrt{1-\mathrm{k}^2}$

$\tan\theta = \frac{\cos\theta}{\sin\theta} = \frac{\left(\sqrt{1-\mathrm{k}^2}\right)}{\mathrm{k}}$

18. $\sin \mathrm{A} + \sin \mathrm{B} = \mathrm{A} + \mathrm{C} = 180^\circ, \mathrm{A} = 180^\circ - \mathrm{C}$

$\mathrm{B} + \mathrm{D} = 180^\circ, \mathrm{B} = 180^\circ - \mathrm{D}$

$\sin A = \sin(180^\circ - C)$

$\sin \mathrm{B} = \sin(180^\circ - 0)$

$\sin \mathrm{A} = -\sin \mathrm{C}$

$\sin \mathrm{B} = -\sin \mathrm{D}$

$\sin \mathrm{A} + \sin \mathrm{B} = \sin \mathrm{C} + \sin \mathrm{D}$

19. $\sqrt{1 - \sin^2 100}\cdot \sec 100 = -\mathrm{x}$

$|\cos 100^\circ| \cdot \sec 100^\circ = -\mathrm{x}$

In 2 nd quadrant $\cos\theta$ is - ve

$-\cos 100^\circ \cdot \sec 100^\circ = -x$

$-1 = -\mathrm{x}, \mathrm{x} = 1$

20. In quadrilateral $\lfloor A + \lfloor B + \lfloor C + \lfloor \underline{D} = 360^\circ$

$\mathrm{A} + \mathrm{B} = 360^\circ - (\mathrm{C} + \mathrm{D})$

$\frac{\mathrm{A}+\mathrm{B}}{4} = \frac{360^\circ}{4} - \left(\frac{\mathrm{C}+\mathrm{D}}{4}\right) \Rightarrow \frac{\mathrm{A}+\mathrm{B}}{4} = 90^\circ - \left(\frac{\mathrm{C}+\mathrm{D}}{4}\right) \Rightarrow \tan\left(\frac{\mathrm{A}+\mathrm{B}}{4}\right) = \tan\left(90^\circ - \left(\frac{\mathrm{C}+\mathrm{D}}{4}\right)\right)$

$\tan\left(\frac{\mathrm{A}+\mathrm{B}}{4}\right) = \cot\left(\frac{\mathrm{C}+\mathrm{D}}{4}\right)$

21. $\frac{\cos(90+\theta)\cdot\sec(-\theta)\cdot\tan(180-\theta)}{\sec(360-\theta)\cdot\sin(180-\theta)\cdot\cot(90-\theta)} \Rightarrow \frac{(-\sin\theta)(+\sec\theta)(-\tan\theta)}{\sec\theta\cdot\sin\theta\cdot\tan\theta} = +1$

22. $\sqrt{1-\cos^2 215^\circ}\cdot \operatorname{cosec} 215^\circ \Rightarrow |\sin 215^\circ| \cdot \operatorname{cosec} 215^\circ$

In 3rd quadrant sin is negative

$-\sin 215^\circ \cdot \operatorname{cosec} 215^\circ = -1$

23. $\cos\frac{3\pi}{19}+\cos\frac{7\pi}{19}+\cos\frac{12\pi}{19}+\cos\frac{16\pi}{19}$

$\cos\frac{3\pi}{19}+\cos\frac{7\pi}{19}+\cos\left(\pi-\frac{7\pi}{19}\right)+\cos\left(\pi-\frac{3\pi}{19}\right)\Rightarrow\cos\frac{3\pi}{19}+\cos\frac{7\pi}{19}-\cos\frac{7\pi}{19}+\cos\frac{3\pi}{19}=0$

24. $\tan\theta=\frac{p}{q}\Rightarrow\sin\theta=\frac{p}{\sqrt{p^2+q^2}}\Rightarrow\cos\theta=\frac{q}{\sqrt{p^2+q^2}}$

$$\frac{P\sin\theta-q\cos\theta}{P\sin\theta+q\cos\theta}=\frac{\frac{p^2}{\sqrt{p^2+q^2}}-\frac{q^2}{\sqrt{p^2+q^2}}}{\frac{p^2}{\sqrt{p^2+q^2}}+\frac{q^2}{\sqrt{p^2+q^2}}}$$

$$\frac{p^2-q^2}{\sqrt{p^2+q^2}}\times\frac{\sqrt{p^2+q^2}}{p^2+q^2}=\frac{p^2-q^2}{p^2+q^2}$$

25. $\text{Tan}\,\theta=\frac{-4}{3}\theta\in Q_4$

Tan is negative in Q_2 by given question Sin, cosec will be tve, cos, sec will be-ve in Q_2

$$\sin\theta=\frac{4}{5}\ \text{cosec}\,\theta=\frac{5}{4}\ \cot\theta=-\frac{3}{4}\ \cos\theta=-\frac{3}{5}\ \sec\theta=-\frac{5}{3}$$

$5\sin\theta+10\cos\theta+9\sec\theta+16\text{cose}\,\theta+4\cot\theta$

$=5\left(\frac{4}{5}\right)+10\left(\frac{-3}{5}\right)+9\left(-\frac{5}{3}\right)+16\left(\frac{5}{4}\right)+4\left(-\frac{3}{4}\right)=4-6-15+20-3=0$

26. If a,b,c are in G.P then $b2=ac$

$$\sin^2A=\text{Cot}\,A\times\text{Cos}\,A\Rightarrow\text{Sin}^2A=\frac{\text{Cos}^2A}{\text{Sin}\,A}\Rightarrow\text{Sin}^3A=\text{Cos}^2A$$

$$\text{Tan}^6A-\text{Tan}^2A\Rightarrow\frac{\text{Sin}^6A}{\text{Cos}^6A}-\frac{\text{Sin}^2A}{\text{Cos}^2A}\Rightarrow\frac{\text{Cos}^4A}{\text{Cos}^6A}-\frac{\text{Sin}^2A}{\text{Con}^2A}\Rightarrow\frac{1}{\text{Cos}^2A}-\frac{\text{Sin}^2A}{\text{Cos}^2A}=\frac{1-\text{Sin}^2A}{\text{Cos}^2A}$$

LEVEL – III

27. $a^2+b^2=\text{Sin}^2A+\text{Sin}^2B\Rightarrow A+B=90^\circ\Rightarrow A=90^\circ-B$

$=\text{Sin}^2(90^\circ-B)+\text{Sin}^2B=\text{Cos}^2B+\text{Sin}^2B=1$

28. $\frac{\text{Cos}^0\,20^\circ+\text{Cos}^2\,70^0}{\text{Sin}^2\,59^0+\text{Sin}^2\,31^0}+\text{Sin}\,35^0\cdot\text{Sec}\,55^0=$

$\text{Cos}^2\,70^\circ=\text{Cos}^2(90^\circ-20^\circ)=\text{Sin}^2\,20^\circ$

$\text{Sin}^2\,31^0=\text{Sin}^2(90^\circ-59^\circ)=\text{Cos}^2\,59^0$

$\text{Sin}\,35^\circ=\text{Sin}(90^\circ-55^\circ)=\text{Cos}\,55^\circ$

$$\Rightarrow\frac{\text{Cos}^2\,20^\circ+\text{Sin}^2\,20^\circ}{\text{Sin}^2\,59^0+\text{Cos}^2\,59^0}+\text{Cos}\,55^0\cdot\text{Sec}\,55^0\Rightarrow\frac{1}{1}+1\Rightarrow1+1=2$$

29. a) $\text{Sec}\left(\frac{13\pi}{3}\right)\Rightarrow\text{sec}\left(\frac{13\times180^\circ}{3}\right)\Rightarrow\text{Sec}(60^\circ\times13)\Rightarrow\text{Sec}(780^\circ)\Rightarrow\text{Sec}(720^\circ+60^\circ)$

$\Rightarrow\text{Sec}((8)90^\circ+60^\circ)\Rightarrow\text{Sec}\,60^\circ=2/\sqrt{3}$

b) $\text{Tan}\left(\frac{4\pi^0}{3}\right) \Rightarrow \text{Tan}\left(\frac{4\times180}{3}\right) \Rightarrow \text{Tan}\,240° \Rightarrow \text{Tan}(270° - 30°) \Rightarrow \text{Cot}\,30° \Rightarrow \sqrt{3}$

c) $\text{Sec}(270° - \theta) \Rightarrow -\text{cosec}\,\theta$

d) $\sqrt{3}\text{Sec}\,750° \Rightarrow \sqrt{3} \cdot \text{Sec}(720° + 30°) \Rightarrow \sqrt{3} \cdot \text{Sec}\,30° \Rightarrow \sqrt{3} \cdot \frac{2}{\sqrt{3}} \Rightarrow 2$

30. $\text{Cos}(180 + A) + \text{Cos}(180 - B) + \text{Cos}(180 - c) - \text{Sin}(90 - D)$

$\Rightarrow -\text{Cos}\,A + (-\text{Cos}\,B) + (-\text{Cos}\,C) + (-\text{Cos}\,D) \Rightarrow -[\text{Cos}\,A + \text{Cos}\,B + \text{Cos}\,C + \text{Cos}\,D]$

$\Rightarrow -[0] \Rightarrow 0$

31. $\text{Tan}\,A + \text{Tan}(90° + A) + \text{Tan}(180 - A) + \text{Tan}(90 - A) \Rightarrow \text{Tan}\,A + (-\text{Cot}\,A) + (-\text{Tan}\,A) + \text{Cot}\,A \Rightarrow 0$

32 a) $\sin^2\frac{2\pi}{3} + \cos^2\frac{5\pi}{6} - \tan^2\frac{3\pi}{4} \Rightarrow \sin^2 120° + \cos^2 150° - \tan^2 135°$

$\rightarrow \sin^2(90° + 30°) + \cos^2(90° + 60°) - \tan^2(90° + 45°)$

$\rightarrow \cos^2 30° + (-\sin^2 60°) - (-\cot^2 45°)$

$\Rightarrow \cos^2 30° + \sin^2 60° + \cot^2 45°$

$\rightarrow \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2$

$\rightarrow \frac{3}{4} + \frac{3}{4} + 1$

$\rightarrow \frac{10}{4}$

$\rightarrow \frac{5}{2}$

b) $\sin^2\frac{11\pi}{6} + \cos^2\frac{7\pi}{6} - \tan^2\frac{5\pi}{4} \rightarrow \sin^2 330° + \cos^2 210° - \tan^2 225°$

$\rightarrow \sin^2(360° - 30°) + \cos^2(180° + 30°) - \tan^2(270° - 45°)$

$\rightarrow (-\sin^2 30°) + (-\cos^2 30°) - \cot^2 45° \rightarrow \frac{1}{4} + \frac{3}{4} - 1 \rightarrow \frac{1 + 3 - 4}{4} \rightarrow \frac{0}{4} = 0$

c) $\sin^2\frac{3\pi}{4} + \sec^2\frac{5\pi}{3} + \tan^2\frac{2\pi}{3}$

$\rightarrow \sin^2 135° + \sec^2 300° + \tan^2 120°$

$\rightarrow \sin^2(90° + 45°) + \sec^2(360° - 60°) + \tan^2(90° - 30°)$

$\rightarrow \cos^2 45° + \sec^2 60° + (-\cot^2 30°)$

$\rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 + (2)^2 + (\sqrt{3})^2 \rightarrow \frac{1}{2} + 4 + 3 \rightarrow \frac{2 + 16 + 12}{4} \rightarrow \frac{30}{4} = \frac{15}{2}$

d) $\cos^2\frac{2\pi}{3} + \sin^2\frac{5\pi}{3} + \tan^2\frac{3\pi}{4} \rightarrow \cos^2 120° + \sin^2 300° + \tan^2 135°$

$\rightarrow \cos^2(90° + 30°) + \sin^2(270° + 30°) + \tan^2(90° + 45°)$

$\rightarrow (-\sin^2 30°) + (-\cos^2 30°) + (-\cot^2 45°)$

$\rightarrow \sin^2 30° + \cos^2 30° + \cot^2 45° \Rightarrow \frac{1}{4} + \frac{3}{4} + 1 = \frac{1+3+4}{4} = \frac{8}{4} = 2$

33. i) A, B, C are angles of a $\triangle ABC$,

$\rightarrow \cos\left(\frac{3A+2B+C}{2}\right) + \cos\left(\frac{A-C}{2}\right) \rightarrow \cos\left(\frac{2A+B+180°}{2}\right) + \cos\left(\frac{A-C}{2}\right)$

$\rightarrow \cos\left(90° + \frac{180° - C + A}{2}\right) + \cos\left(\frac{A-C}{2}\right) \rightarrow \cos\left(180° + \frac{A-C}{2}\right) + \cos\left(\frac{A-C}{2}\right)$

$\rightarrow -\cos\left(\frac{A-C}{2}\right) + \cos\left(\frac{A-C}{2}\right)$

$\rightarrow 0$

34. $\cos\theta + \cos(\pi + \theta) + \cos(2\pi + \theta) + \cdots \ldots + \cos(20\pi + \theta) =$

$\rightarrow \cos\theta - \cos\theta + \cos\theta - \cos\theta + \cdots \ldots + \cos\theta = \cos\theta$

35. a) $x\tan^2 120° + 4\cos^2 150 = 9^0$ then $x = ?$

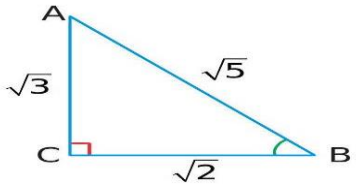
$\Rightarrow x\cot^2 30° + 4(\sin^2 60°) = 9^0$

$\Rightarrow 3x + 3 = 9$

$\Rightarrow 3x = 6$

$\Rightarrow x = 2$

b) $3\sin^2 x + 3\cos^2 x = 5, \cot x = ?$



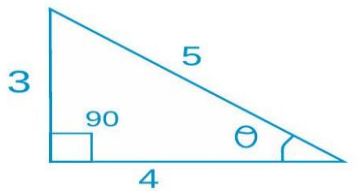
$5\sin^2 x + 3[1] = 5$

$\sin^2 x = \frac{2}{5}$

$\sin x = \frac{\sqrt{2}}{\sqrt{5}}$

$\cos x = \frac{\sqrt{3}}{\sqrt{5}}$

$\cot x = \frac{\sqrt{3}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}} = \pm\sqrt{\frac{3}{2}}$



c) $\sqrt{1 - \cos^2 100°} \cdot \operatorname{cosec} 100° \Rightarrow |\sin 100°| \cdot \operatorname{cosec} 100° \Rightarrow 1$

d) $\sin\alpha = \frac{3}{5}, \sin\alpha\cos\beta - \cos\alpha\sin\beta$

$\alpha + \beta = 90°$

$\Rightarrow \sin\alpha\cos(90 - \alpha) - \cos\alpha\sin(90 - \alpha) \Rightarrow \sin^2\alpha - \cos^2\alpha$

$\cos\alpha = ?$

$\cos\alpha = \frac{4}{5} \Rightarrow \left(\frac{3}{5}\right)^2 - \left(\frac{4}{5}\right)^2 \Rightarrow \frac{9}{25} - \frac{16}{25} \Rightarrow -\frac{7}{25}$

WORKSHEET – 4

LEVEL – I

1. A	2. C	3. B	4. D	5. A
6. C	7. A	8. D	9. A	10. A

LEVEL – II

1. C	2. C	3. C	4. A	5. A	6. A	7. B
8. B	9. D	10. A	11. A	12. C	13. B	14. A
15. A	16. B	17. A	18. D	19. B	20. D	21. B
22. 7						

LEVEL – III

23. B, C, D	24. A, D	25. B, C	26. D	27. A	28. A
29. a-r,s ,b-q,c-t,d-p	30. A	31. A	32. A, B, C, D		33. B
34. D					

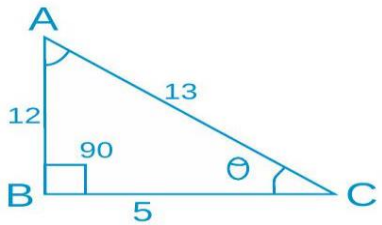
LEVEL – I

1. $\sin^2\theta + \cos^2\theta = 1$
2. $\sec^2\theta - \tan^2\theta = 1$

   $\sec^2\theta = 1 + \tan^2\theta$

   $\sec\theta = \pm\sqrt{1 + \tan^2\theta}$
3. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$

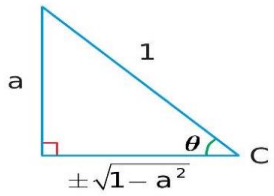
   $= 1 + 2\sin\theta\cos\theta$
4. $\cos\theta = \frac{\sqrt{3}}{2}; \sin^2\theta + \cos^2\theta = 1$
5. $\sin\theta = \frac{12}{13}; \tan\theta = ?$



$\cos\theta = \frac{5}{13} \Rightarrow \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{12/13}{5/13} = \frac{12}{5}$

6. $(1-\sin^2\theta)\sec^2\theta = (\cos^2\theta)\cdot\sec^2\theta = 1$
7. $(1+\tan^2\theta)(1-\sin\theta)(1+\sin\theta) = (\sec^2\theta)(1-\sin^2\theta) = (\sec^2\theta)(\cos^2\theta) = 1$
8. $(\sec^2\theta-1)(\operatorname{cosec}^2\theta-1) = (\tan^2\theta)(\cot^2\theta) = 1$
9. $(1+\cot^2\theta)\sin^2\theta = (\operatorname{cosec}^2\theta)(\sin^2\theta) = 1$
10. If $\sin\theta = a$; $\cos\theta =$

$1^2 = a^2 + x^2$
$x^2 = 1 - a^2$
$x = \pm\sqrt{1-a^2} \Rightarrow \cos\theta = \pm\sqrt{1-a^2}$



LEVEL – II

1. $\cot\theta - \tan\theta = \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = \frac{\cos^2\theta-\sin^2\theta}{\sin\theta\cos\theta} = \frac{2\cos^2\theta-1}{\sin\theta\cos\theta}$
2. $\tan\theta = \sqrt{3} \Rightarrow \tan\theta = \tan 60^\circ \Rightarrow \theta = 60^\circ \Rightarrow \sin 60^\circ = \frac{\sqrt{3}}{2}$
3. $\cot^2\theta = 5; \operatorname{cosec}\theta \Rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1$

   $\Rightarrow \operatorname{cosec}^2\theta - 5 = 1 \Rightarrow \operatorname{cosec}\theta = \sqrt{6}$
4. $x = \sec\theta + \tan\theta; y = \sec\theta - \tan\theta \Rightarrow xy = (\sec\theta+\tan\theta)(\sec\theta-\tan\theta)$ $xy = \sec^2\theta - \tan^2\theta = 1 \Rightarrow xy = 1$
5. $(\operatorname{cosec}\theta+\cot\theta)(\operatorname{cosec}\theta-\cot\theta) \Rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1$
6. $\left(1+\frac{1}{\tan^2\theta}\right)\left(1+\frac{1}{\cot^2\theta}\right) \Rightarrow \left(\frac{\tan^2\theta+1}{\tan^2\theta}\right)\left(\frac{\cot^2\theta+1}{\cot^2\theta}\right)$

   $\Rightarrow (\sec^2\theta)(\operatorname{cosec}^2\theta)$
7. $(\sin\theta+\cos\theta)^2 + (\sin\theta-\cos\theta)^2$

   $\Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta \Rightarrow 2[\sin^2\theta + \cos^2\theta]$

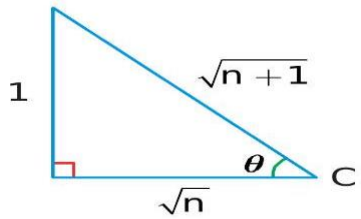
   $\Rightarrow 2[1] \Rightarrow 2$
8. $(\operatorname{cosec}\theta+\cot\theta) = P$ then $(\operatorname{cosec}\theta-\cot\theta) = x$

   $(\operatorname{cosec}\theta+\cot\theta)(\operatorname{cosec}\theta-\cot\theta) = 1$

   $P(x) = 1 \Rightarrow x = \frac{1}{P}$

   $\therefore (\operatorname{cosec}\theta-\cot\theta) = \frac{1}{P}$

9.



$$\Rightarrow \cos\theta = \frac{\sqrt{n}}{\sqrt{n+1}} \Rightarrow \cos\theta = \sqrt{\frac{n}{n+1}}$$

10. $x = \frac{\tan\theta}{\sec\theta}; y = \frac{\cot\theta}{\operatorname{cosec}\theta} \Rightarrow x^2 + y^2 = \frac{\tan^2\theta}{\sec^2\theta} + \frac{\cot^2\theta}{\operatorname{cosec}^2\theta}$

$= \frac{\sin^2\theta}{\cos^2\theta} \times \cos^2\theta + \frac{\cos^2\theta}{\sin^2\theta} \times \sin^2\theta = \sin^2\theta + \cos^2\theta = 1$

11. $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \sqrt{\frac{1+\cos\theta}{1-\cos\theta} \times \frac{1+\cos\theta}{1+\cos\theta}} = \sqrt{\frac{(1+\cos\theta)^2}{1-\cos^2\theta}} = \sqrt{\frac{(1+\cos\theta)^2}{\sin^2\theta}} = \frac{1}{\sin\theta}(1+\cos\theta)$

$= \frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} = \operatorname{cosec}\theta + \cot\theta$

12. $\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \sqrt{\frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}} = \sqrt{\frac{\sin^2\theta+\cos^2\theta}{\cos^2\theta\sin^2\theta}} = \frac{1}{\cos\theta\sin\theta}$

$= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \frac{\sin^2\theta}{\sin\theta\cos\theta} + \frac{\cos^2\theta}{\sin\theta\cos\theta} = \tan\theta + \cot\theta$

13. $\sin\theta + \sin^2\theta = 1 \Rightarrow \sin\theta = 1 - \sin^2\theta \Rightarrow \sin\theta = \cos^2\theta$

then $\cos^2\theta + \cos^4\theta = \sin\theta + \sin^2\theta = 1$

14. $\tan^2 1^\circ \cdot \tan^2 2^\circ \tan^2 89^\circ$

$= \tan^2 1^0.22^\circ \tan^2 45^\circ \tan^2 88^\circ \cdot \tan^2 89^\circ$

$= \tan^2 1^0.22^\circ (1) \cot^2 2 \cdot \cot^2 1$

$= 1$

15. $\cos\theta + \sin\theta = \sqrt{2}$

$(\cos\theta + \sin\theta)^2 = (\sqrt{2})^2$

$\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta = 2$

$1 + 2\cos\theta\sin\theta = 2 \Rightarrow \cos\theta\sin\theta = \frac{1}{2}$

$(\cos\theta - \sin\theta)^2 = (\cos\theta + \sin\theta)^2 - 4\cos\theta\sin\theta$

$(\cos\theta - \sin\theta)^2 = (\sqrt{2})^2 - 4\left(\frac{1}{2}\right) \Rightarrow (\cos\theta - \sin\theta)^2 = 2 - 2 \Rightarrow \cos\theta - \sin\theta = 0$

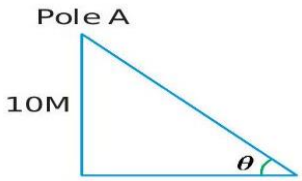
16. $x = a\sec\theta\ y = b\tan\theta$

$\frac{x}{a} = \sec\theta\ \frac{y}{b} = \tan\theta$

$\frac{x^2}{a^2} = \sec^2\theta\ \frac{y^2}{b^2} = \tan^2\theta$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

17. Height of pole = 10mt



Shadow = $10\sqrt{3}$mts

length of shadow = $10\sqrt{3}$mt $\Rightarrow \tan\theta = \frac{10}{10\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \tan\theta = \tan 30° \Rightarrow \theta = 30°$

18. $x = 2\text{cosec}\,\theta\ y = 2\cot\theta$

$x^2 - y^2 = 4\text{cosec}^2\,\theta - 4\cot^2\theta = 4(\text{cosec}^2\,\theta - \cot^2\theta) = 4$

19. $\sin^2 1° + \sin^2 2° + \cdots \ldots \sin^2 90°$

$\sin^2 1° + \sin^2 2° + \cdots \ldots . \sin^2 45° + \cdots \ldots . \sin^2 88° + \sin^2 89° + 1$

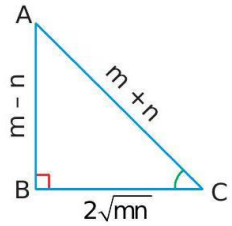
$= \sin^2 1^0 + \sin^2 2° + \cdots \ldots \ldots \ldots + \frac{1}{2} + \cdots \ldots \ldots + \cos^2 2° + \cos^2 1^0 + 1 = 44 + \frac{1}{2} + 1$

$= 45 + \frac{1}{2} = \frac{91}{2}$

20. $\cos^2 1^0 + \cos^2 2° + \cdots \ldots . \cos^2 45° + \cdots \ldots + \cos^2 88° + \cos^2 89° + 0$

$= \cos^2 1^0 + \cos^2 2° + \cdots \ldots \ldots . . \frac{1}{2} + \cdots \ldots . . \sin^2 2° + \sin^2 1^0 = 44 + \frac{1}{2} = \frac{89}{2}$

21. $\sec\theta = \frac{m+n}{2\sqrt{mn}}$



$\cos\theta = \frac{2\sqrt{mn}}{m+n} \Rightarrow AB^2 + BC^2 = AC^2 \Rightarrow AB = \sqrt{(m+n)^2 - (2\sqrt{mn})^2} = m - n$

$\sin\theta = \frac{m-n}{m+n}$

22. $(\sin\alpha + \text{cosec}\,\alpha)^2 + (\cos\alpha + \sec\alpha)^2 - \tan^2\alpha - \cot^2\alpha$

$\sin^2\alpha + \text{cosec}^2\,\alpha + 2\sin\alpha\text{cosec}\,\alpha + \cos^2\alpha + \sec^2\alpha + 2\cos\alpha\sec\alpha - \tan^2\alpha - \cot^2\alpha\ 1 +$

$1 + 1 + 2 + 2 = 7$

LEVEL – III

23. $\sec\theta = \sec^2\theta - \tan^2\theta = 1 = \sec^2\theta = 1 + \tan^2\theta = \sec\theta = \sqrt{1 + \tan^2\theta}$

24. $\sin^2 A \cdot \cos^2 B - \cos^2 A\sin^2 B = \sin^2 A(1 - \sin^2 B) - (1 - \sin^2 A)(\sin^2 B)$

$= \sin^2 A - \sin^2 A\sin^2 B - \sin^2 B + \sin^2 A\sin^2 B = \sin^2 A - \sin^2 B$

25. $(\operatorname{cosec}\theta - \cot\theta)(1 + \cos\theta) =$

$= \left(\frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta}\right)(1 + \cos\theta) = \frac{(1 - \cos\theta)(1 + \cos\theta)}{\sin\theta} = \frac{1 - \cos^2\theta}{\sin\theta} = \frac{\sin^2\theta}{\sin\theta} = \sin\theta$

26. $\sin A \cdot \cos A =$

$\tan A + \cot A = 2$

$\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = 2 \sin A\cos A = \frac{1}{2}$

$\frac{\sin^2 A + \cos^2 A}{\sin A\cos A} = 2 \Rightarrow \frac{1}{\sin A\cos A} = 2$

27. The value of $\sec A \cdot \operatorname{cosec} A = -\frac{1}{\sin A\cos A} = 2$

28. $\sin^2 A - \cos^2 A$

$(\cos A + \sin A)^2 = 2 \Rightarrow 1 + 2\sin A\cos A = 2$

$\sin A\cos A = \frac{1}{2}$

$(\cos A - \sin A)^2 = (\cos A + \sin A)^2 - 4\sin A\cos A$

$(\cos A - \sin A)^2 = 2 - 4\left(\frac{1}{2}\right)$

$\cos A - \sin A = 0$

$\sin^2 A - \cos^2 A = (\sin A + \cos A)(\sin A - \cos A) = (\sqrt{2})(0)$

29. (a) $\sin^4 A - \cos^4 A$

$(\sin^2 A)^2 - (\cos^2 A)^2$

$= (\sin^2 A + \cos^2 A)(\sin^2 A - \cos^2 A) = (1)(\sin^2 A - \cos^2 A) = \sin^2 A - \cos^2 A$

$= 1 - 2\cos^2 A = 2\sin^2 A - 1$

(b) $1 - \frac{\cos^2 A}{1+\sin A} = \frac{1+\sin A-1+\sin^2 A}{1+\sin A} = 1 - \frac{\sin A(1+\sin A)}{1+\sin A} = \sin A$

(c) $\cos A + \sin A = \sqrt{2}$

then $\frac{\cos A-\sin A}{\sqrt{2}} = ?$

$= \cos^2 A + \sin^2 A + 2\sin A\cos A = 2 = \sin A\cos A = \frac{1}{2}$

$= (\sin A - \cos A)^2 = (\sin A + \cos A)^2 - 4\sin A\cos A$

$= (\sin A - \cos A)^2 = 2 - 2 = \cos A - \sin A = 0 = \frac{\cos A - \sin A}{\sqrt{2}} = \frac{0}{\sqrt{2}} = 0$

(d) $\tan^2 A + \tan^4 A = \tan^2 A[1 + \tan^2 A] = \tan^2 A(\sec^2 A)$

$= \frac{\sin^2 A}{\cos^2 A}\left[\frac{1}{\cos^2 A}\right] = \frac{\sin^2 A}{\cos^4 A} = \sec^4 A - \sec^2 A$

30. Statement-I: right angle; $\sin^2 A + \sin^2 B + \sin^2 C = 2$

in a right angle triangle

one angle is $90°$

let $C = 90°$

$\sin^2 C = \sin^2 90° = 1^2 = 1$

$A + B = 90°$

$\sin B = \sin(90° - A°)$

$\sin B = \cos A$

$\sin^2 A + \cos^2 A + \sin^2 C = 2$

$1 + 1 = 2$

$2 = 2$

Statement-II: α and β are complementary angles

$\alpha + \beta = 90°$

$\beta = 90° - \alpha \sin^2 \alpha + \sin^2 \beta = 1$

$\sin^2 \alpha + \cos^2 \alpha = 1 \sin^2 \alpha + \sin^2(90° - \alpha) = 1$

$1 = 1$

31. Statement-I: $\frac{1-\tan^2 30°}{1+\tan^2 30°} = \frac{1}{2} \Rightarrow \frac{1-\left(\frac{1}{\sqrt{3}}\right)^2}{1+\left(\frac{1}{\sqrt{3}}\right)^2} = \frac{1}{2} \Rightarrow \frac{\frac{3-1}{3}}{\frac{3+1}{3}} = \frac{2}{4} = \frac{1}{2} = \frac{1}{2}$

Statement-II $\cos 2\theta = \frac{1-\tan^2 \theta}{1+\tan^2 \theta}$

32. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1 = \frac{1-\tan^2 A}{1+\tan^2 A}$:

33. $9\sec^2 A - 9\tan^2 A = 9(\sec^2 A - \tan^2 A) = 9(1) = 9$

34. $\frac{1+\tan^2 A}{1+\cot^2 A} = \frac{\sec^2 A}{\operatorname{cosec}^2 A} = \frac{1}{\cos^2 A} \times \frac{\sin^2 A}{1} = \tan^2 A$

WORKSHEET – 5

LEVEL – I

1. A	2. C	3. B	4. A	5. D	6. B	7. A
8. A	9. B	10. A				

LEVEL – II

1. B	2. A	3. C	4. C	5. B	6. B	7. A
8. B	9. C	10. D	11. A	12. C	13. B	14. C
15. B	16. A	17. B	18. A	19. A	20. C	21. A
22. A	23. A	24. ± A				

LEVEL – III

25. B 26. C 27. C 28. B 29. A 30. a-s,b-q,c-r,d-p

31. A,B,C,D 32. a-r,b-s,c-p,d-q 33. A, C, D 34. A

LEVEL – I

1. $\sin(A+B) = \sin A\cos B + \cos A\sin B$
2. $\sin(A-B) = \sin A\cos B - \cos A\sin B$
3. $\cos(A+B) = \cos A\cos B - \sin A\sin B$
4. $\cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$
5. $\cos(\mathrm{A}-\mathrm{B}) = \cos \mathrm{A}\cos \mathrm{B} + \sin \mathrm{A}\sin \mathrm{B}$
6. $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$
7. $\mathrm{Tan}\, 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3}$
8. $\sin^2 A - \sin^2 B = \sin(A+B)\cdot\sin(A-B)$
9. $\cos^2 A - \sin^2 B = \cos(A+B)\cdot\cos(A-B)$
10. $\cot 15^\circ = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3}$

LEVEL – II

1. $\sin 20^\circ\cos 40^\circ + \cos 20^\circ\sin 40^\circ = \sin(20^\circ+40^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$
2. $\sin(x+y)+\sin(x-y) = (\sin x\cos y + \cos x\cdot\sin y) + (\sin x\cos y - \cos x\sin y)$
3. $\cos 100^\circ\cos 40^\circ + \sin 100^\circ\sin 40^\circ = \cos(100-40^\circ) = \cos 60^\circ = \frac{1}{2}$
4. $\cos A = \frac{5}{13}, \sin B = \frac{4}{5} \Rightarrow \sin A = \frac{12}{13}, \cos B = \frac{3}{5}$

   $\sin(A+B) = \sin A\cos B + \cos A\cdot\sin B$
5. $\cos(60+\theta)+\cos(60-\theta) = 2\cos 60\cdot\cos\theta = 2\times\frac{1}{2}\times\cos\theta = \cos\theta$
6. $\tan(45+\theta)\tan(45-\theta) \Rightarrow \left(\frac{1+\mathrm{Tan}\,\theta}{1-\mathrm{Tan}\,\theta}\right)\left(\frac{1-\mathrm{Tan}\,\theta}{1+\mathrm{Tan}\,\theta}\right) = 1$
7. $\sin 60^\circ\cdot\cos 30^\circ + \cos 60^\circ\sin 30^\circ = \sin(60^\circ+30^\circ) = \sin 90^\circ = 1$
8. $\sin^2 45^\circ - \sin^2 15^\circ = \sin(45^\circ+15^\circ)\cdot\sin(45^\circ-15^\circ) = \sin 60^\circ\cdot\sin 30^\circ = \frac{\sqrt{3}}{2}\cdot\frac{1}{2} = \frac{\sqrt{3}}{4}$
9. $\frac{\tan 23^\circ+\tan 22^\circ}{1-\tan 23^\circ\tan 22^\circ} = \mathrm{Tan}(23^\circ+22^\circ) = \mathrm{Tan}\,45^\circ = 1$
10. $\sec\theta+\tan\theta = P \Rightarrow \sec\theta-\tan\theta = \frac{1}{P} \Rightarrow 2\sec\theta = \frac{P^2+1}{P}, 2\tan\theta = \frac{P^2-1}{P}$

    $\sec\theta = \frac{P^2+1}{2P}\tan\theta = \frac{P^2-1}{2P} \Rightarrow \sin\theta = \frac{\tan\theta}{\sec\theta} = \frac{p^2-1}{p^2+1}$

11. $\cos A = \frac{3}{5} \sin B = \frac{5}{13} \Rightarrow \sin A = \frac{4}{5} \cos B = \frac{12}{13}$

$$\tan A = \frac{4}{3} \tan B = \frac{5}{12} \Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3} \cdot \frac{5}{12}} = \frac{\frac{48 + 15}{36}}{\frac{36 - 20}{36}} = \frac{63}{16}$$

12. $\cos^2 45° - \cos^2 15° = \sin(A + B) \cdot \sin(A - B) = \sin 60 \cdot \sin(-30°)$

$$= \sin 60°(-\sin 30°) = \frac{\sqrt{3}}{2} \times \frac{-1}{2} = \frac{-\sqrt{3}}{4}$$

13. $A + B = 45° \Rightarrow \tan(A + B) = \tan 45° \Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$

$1 - \tan A \tan B = \tan A + \tan B$

$\tan A + \tan B + \tan A \tan B = 1$

$1 + \tan A + \tan B + \tan A \tan B = 2$

$(1 + \tan A)(1 + \tan B) = 2$

14. $55° = 20° + 35° \Rightarrow \text{Tan } 55° = \frac{\text{Tan } 20° + \text{Tan } 35°}{1 - \text{Tan } 20° \text{Tan } 35°}$

$\text{Tan } 55°(1 - \text{Tan } 20° \text{Tan } 35°) = \text{Tan } 20° + \text{Tan } 35°$

$\text{Tan } 55° - \text{Tan } 55° \text{Tan } 20° \text{Tan } 35° = \text{Tan } 20° + \text{Tan } 35°$

$\text{Tan } 55° - 2\text{Tan } 20° - \text{Tan } 35° = 0, \text{ x} = 55°$

15. $A + B = 135° \Rightarrow \cot(A + B) = \cot(135°) \Rightarrow \frac{\cot A \cot B - 1}{\cot B + \cot A} = -1$

$\cot A \cdot \cot B - 1 = -\cot B - \cot A \Rightarrow \cot A \cot B + \cot A + \cot B = 1$

$1 + \cot A \cot B + \cot A + \cot B = 2 \Rightarrow (1 + \cot A)(1 + \cot B) = 2$

16. $(1 + \tan 13)(1 + \tan 32) = 2$

If $A + B = 45° \Rightarrow (1 + \tan A)(1 + \tan B) = 2$

$(1 + \tan 12)(1 + \tan 33) = 2 \Rightarrow \frac{2}{2} = 1$

17. $\cos^2 48 - \sin^2 12 \Rightarrow \cos(A + B)\cos(A - B) \Rightarrow \cos(60°) \cdot \cos(36°)$

$$\left(\frac{1}{2}\right)\left(\frac{\sqrt{5} + 1}{4}\right) = \frac{\sqrt{5} + 1}{8}$$

18. $\tan \text{A} = \frac{1}{2}, \tan \text{B} = \frac{1}{3}$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1 \Rightarrow \tan(A + B) = 1$$

19. $\cot 20° = \text{Ptan } 20° = \frac{1}{\text{P}} \Rightarrow \tan 160° = \tan(180 - 20°) = -\tan 20° = -\frac{1}{\text{P}}$

$$\tan 110° = \tan(90 + 20°) = -\cot 20 = -\text{P} \Rightarrow \frac{-\frac{1}{P} + \text{P}}{1 + \left(\frac{-1}{P}\right)(-\text{P})} = \frac{\frac{\text{P}^2 - 1}{\text{P}}}{2} = \frac{\text{P}^2 - 1}{2\text{P}}$$

20. $\frac{\cos(A-B)}{\cos(A+B)}+\frac{\cos(C+D)}{\cos(C-D)}=0$

$\Rightarrow \cos(A-B)\cos(C-D)+\cos(C+D)\cos(A+B)=0$

Let $\cos A\cos B=a, \sin A\sin B=b\ \cos C\cos D=c, \sin C\sin D=d$

$(a+b)(c+d)+(a-b)(c-d)=0$

$ac+ad+bc+bd+ac-ad-bc+bd=0$

$2(\mathrm{ac}+\mathrm{bd})=0$

$\cos A\cos B\cos C\cos D=-\sin A\sin B\sin C\sin D$

$\mathrm{Tan\ ATan}\ B\mathrm{TanC\ Tan\ D}=-1$

21. $\sin^2(51^\circ-x)+[\sin 90-(51-x)]^2$

$\sin^2(51^0-x)+\cos^2(51-x)=1$

22. $A+B=300^\circ \Rightarrow \cot(A+B)=\cot 300^\circ \Rightarrow \frac{\cot A\cot B-1}{\cot B+\cot A}=\cot(360^\circ-60^\circ)$

$\frac{\cot A\cot B-1}{\cot B+\cot A}=-\cot 60^\circ \Rightarrow \frac{\cot A\cot B-1}{\cot B+\cot A}=\frac{-1}{\sqrt{3}}$

$\sqrt{3}\cot A\cot B-\sqrt{3}=-\cot B-\cot A$

$3\cot A\cot B+\sqrt{3}\cot A+\sqrt{3}\cot B=3$

$1+\sqrt{3}\cot A+\sqrt{3}\cot B+3\cot A\cot B=3+1$

$(1+\sqrt{3}\cot A)(1+\sqrt{3}\cot B)=4$

23. $\frac{\sin(A-B)}{\cos A\cos B}\Rightarrow\frac{\sin A\cos B-\cos A\cdot\sin B}{\cos A\cos B}=\mathrm{Tan}\ A-\mathrm{Tan}\ B$

Similarly $\Rightarrow \mathrm{Tan}\ A-\mathrm{Tan}\ B+\mathrm{Tan}\ B-\mathrm{Tan}\ C+\mathrm{Tan}\ C-\mathrm{Tan}\ A=0$

24. $\mathrm{Let}(\sec A+\tan A)(\sec B+\tan B)(\sec C+\tan C)=k$

$=(\sec A-\tan A)(\sec B-\tan B)(\sec C-\tan C)\Rightarrow k^2=1\Rightarrow k=\pm 1$

LEVEL – III

25. $\tan(\mathrm{A}-\mathrm{B})=\frac{\tan \mathrm{A}-\tan \mathrm{B}}{1+\tan \mathrm{A}\tan \mathrm{B}}=\frac{\frac{1}{35}-\frac{18}{17}}{1+\frac{18}{1735}\frac{1}{}}=\frac{-613}{613}=-1$

26. $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{\frac{m}{n}+\frac{n-m}{n+m}}{1-\frac{mn-m}{n\,n+m}}$

$=\frac{mn+m^2+n^2-mn}{n(n+m)}\times\frac{n(n+m)}{n^2+mn-mn+m^2}=1$

$\tan(A+B)=1\Rightarrow A+B=\pi/4$

$\tan(A+B)=\frac{\frac{1}{2}+\frac{2}{3}}{1-\frac{1}{2}\cdot\frac{2}{3}}=\frac{\frac{3+4}{6}}{\frac{6-2}{6}}=\frac{7}{4}$

27. $\mathrm{A}+\mathrm{B}=\frac{\pi}{4}\Rightarrow\frac{\tan \mathrm{A}+\tan \mathrm{B}}{1-\tan \mathrm{A}\tan \mathrm{B}}=1$

$\tan A + \tan B = 1 - \tan A \tan B \Rightarrow \tan A + \tan B + \tan A \tan B = 1$

$1 + \tan A + \tan B + \tan A \tan B = 2 \Rightarrow (1 + \tan A)(1 + \tan B) = 2$

28. $\frac{\cot A \cot B - 1}{\cot B + \cot A} = 1 \Rightarrow \cot A \cdot \cot B - 1 = \cot A + \cot B$

$\cot A \cot B - \cot A - \cot B = 1 \Rightarrow (1 - \cot A)(1 - \cot B) = 2$

29. $\tan(A + B) = \tan\frac{\pi}{4} = 1$

30. (a) $\sin(A + B) + \sin(A - B)$

$\sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B = 2 \sin A \cos B$

(b) $\sin(A + B) - \sin(A - B)$

$\sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B = 2 \cos A \sin B$

(c) $\cos(A + B) + \cos(A - B)$

$\cos A \cos B - \sin A \sin B + \cos A \cos B + \sin A \sin B = 2 \cos A \cos B$

(d) $\cos(A - B) - \cos(A + B)$

$\cos A \cos B + \sin A \sin B - \cos A \cos B + \sin A \sin B = 2 \sin A \sin B$

31. $\cos\alpha = \frac{3}{5}; \cos\beta = \frac{5}{13} \Rightarrow \sin\alpha = \frac{4}{5}; \sin\beta = \frac{12}{13}$

$\cos(\alpha + \beta) = \cos\alpha \cdot \cos\beta - \sin\alpha \cdot \sin\beta = \frac{-33}{65}; \ \cos(\alpha - \beta) = \frac{15}{65} + \frac{48}{65} = \frac{63}{65}$

$\sin(\alpha + \beta) = \sin\alpha \cdot \cos\beta + \cos\alpha \cdot \sin\beta = \frac{20}{65} + \frac{36}{65} = \frac{56}{65}$

$\sin^2\left(\frac{\alpha - \beta}{2}\right) = \frac{1 - \cos(\alpha - \beta)}{2} = \frac{1 - \frac{63}{65}}{2} = \frac{1}{65}$

32. (a) $\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} \cos 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$\sin 15^\circ \cos 15^\circ = \frac{(\sqrt{3})^2 - (1)^2}{(2\sqrt{2})^2} = \frac{3-1}{8} = \frac{2}{8} = \frac{1}{4}$

(b) $\text{Tan}\, 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}$

(c) $\text{Tan}\, 22\frac{1}{2} = \sqrt{2} - 1$

(d) $\text{Tan}\, 15^\circ - \cot 15^\circ = 2 - \sqrt{3} - 2 - \sqrt{3} = -2\sqrt{3}$

33. $\cos(\theta - \alpha) = a; \sin(\theta - \beta) = b$

$\sin(\theta - \alpha) = \sqrt{1 - \cos^2(\theta - \alpha)} = \sqrt{1 - a^2}, \cos(\theta - \beta) = \sqrt{1 - \sin^2(\theta - \beta)} = \sqrt{1 - b^2}$

$[\cos(\alpha - \beta + \theta - \theta)]^2 + 2\text{ab}[\sin(\alpha - \beta - \theta + \theta)]$

$\left[a\sqrt{1 - b^2} + b\sqrt{1 - a^2}\right]^2 + 2ab\left[-\sqrt{1 - a^2}\sqrt{1 - b^2} + ab\right]$

$= a^2(1 - b^2) + b^2(1 - a^2) + 2ab\sqrt{(1 - a^2)(1 - b^2)} - 2ab\sqrt{(1 - a^2)(1 - b^2)} + 2a^2b^2$

$= a^2 - a^2b^2 + b^2 - a^2b^2 + 2a^2b^2 = a^2 + b^2$

34. $\tan A = x\tan B \Rightarrow \frac{\tan A}{\tan B} = x \Rightarrow \frac{\sin A}{\cos A} \times \frac{\cos B}{\sin B} = x \Rightarrow \frac{\sin A}{\cos A} \times \frac{\cos B}{\sin B} = x$

$\frac{\sin A\cos B - \cos A\sin B}{\sin A\cos B + \cos A\sin B} = \frac{x-1}{x+1} \Rightarrow \frac{\sin(A-B)}{\sin(A+B)} = \frac{x-1}{x+1}$

WORKSHEET – 6

LEVEL – I

1. D	2. B	3. D	4. C	5. B	6. B	7. A
8. B	9. B	10. C				

LEVEL – II

1. B	2. C	3. C	4. A	5. A	6. A	7. A
8. C	9. C	10. A	11. A	12. B	13. C	14. A
15. A	16. A	17. 0	18. D	19. C	20. A	

LEVEL – II

21. A, B, C, D	22. C	23. A	24. C	25. B	
26. a-r, b-s, c-q, d-p	27. B	28. D	29. C	30. A, C	31. A, C
32. A	33. B	34. D			

LEVEL – I

1. Sin 2 A = 2 Sin A Cos A
2. Cos 2 A = $\cos^2 A$- $\sin^2 A$
3. Tan 2 A = $\frac{2\tan A}{1-\tan^2 A}$
4. $\sin 3A = 3\sin A - 4\sin^3 A$
5. $\cos 3A = 4\cos^3 A - 3\cos A$
6. $\sin 18° = \frac{\sqrt{5}-1}{4}$
7. $\cos 36° = \frac{\sqrt{5}+1}{4}$
8. $\sin\theta = \frac{1}{\sqrt{3}} \Rightarrow \cos\theta = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \cos 2\theta = \cos^2\theta - \sin^2\theta = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$
9. $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

   $\sin^2\theta + \cos^2\theta + \underline{2}\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - \underline{2}\sin\theta\cos\theta = 1 + 1 = 2$
10. $\cos^4\theta - \sin^4\theta = [\cos^2\theta]^2 - [\sin^2\theta]^2 = (\cos^2\theta + \sin^2\theta)(\cos^2\theta - \sin^2\theta)$

    $= 1 \times (\cos^2\theta - \sin^2\theta) = \cos 2\theta$

LEVEL – II

1. $\sin A = \frac{3}{5} \cos A = \frac{4}{5} \Rightarrow \sin 2A = 2\sin A\cos A = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$
2. $\tan A = \frac{4}{3} \Rightarrow \cos 2\ A = \frac{1-\tan^2 A}{1+\tan^2 A} = \frac{1-\frac{16}{9}}{1+\frac{16}{9}} = \frac{-7}{25}$

3. $3\sin 20^\circ - 4\sin^3 20 = \sin 3(20^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$

4. $4\cos^3 30^\circ - \cos 30^\circ = \cos 3(30^\circ) = \cos 90^\circ = 0$

5. $8\cos^3 20 - 6\cos 20 \Rightarrow 2(4\cos^3 20^\circ - 3\cos 20^\circ) \Rightarrow 2\cos 3(20^\circ) = 2 \cdot \cos 60^\circ = 2 \times \frac{1}{2} = 1$

6. $8\sin\theta\cos\theta\cos 2\theta\cos 4\theta \Rightarrow 4[2\sin\theta\cos\theta]\cos 2\theta\cos 4\theta$

   $4\sin 2\theta\cos 2\theta\cos 4\theta \Rightarrow 2[2\sin 2\theta\cos 2\theta]\cos 4\theta$

   $2\sin 4\theta\cos 4\theta \Rightarrow \sin 2(4\theta) = \sin 8\theta$

7. $\frac{\sin 3\theta\cos\theta - \cos 3\theta \cdot \sin\theta}{\sin\theta \cdot \cos\theta} \Rightarrow \frac{\sin(3\theta-\theta)}{\sin\theta\cos\theta} = \frac{\sin 2\theta}{2\sin\theta\cos\theta} \times 2 = 2\frac{\sin 2\theta}{\sin 2\theta} = 2$

8. $\cot A + \tan A$

   $$\frac{\cos A}{\sin A} + \frac{\sin A}{\cos A} = \frac{\cos^2 A + \sin^2 A}{\sin A\cos A} = \frac{1}{\sin A\cos A} \times \frac{2}{2} = \frac{2}{\sin 2A}$$

   $= 2\operatorname{cosec} 2A$

9. $\left[2[3\sin 20^\circ - 4\sin^3 20^\circ]\right]^2 \Rightarrow \left[2[\sin 60^\circ]\right]^2 = \left[2 \cdot \frac{\sqrt{3}}{2}\right]^2 = 3$

10. $\frac{3\sin A - 4\sin^3 A}{1+2[1-2\sin^2 A]} \Rightarrow \frac{3\sin A - 4\sin^3 A}{1+2-4\sin^2 A} = \frac{\sin A(3-4\sin^2 A)}{3-4\sin^2 A} = \sin A.$

11. $\sin A\sin(60+A)\sin(60-A) \Rightarrow \sin A = (\sin^2 60 - \sin^2 A)$

    $$\sin A\left(\frac{3}{4} - \sin^2 A\right) \Rightarrow \sin A\left(\frac{3-4\sin^2 A}{4}\right) \Rightarrow \frac{3\sin A - 4\sin^3 A}{4} = \frac{\sin 3A}{4}$$

12. $\cos A\cos(60+A)\cos(60+A) \Rightarrow \cos A(\cos^2 60 - \sin^2 A)$

    $$\cos A\left(\frac{1}{4} - (1-\cos^2 A)\right) \Rightarrow \cos A\left(\frac{1}{4} - 1 + \cos^2 A\right)$$

    $$\cos A\left(\cos^2 A - \frac{3}{4}\right) \Rightarrow \cos A\left(\frac{4\cos^2 A - 3}{4}\right) \Rightarrow \frac{4\cos^3 A - 3\cos A}{4} = \frac{1}{4}\cos 3A$$

13. $\tan A \cdot \tan(60-A)\tan(60+A) = \tan 3\ A \Rightarrow \tan 20^\circ \cdot \tan 40^\circ \tan 80^\circ \cdot \tan 60^\circ$

    $\tan 3(20^\circ)\tan 60^\circ \Rightarrow \sqrt{3} \cdot \sqrt{3} = 3$

14. $\frac{1-\tan^2(45-\theta)}{1+\tan^2(45-\theta)} \Rightarrow \cos 2(45-\theta) \Rightarrow \cos(90-2\theta) \Rightarrow \sin 2\theta$

15. $\frac{1-\left(\frac{1-\tan^2 A}{1+\tan^2 A}\right)}{1+\left(\frac{1-\tan^2 A}{1+\tan^2 A}\right)} = \frac{-}{\frac{1+\tan^2 A+1-\tan^2 A}{1+\tan^2 A}} = \frac{2\operatorname{Tan}^2 A}{2} = \operatorname{Tan}^2 A$

16. $\cos\theta = 2\cos^2\frac{\theta}{2} - 1$

    $$\sqrt{\frac{1+\cos\theta}{2}} = \cos\theta/2 \quad \sin\theta = \frac{-3}{5}$$

    $$\sqrt{\frac{1-4/5}{2}} = \cos\theta/2 \quad \cos\theta = \frac{-4}{5}$$

$\sqrt{\frac{1}{10}} = \cos\theta/2 \Rightarrow \cos\theta/2 = \frac{1}{\sqrt{10}}$

17. $\cos\theta + \cos(180 - (60 - \theta)) + \cos(180 - (60 + \theta))$

$\cos\theta - \cos(60 - \theta) - \cos(60 + \theta)$

$\cos\theta - [\cos(60 - \theta) - \cos(60 + \theta)]$

$\cos\theta - [2\cos 60 \cdot \cos\theta] \Rightarrow \cos\theta - \left(2 \cdot \frac{1}{2}\cos\theta\right) \Rightarrow \cos\theta - \cos\theta = 0$

18. $\text{Tan}\left(\frac{\pi}{4} + \frac{A}{2}\right) = \frac{1+\text{Tan}\,A/2}{1-\text{Tan}\,A/2} = \frac{\cos A/2+\sin A/2}{\cos A/2-\sin A/2} \times \frac{(\cos A/2+\sin A/2)}{\cos A/2+\sin A/2}$

$= \frac{1 + \sin A}{\cos A} = \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \text{Tan}\,A$

19. $\frac{1+\text{Tan}\,A}{1-\text{Tan}\,A} + \frac{1-\text{Tan}\,A}{1+\text{Tan}\,A}$

$\frac{(1 + \text{Tan}\,A)^2 + (1 - \text{Tan}\,A)^2}{1 - \text{Tan}^2\,A} = \frac{2(1 + \text{Tan}^2\,A)}{1 - \text{Tan}^2\,A} \Rightarrow 2 \times \frac{1}{\cos 2A} = 2\sec 2\ \text{A}$

20. $x + \frac{1}{x} = 2\cos 20^\circ$

$x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x}\left(x + \frac{1}{x}\right) = 8\cos^3 20^\circ \Rightarrow x^3 + \frac{1}{x^3} + 3(2\cos 20^\circ) = 8\cos^3 20^\circ$

$x^3 + \frac{1}{x^3} = 8\cos^3 20^\circ - 6\cos 20^\circ = 2\left(4\cos^3 20^\circ - 3\cos 20^\circ = 2\cos 60^\circ = 2 \times \frac{1}{2} = 1\right.$

LEVEL – III

21. $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \frac{1-\text{Tan}^2\,\theta}{1+\tan^2\theta}$

22. $\cos 18^\circ = \sin 72^\circ$

23. $3\sin 30^\circ - 4\sin^3 30^\circ = \sin 3(30^\circ) = \sin 90^\circ = 1$

24. $4\cos^3 10 - 3\cos 10^\circ = \cos(3)10^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$

25. $4\cos^3 12^\circ - 3\sin 78^\circ = 4\cos^3 12^\circ - 3\cos 12 = \cos 3(12^\circ) = \cos 36^\circ = \frac{\sqrt{5}+1}{4}$

26. a) $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$

b) $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$

C) $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$

d) $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$

27. $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ$

$= \sin 20^\circ \cdot \sin 40^\circ \cdot \sin 80^\circ \cdot \frac{\sqrt{3}}{2} = \frac{1}{4}\sin 3(20^\circ) \times \frac{\sqrt{3}}{2} = \frac{1}{4} \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{3}{16}$

28. $\cos 10^\circ \cos 50^\circ \cos 70^\circ$

$= \cos 10^\circ \cdot \cos(60^\circ - 10^\circ) \cdot \cos(60^\circ + 10^\circ) = \frac{1}{4}\cos 3(10^\circ) = \frac{1}{4} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{8}$

29. $\text{Tan } 20^\circ \cdot \text{Tan } 40^\circ \cdot \text{Tan } 80^\circ \cdot \cot 30^\circ = \tan 20^\circ \cdot \tan(60^\circ - 20^\circ) \cdot \tan(60 + 20^\circ) \cdot \sqrt{3}$

$= \tan 3(20^\circ) \cdot \sqrt{3} = \tan 60^\circ \cdot \sqrt{3} = \sqrt{3} \times \sqrt{3} = 3$

30. $\left(\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}\right)^2$

$= \left(\frac{1 - \cos\theta + \sin\theta}{1 + \cos\theta + \sin\theta}\right)^2 = \left(\frac{2\sin^2\frac{\theta}{2} + 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2} + 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}\right)^2 = \left(\text{—}\right)^2$

$= \tan^2\frac{\theta}{2} = \tan^2\frac{\theta}{2} = \frac{\sin^2\frac{\theta}{2}}{\cos^2\frac{\theta}{2}} = \frac{1 - \cos\theta}{1 + \cos\theta}$

31. $\sin A = \frac{-7}{25}$ $A \in$ 4 th Quadrant

$\cos A = \frac{24}{25} \Rightarrow \sin 2A = 2\sin A\cos A = 2 \times \frac{-7}{25} \times \frac{24}{25} = -\frac{168}{625} \times 2 = \frac{-336}{625}$

$\cos\frac{A}{2} = \sqrt{\frac{1 + \cos A}{2}} = \sqrt{\frac{1 + \frac{24}{25}}{2}} = \sqrt{\frac{49}{50}} = \frac{7}{5\sqrt{2}}$

$\sin\frac{A}{2} = \sqrt{\frac{1-\cos A}{2}} = \sqrt{\frac{1-\frac{24}{25}}{2}} = \sqrt{\frac{1}{50}} = -\frac{1}{5\sqrt{2}}$ ($\therefore A \in$ 4 th Quadrant)

$\tan\frac{A}{2} = \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}} = \frac{-1}{7}$

32. $\sin\alpha + \sin\beta = \frac{1}{4}$

$2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{4}$

$\cos\alpha + \cos\beta = \frac{1}{3}$

$2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{3}$

Dividing Eq.(ii) by Eq.(iv), we have

$\tan\left(\frac{\alpha+\beta}{2}\right) = \frac{3}{4} \Rightarrow \sin(\alpha+\beta) = \frac{2\tan\left(\frac{\alpha+\beta}{2}\right)}{1 + \tan^2\left(\frac{\alpha+\beta}{2}\right)} = \frac{2 \times \frac{3}{4}}{1 + \left(\frac{3}{4}\right)^2} = \frac{24}{25}$

33. $\cos(\alpha+\beta) = \frac{7}{25}$ 34. $\tan(\alpha+\beta) = \frac{24}{7}$

WORKSHEET – 7

LEVEL – I

1. B	2. D	3. A	4. B	5. A	6. D	7. A
8. B	9. A	10. B				

LEVEL – II

1. B	2.A	3.C	4.A	5.B	6.A	7.A
8. B	9.A	10.C	11.C	12.A	13.D	14.B
15.C	16.A	17.A	18.A	19.C	20. 0	21. B
22.B	23.A					

LEVEL – III

24. A, B, C	25.A, B, C	26. A	27.A	28. C	29.a-r, b-s, c-q, d-p
30. B, C	31. C	32. A	33. B	34.D	35. a-s, b-r, c-p, d-q

LEVEL – I

1. $\text{Sin}(A+B)+\text{Sin}(A-B)=2\text{Sin}\,A\text{Cos}\,B$
2. $\text{Sin}(A+B)-\text{Sin}(A-B)=2\text{Cos}\,A\text{Sin}\,B$
3. $\cos(A+B)+\cos(A-B)=2\cos A\cos B$
4. $\text{Cos}(A-B)-\text{Cos}(A+B)=2\text{Sin}\,A\text{Sin}\,B$
5. $\sin C+\sin D=2\sin\left(\frac{C+D}{2}\right)\cdot\cos\left(\frac{C-D}{2}\right).$
6. $\cos C-\cos D=2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$
7. $\text{Cos}\,13\theta+\text{Cos}\,9\theta=2\cos\left(\frac{13\theta+9\theta}{2}\right)\cos\left(\frac{13\theta-9\theta}{2}\right)$

   $=2\cos 11\theta\cdot\cos 2\theta$
8. $2\sin 6^0\sin 66^\circ=2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)(Lct)$

   $\frac{C+D}{2}=6\ \frac{D-C}{2}=66$

   $C+D=2----(1)\ D-C=132$

   $(1)+(2)\Rightarrow 2\text{D}=144\Rightarrow D=72^\circ\Rightarrow C=-60^\circ$

   $2\sin 6^0\sin 66^\circ=\cos(-60)^0-\cos(72)^0=\cos 60^\circ-\cos 72^\circ=\frac{1}{2}-\cos 72^0$
9. $\text{Sin}\,70^\circ-\text{Cos}\,40^\circ$

   $=\text{Sin}\,70^\circ-\sin 50^\circ=2\cos\left(\frac{70+50}{2}\right)\sin\left(\frac{70-50}{2}\right)=2\cos 60^\circ\cdot\sin 10^\circ=\sin 10^\circ$
10. $\cos 70^\circ+\sin 40^\circ=\cos 70^\circ+\sin 50^\circ=2\cos\frac{70^\circ+50^\circ}{2}\cos\frac{70^\circ-50^\circ}{2}$

$= 2\cos 60°\cos 10° = 2 \times \frac{1}{2} \times \cos 10° = \cos 10°$

LEVEL – II

1. $2\text{Cos}\left(\frac{50°+70°}{2}\right) \cdot \sin\left(\frac{50°-70°}{2}\right) + \sin 10° = 2\text{Cos}\, 60°\sin\left(\frac{-20°}{2}\right) + \sin 10°$

   $= 2\text{Cos}\, 60°\sin(-10°) + \sin 10° = -\sin 10° + \sin 10° = 0$

2. $(2\cos^2 3\theta - 1) \cdot \cos 5\theta = \cos 6\theta\cos 5\theta$

   $= \frac{1}{2}[2\cos 6\theta\cos 5\theta] = \frac{1}{2}[\cos(6\theta + 5\theta) + \cos(6\theta - 5\theta)] = \frac{1}{2}[\cos(11\theta) + \cos\theta]$

3. $\sin(120° + \theta) + \sin(120° - \theta) = 2 \cdot \sin 120°\cos\theta = 2 \cdot \frac{\sqrt{3}}{2}\cos\theta = \sqrt{3}\cos\theta$

4. $\text{Cos}(60° + \theta) + \cos(60° - \theta) = 2\cos 60°\cos\theta = 2 \cdot \frac{1}{2}\cos\theta = \cos\theta$

5. $\frac{\sin 70°+\cos 40°}{\cos 70°+\sin 40°} = \frac{\cos 20°+\cos 40°}{\sin 20°+\sin 40°} = \frac{2\cos 30°\cos 10°}{2\sin 30°\cos 10°} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$

6. $\frac{\sin(90°-25°)+\sin(90°-65°)}{\cos 65°+\cos 25°} = \frac{\cos 25°+\cos 65°}{\cos 65°+\cos 25°} = 1$

7. $\cos 35° + \cos 85° + \cos 155° = 2\cos 60°\cos 25° + \cos 155° = 2\frac{1}{2}\cos 25° + \cos(180 - 25°)$

   $= \cos 25° - \cos 25° = 0$

8. $2 \cdot \sin\left(-\right)\cos\left(\frac{240-\theta-240-\theta}{2}\right) = 2 \cdot \sin 240°\cos\theta$

   $= 2 \cdot \sin(180 + 60°)\cos\theta = -2 \cdot \frac{\sqrt{3}}{2}\cos\theta = -\sqrt{3}\cos\theta$

9. $\sin(5\alpha + 2\alpha) - \sin(5\alpha - 2\alpha) = \sin 7\alpha - \sin 3\alpha$

10. $2(1 - 2\sin^2 7\theta) \cdot \sin 3\theta = 2\cos 14\theta \cdot \sin 3\theta = \sin(14\theta + 3\theta) - \sin(14\theta - 3\theta) =$

    $\sin 17\theta - \sin 11\theta$

11. $2[\cos 75° + \cos 15°]^2 = 2[2\cos 45° \cdot \cos 30°]^2 = 2[\quad]^2 = 2 \times \frac{3}{2} = 3$

12. $\frac{2\cos 20°\cos 40° - 2\sin 5°\sin 25°}{2} = \frac{[\cos 60°+\cos 20°]-[\cos 20°-\cos 30°]}{2}$

    $= \frac{\frac{1}{2} + \frac{\sqrt{3}}{2}}{2} = \frac{1 + \sqrt{3}}{4}$

13. $\frac{2\cos\left(\frac{7\theta+5\theta}{2}\right)\cdot\sin\left(\frac{7\theta-5\theta}{2}\right)}{2\cos\left(\frac{7\theta+5\theta}{2}\right)\cos\left(\frac{7\theta-5\theta}{2}\right)} = \frac{\sin\theta}{\cos\theta} = \tan\theta$

14. $\frac{2\sin\left(\frac{A+2A}{2}\right)\cos\left(\frac{2A-A}{2}\right)}{2\sin\left(\frac{A+2A}{2}\right)\cdot\sin\left(\frac{2A-A}{2}\right)} = \frac{\text{Cos}\, A/2}{\text{Sin}\, A/2} = \text{Cot}\, A/2$

15. $\frac{2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)}{2\cos\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)} = \text{Tan}\left(\frac{A+B}{2}\right)$

16. $\frac{[\quad]\cdot[\quad]}{[\quad][\quad]}$

$$\frac{\sin 2\theta \cdot \sin\theta \cdot \sin 5\theta \cos 3\theta}{\cos 3\theta \cdot \sin 2\theta \sin 5\theta \sin\theta} = 1$$

17. $\text{Cos}(-89^0) + \text{Cos}(-87^0) + \text{Cos}(-85)^0 + \cdots \ldots + \text{Cos}(85^\circ) + \cos(87^0) + \text{Cos}(89^0)$

$= 2(\text{Cos}\,1^0 + \text{Cos}^0 + \text{Cos}^0 + \cdots .. + \text{Cos}\,89^\circ) = 2\dfrac{\sin(45^\circ)}{\sin(1^0)}\left\{\cos\left[\dfrac{1^0 + 89^0}{2}\right]\right\}$

$\dfrac{= 2\sin 45^\circ \cos 45^\circ}{\sin 1^\circ} = \dfrac{\sin 90^\circ}{\sin 1^\circ} = \text{cosec}\,1^0$

18. Given

$\text{Sin}(A+B) + \sin(A-B) + \sin(B+C) + \sin(B-C) + \sin(C+A) + \sin(C-A)$

$= \sin A + \sin B + \sin C \Rightarrow \sin(A-B) + \sin(B-C) + \sin(C-A) = 0$

$\Rightarrow 4\sin\frac{A-B}{2}\sin\frac{B-C}{2}\sin\frac{C-A}{2} = 0 \Rightarrow \therefore \Delta$ is isosceles

19. $\text{Cos}\,A, \cos B$ and $\text{Cos}\,C$ are the roots of the cubic equation $x^3 + ax^2 + bx + c = 0$

$\Rightarrow C$ os $A + \cos B + \cos C = -a$

$\cos A\cos B + \cos B\cos C + \cos C\cos A = b \Rightarrow \text{Cos}\,A\text{Cos}\,B\text{Cos}\,C = c$

Now $(\cos A + \cos B + \cos C)^2 = (\sum\cos^2 A) + 2(\sum\cos A\cos B)$

$\therefore \cos^2 A + \cos^2 B + \cos^2 C = a^2 - 2b \Rightarrow \ \ \therefore 3 + \cos 2A + \cos 2B + \cos 2C = 2a^2 - 4b$

$\therefore 3 - 1 - 4\cos A\cos B\cos C = 2a^2 - 4b \ \ \therefore 1 + 2c = a^2 - 2b \ \ \therefore a^2 - 2b - 2c = 1$

20. $4\cos 18^\circ - \frac{3}{\cos 18^0} - 2\tan 18^\circ = \frac{4\cos^2 18^\circ - 3 - 2\sin 18^\circ}{\cos 18^\circ}$

$= \dfrac{2(1+\cos 36^\circ) - 2\sin 18^\circ - 3}{\cos 18^\circ} = \dfrac{2(1+\cos 36^\circ - \sin 18^\circ) - 3}{\cos 18^\circ} = 0$

21. $\cos^2\frac{\pi}{11} + \cos^2\frac{2\pi}{11} + \cos^2\frac{3\pi}{11} + \cos^2\frac{4\pi}{11} + \cos^2\frac{5\pi}{11}$

$= \dfrac{1}{2}\left[\left[1+\cos\dfrac{2\pi}{11}\right] + \left[1+\cos\dfrac{4\pi}{11}\right] + \left[1+\cos\dfrac{6\pi}{11}\right] + \cdots \ldots + \left[1+\cos\dfrac{10\pi}{11}\right]\right]$

$= \dfrac{1}{2}\left[5 + \left[\cos\dfrac{2\pi}{11} + \cos\dfrac{4\pi}{11} + \cos\dfrac{6\pi}{11} + \cdots \ldots + \cos\dfrac{10\pi}{11}\right]\right]$

$= \dfrac{1}{2}\left[5 + \dfrac{2\sin\frac{5\pi}{11}}{\sin\frac{\pi}{11}}\cos\left(\dfrac{\frac{2\pi}{11} + \frac{10\pi}{11}}{2}\right)\right] = \dfrac{1}{2}\left[5 + \dfrac{\sin\frac{5\pi}{11}}{\sin\frac{\pi}{11}}\cos\dfrac{6\pi}{11}\right]$

$= \dfrac{1}{2}\left[5 - \dfrac{2\sin\frac{5\pi}{11}\cos\frac{5\pi}{11}}{1\sin\frac{\pi}{11}}\right] = \dfrac{1}{2}\left[5 - \dfrac{\sin\frac{10\pi}{11}}{2\sin\frac{\pi}{11}}\right] = \dfrac{1}{2}\left(5 - \dfrac{1}{2}\right) = \dfrac{9}{4}$

22. $\text{A} + \text{B} + \text{C} + \text{D} = 2\pi \Rightarrow \therefore \text{A} + \text{B} = 2\pi - (\text{C} + \text{D})$

$\therefore \tan(A+B) = \tan[2\pi - (C+D)] = -\tan(C+D)$

$\Rightarrow \dfrac{\tan A + \tan B}{1 - \tan A\tan B} = -\dfrac{\tan C + \tan D}{1 - \tan C\tan D}$

$\Rightarrow (\tan A + \tan B)(1 - \tan C\tan D) = -(\tan C + \tan D)(1 - \tan A\tan B)$

$\Rightarrow \tan A + \tan B - \tan A\tan C\tan D - \tan B\tan C\tan D =$

$-[(\tan C + \tan D - \tan A\tan B\tan C - \tan A\tan B\tan D)]$

$\Rightarrow \tan A + \tan B + \tan C + \tan D = \tan A\tan B\tan C + \tan A\tan C\tan D + \tan A\tan B\tan D + \tan B\tan C\tan D$

Dividing both sides by $\tan A\tan B\tan C\tan D$, we get

$$\frac{\tan A + \tan B + \tan C + \tan D}{\tan A\tan B\tan C\tan D} = \frac{1}{\tan D} + \frac{1}{\tan B} + \frac{1}{\tan C} + \frac{1}{\tan A}$$

$$= \cot A + \cot B + \cot C + \cot D$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C + \tan D}{\cot A + \cot B + \cot C + \cot D} = \tan A\tan B\tan C\tan D$$

23. $\frac{\cot 84^\circ \cot 48^\circ}{\cot 66^\circ \cot 78^\circ} = \frac{(2\cos 84^\circ \sin 66^\circ)(2\sin 78^\circ \cos 48^\circ)}{(2\sin 84^\circ \cos 66^\circ)(2\cos 78^\circ \sin 48^\circ)}$

$$= \frac{(\sin 150^\circ - \sin 18^\circ)(\sin 126^\circ + \sin 30^\circ)}{(\sin 150^\circ + \sin 18^\circ)(\sin 126^\circ - \sin 30^\circ)} = \frac{(\sin 30^\circ - \sin 18^\circ)(\sin 54^\circ + \sin 30^\circ)}{(\sin 30^\circ + \sin 18^\circ)(\sin 54^\circ - \sin 30^\circ)}$$

$$= \frac{\left[\frac{1}{2} - \frac{\sqrt{5}-1}{4}\right]\left[\frac{\sqrt{5}+1}{4} + \frac{1}{2}\right]}{\left[\frac{1}{2} + \frac{\sqrt{5}-1}{4}\right]\left[\frac{\sqrt{5}+1}{4} - \frac{1}{2}\right]} = \frac{(3-\sqrt{5})(3+\sqrt{5})}{(\sqrt{5}+1)(\sqrt{5}-1)} = \frac{9-5}{5-1} = \frac{4}{4} = 1$$

LEVEL – II

24. $2\cos\left(\frac{80^\circ+40^\circ}{2}\right)\cdot\cos\left(\frac{80^\circ-40^\circ}{2}\right) - \cos 20^\circ \Rightarrow 2\cos 60^\circ \cdot \cos 20^\circ - \cos 20^\circ = 0$

25. $\sin\alpha - \sin\beta = \frac{1}{3} \Rightarrow 2\sin\left(\frac{\alpha-\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right) = \frac{1}{3}$

$$\cos\beta - \cos\alpha = \frac{1}{2} \Rightarrow 2\sin\left(\frac{\alpha-\beta}{2}\right)\cdot\sin\left(\frac{\alpha+\beta}{2}\right) = \frac{1}{2}$$

$$\frac{(1)}{(2)} \Rightarrow \cot\frac{\alpha+\beta}{2} = \frac{2}{3}$$

26. $\sin(5\alpha + 3\alpha) - \sin(5\alpha - 3\alpha) = \sin 8\alpha - \sin 2\alpha$

27. $\sin 6\theta + \sin 4\theta = 2\sin 5\theta \cdot \cos\theta$

28. $2\sin A \cdot \cos B = \sin(A+B) + \sin(A-B)$

29. a-r,b-s,c-q,d-p

30. $2\cos 30^\circ\cos\theta = 2\frac{\sqrt{3}}{2}\cos\theta = \sqrt{3}\cos\theta$

31. $\sin(120^\circ + \theta) - \sin(120^\circ - \theta) = 2\cos 120^\circ\sin\theta = 2(-\sin 30^\circ)\sin\theta$

$$= 2\left(-\frac{1}{2}\right)\sin\theta = -\sin\theta$$

32. $\left(\frac{\cos 30^\circ+\cos 15^\circ}{\sin 30^\circ-\sin 15^\circ}\right)^3 + \left(\frac{\sin 30^\circ+\sin 15^\circ}{\cos 30^\circ-\cos 15^\circ}\right)^3 = 0$ (n is odd)

33. $\left(\frac{\cos 45^\circ+\cos 15^\circ}{\sin 45^\circ-\sin 15^\circ}\right)^2 + \left(\frac{\sin 45^\circ+\sin 15^\circ}{\cos 45^\circ-\cos 15^\circ}\right)^2 = 2\cot^2\left(\frac{45^\circ-15^\circ}{2}\right) = 2\cot^2 15^\circ$ (n is even)

34. $\left(\frac{\cos\alpha+\cos\beta}{\sin\alpha-\sin\beta}\right)^{1009}+\left(\frac{\sin\alpha+\sin\beta}{\cos\alpha-\cos\beta}\right)^{1009}=0$ (n is odd)

35. a) $\cos 20^\circ+\cos 80^\circ-\sqrt{3}\cos 50^\circ=2\cos 30^\circ\cos 50^\circ-\sqrt{3}\cos 50^\circ=\sqrt{3}\cos 50^\circ-\sqrt{3}\cos 50^\circ=0$

b) $\cos 0+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$

$=1+\left[\cos\frac{\pi}{7}+\cos\frac{6\pi}{7}\right]+\left[\cos\frac{2\pi}{7}+\cos\frac{5\pi}{7}\right]+\left[\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}\right]$

$=1+\left[\cos\frac{\pi}{7}+\cos\left[\pi-\frac{\pi}{7}\right]\right]+\left[\cos\frac{2\pi}{7}+\cos\left[\pi-\frac{2\pi}{7}\right]\right]+\left[\cos\frac{3\pi}{7}+\cos\left[\pi-\frac{3\pi}{7}\right]\right]$

$=1+0+0+0=1$

c) $\cos 20^\circ+\cos 40^\circ+\cos 60^\circ-4\cos 10^\circ\cos 20^\circ\cos 30^\circ$

$=2\cos 30^\circ\cos 10^\circ+2\cos^2 30^\circ-1-4\cos 10^\circ\cos 20^\circ\cos 30^\circ$

$=2\cos 30^\circ(\cos 10^\circ+\cos 30^\circ)-1-4\cos 10^\circ\cos 20^\circ\cos 30^\circ$

$=2\cos 30^\circ(2\cos 10^\circ\cos 20^\circ)-1-4\cos 10^\circ\cos 20^\circ\cos 30^\circ=-1$

d) $\cos 20^\circ\cos 100^\circ+\cos 100^\circ\cos 140^\circ-\cos 140^\circ\cos 200^\circ$

$=\frac{1}{2}(\cos 120^\circ+\cos 80^\circ+\cos 240^\circ+\cos 40^\circ-\cos 340^\circ-\cos 60^\circ)$

$=\frac{1}{2}\left(\frac{-1}{2}+\cos 80^\circ-\frac{1}{2}+\cos 40^\circ-\cos 340^\circ-\frac{1}{2}\right)=\frac{1}{2}\left(\frac{-3}{2}+\cos 80^\circ\cos 40^\circ-\cos 20^\circ\right)$

$=\frac{1}{2}\left(\frac{-3}{2}+2\cos 60^\circ+\cos 20^\circ-\cos 20^\circ\right)=\frac{1}{2}\left(\frac{-3}{2}\right)=\frac{-3}{4}$

9. PROGRESSIONS

WORKSHEET – 1

LEVEL – I

1. A	2. B	3. B	4. A	5. B	6. C
7. A	8. D	9. C	10. D		

LEVEL – II

1. D	2. B	3. C	4. A	5. B	6. A
7. C	8. A	9. C	10. D	11. B	12. C
13. D	14. D	15. A	16. A	17. B	18. D
19. B	20. B				

LEVEL – III

21. A, B	22. A	23. A	24. C	25. 19	26. *a – q, b- p, c -r, d - p*
27. A, B, C, D		28. D	29. C	30. A	31. A
32. *a -r, b – p, c -s ,d -q*		33. A	34. B	35. C	

LEVEL – II

1. 3,8,13,18 AP

   $a = 3,\ d = 5$

   $t_{10} = 3 + 9 \times 5 = 48$

2. $a_1 = 3(1) + 2 = 5$

   $a_2 = 8$

   $a_3 = 11$

   $a_4 = 14$

   $a_5 = 17$

3. $A \cdot M = \frac{a+b}{2}$

4. $2b = a + c$

5. $a_n = a + (n-1)d$

   $20 + (n-1)5 = 130$

   $n = 23$

6. $a_n = a + (n-1)d$

7. $s = \frac{n}{2}[2a + (n-1)d]$

   $S_{20} = \frac{20}{2}[2 \times 3 + 19 \times 7] = 1390$

8. $2b = a + c$

   $2x = \frac{2}{7} - \frac{16}{7} \Rightarrow x = 1$

9. $s_n = \frac{n}{2}[a + I]$

   $a_n = a + (n-1)d = 19$

   $n = 15$

   $S_n = \frac{15}{2}[5 + 19] = 180$

10. $a_n = \frac{n(n-2)}{n+3}$

    $a_{20} = \frac{20(18)}{23} = \frac{360}{23}$

11. AP is given 1,3,5 ... .

    $a = 1, d = 2$

    $a + (n-1)d = 55$

    $n = 28$

12. Sum of first n odd natural numbers $= n^2 = 20^2 = 400$

13. $a = 6, d = 5$

    $a_n = a + (n-1)d$

    $= 6 + 10 \times 5 = 56$

14. $a = 5a + 2d = 15$

    $d = 5$

    $S_n = \frac{n}{2}[2a + (n-1)d] = 680$

15. $\frac{2}{3}$, K, $\frac{5}{8}$ K arein AP

    $K - \frac{2}{3} = \frac{5}{8}K - K$

    $\therefore k = \frac{16}{33}$

16. $a_8 = a + 7d = 17$

    $a + 8d = 19$

    $d = 2\ a = 3$

    $a_{25} = a + (n-1)d$

    $= 3 + 24 \times 2 = 51$

17. $a = 30°\ d = 30°$

    $S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{8}{2}[2(30°) + 7 \times 30°] = 1080°$.

18. Sum of the interior angles of a polygon $= (n-2) \times 180$

$\therefore a = 120°\ d = 5°$

$S_n \Rightarrow \frac{n}{2}[2 \times 120 + (n-1)5] = (n-2)(180)$

$n^2 - 25n + 144 = 0$

$\therefore n = 16, n = 9$

$\therefore a_n = a + (n-1)d$

$= 120 + 15 \times 5 = 195°$ [It is not possible because greater than 180°]

$\therefore n = 9$ (correct answers)

19. $S = \frac{n}{2}(a + l) \Rightarrow n = \frac{2s}{a+l}$

$a + (n-1)d = 1$

$d = \frac{l-a}{n-1}$

$\therefore$ from(1)&(2) $\frac{1^2 - a^2}{2s - (1+a)}$

20. $a = 4d = \left(\frac{9}{2} - 4\right) = \frac{1}{2}$

$a_{105} = 4 + 104 \times \frac{1}{2} = 56$

LEVEL – III

21. $a = 3, d = 4 \Rightarrow a_n = 3 + (n-1)4 = 4n - 1$

$S_n = \frac{n}{2}[6 + (n-1)4] = 2n^2 + n$

22. $5(a - 4d) = 8(a + 7d)$

$a + 12d = 0$

$\therefore a_{13}$ term is zero

23. $a = 3x - 5y d = 3x - 9y - (3x - 5y) = -4y$

$a_{90} = 3x - 5y + 89(-4y) = 3x - 361y$

24. $(a + 17d) - (a + 13d) = 32 \Rightarrow d = 8$

25. $\frac{n}{2}[2a + (n-1)d] = 0$

$2 \times 18 + (n-1)(-2) = 0$

$n = 19$

26. *a – q, b- p, c -r, d - p*

27. d = 4 months, a = 8 years, S_n = 168 years

$S_n = \frac{n}{2}[2a + (n-1)d],\ 168 = \frac{n}{2}\left[16 + \frac{n-1}{3}\right]$

$n^2 + 63n - 16n - 1008 = 0$

$n = 16$

28. $a - d + a - a + d = 3$

$(a-d)(a)(a+d) = 8$

$d = \pm 3$

29. $A = a, d = a - b - a = b$

$T_{10} = 20 \Rightarrow a - 9b = 20$

$T_{20} = 10 \Rightarrow a - 19\ b = 10$

$\therefore n^{th}$ term $= a + b(n-1)d$

$= 29 + (n-1)(-1) = 30 - n$

30. $a = 345\ d = 3$

$345 + (n-1)3 = 1002$

$n = 220$

31. By concept $d = \frac{b-a}{n+1}$

(a) $1^2 + 3^2 +$

$+19^2 = (1^2 + 2^2 +$

$19^2) - (2^2 + 4^2$

$= 19\frac{(19+1)(38+1)}{6} - (1 \times 2^2 \times 2^2 \times 2^2 \times 2^2 \times 3^2 + \cdots \ldots + 2^2 \times 9^2)$

$= 2470 - 2^2[1^2 + 2^2 + +9^2]$

$= 2470 - 1140 = 1330$

$1^2 + 2^2 + 3^2 + \cdots \ldots \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(b) $(1^2 + 2^2 + \cdots \ldots \ldots + 13^2 + 14^2 + \cdots$

$+30^2) - (1^2 + n^2 +$

$= \left(\frac{30 \times 31 \times 61}{6}\right) - \left(\frac{13 \times 14 \times 27}{6}\right) = 9455 - 819 = 8636$

(c) $\left(n\frac{(n+1)}{2}\right)^2 = \left(\frac{15 \times 16}{2}\right)^2 = (120)^2 = 14400$

(d) $2(1 + 2 + +99) + 100$

$= 2 \times \frac{99}{2}[1 + 99] + 100 = 2 \times 4950 + 100 = 10000$

33. Sum $^2 = 75 + 76 + 77 +$

$a = 75, d = 1$

$a + (n-1)d = 97 \Rightarrow n = 23$

$\text{sum} = \frac{23}{2}(75 + 97) = \frac{23}{2} \times 172 = (23 \times 86) = 1978$

34. 15,18,21,24 105

$a = 15, d = 3$

$a_n = a + (n-1)d$

n = 31

35. The page are 1,3,5,7

$a = 1,\ d = 2\ l = 1089$

$a_n = a + (n-1)d \Rightarrow n = 545$

WORKSHEET – 2

LEVEL – I

1. B	2. A	3. A	4. D	5. C	6. C	7. A
8. D	9. B	10. C				

LEVEL – II

1. C	2. D	3. A	4. D	5. A	6. B	7. D
8. B	9. C	10. B	11. B	12. C	13.B	14. C
15. A	16. C	17. C	18. C	19. A	20.A	

LEVEL – III

21.A, B, C	22. B	23. C	24. D	25. a-r, b-p, c-s, d-q	
26. Ans:11	27. B, C	28. a-r, b-p, c-s, d-q	29. 6	30. C, D	31. 5
32. a-q, b-s, c-p, d-r					

LEVEL – II

1. For any two in equal numbers a and b, we have A.M. > G.M $\left(\frac{a+b}{2}\right) > \sqrt{ab}$

2. HM = $\frac{2ab}{a+b}$

3. Common ratio = $\frac{t_2}{t_1} = \frac{1}{-3}$

4. $S_\alpha = \frac{a}{1-r}(r < 1)$

$$= \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$$

5. $GM = \sqrt{ab} = GM = \sqrt{6 \times 24}$

$= \sqrt{144}$
$= 112$
$= 12$

6. $a, \frac{3}{5}, \frac{1}{2}$ are in HP

$$\frac{a-\frac{3}{5}}{\frac{3}{5}-\frac{1}{2}}=\frac{a}{\frac{1}{2}} \Rightarrow \frac{a}{\frac{1}{2}}-\frac{3}{5}=\frac{a}{5}$$

$$a=\frac{3}{4}$$

7. $a=2, r=3$

   $a_n = ar^{n-1}$

   $= 2 \times 3^{n-1}$

8. $a_5 = 2, ar^4 = 2$

   Now, The product of first 9 terms

   $a \times ar \times ar^2 \times \ldots\ldots \times ar^8 = (ar^4)^9 = 2^9 = 512$

9. $\frac{2}{b}=\frac{1}{a}+\frac{1}{c}$

   $b=\frac{2ac}{a+c} \Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P

   $\because a, b, c$ are in HP

10. First 5 terms are a, ar, ar^2, ar^3, ar^4

    3rd term $= ar^2 = 4, (ar^2)^6 = 4^5$

11. $ar^6 = 8ar^3$

    $r^3 = 8$

    $r = 2$

    $92^4 = 48 \Rightarrow a = 3$

12. $6, x, y, z, 54$ are in GP.

    $$r=\left(\frac{54}{6}\right)^{\frac{1}{4}} \Rightarrow r = 3^2 a$$

    $r = \sqrt{3} \Rightarrow z = G_3 = ar^3 = 6(\sqrt{3})^3 = 18\sqrt{3}$

13. $a, H_1, H_2, b \rightarrow HP$

    $\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b}$ are in AP

    $$\frac{1}{H_1}-\frac{1}{a}=\frac{1}{b}-\frac{1}{H_2} \Rightarrow \frac{H_1+H_2}{H_1 H_2}=\frac{a+b}{ab}$$

14. $\mathrm{A}=\frac{x+y}{2}$

    $\mathrm{G}=\sqrt{xy}$

    $$\mathrm{H}=\frac{\frac{1}{x}+\frac{1}{y}}{2}=\frac{x+y}{2xy}$$

    $$\frac{A}{H}=\frac{x+y}{2} \times \frac{2xy}{x+y} \Rightarrow \frac{A}{H} = xy$$

$\therefore \frac{A}{H} = G^2$

$A = HG^2$

15 a, b, c are in AP

$2b = a + c$

$3^{2b} = 3^{a+c} \Rightarrow (3^b)^2 = 3^a \times 3^c$

$\therefore 3^a, 3^b, 3^c$ are in G.P

16. $a = 1\, r = \frac{1}{2}$

$S_\alpha = \frac{a}{1-r} \rightarrow S_\alpha = \frac{1}{1-\frac{1}{2}} = 2$

17. x, y, z are in G.P

$y^2 = xz$

$2\log y - \log x + \log z$

$2(1 + \log y) = (1 + \log x) + (1 + \log z)$

$1 + \log x, 1 + \log y, 1 + \log z$ are in AP

$\because \frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z}$ are in HP

18. $\left(\frac{3}{4} + \frac{3}{4^3} + \frac{3}{4^5} + \cdots \ldots \ldots + \alpha\right) - \left[\frac{5}{4^2} + \frac{5}{4^4} + \frac{5}{4^6} + \cdots \ldots \ldots + \alpha\right]$

$= \frac{\frac{3}{4}}{1 - \left(\frac{1}{4}\right)^2} - \frac{\frac{5}{4^2}}{1 - \left(\frac{1}{4}\right)^2}$

$= \frac{12 - 5}{15} = \frac{7}{15}$

19. $an = a \cdot r^{n-1} = \frac{1}{19683} = \frac{1}{3}\left(\frac{1}{3}\right)^{n-1}$

$\Rightarrow \left(\frac{1}{3}\right)^n = \frac{1}{19683} = \left(\frac{1}{3}\right)^9$

$\Rightarrow n = 9$

20. $(2 + 3^1) + (2 + 3^2) + (2 + 3^3) + \cdots \ldots \ldots . + (2 + 3^{11})$

$= 2 + 2 + \cdots\ldots$ upto11 terms $+ (3^1 + 3^2 + 3^3 + \cdots . . + 3^{11})$

$= 11 \times 2 + \frac{3(3^{11} - 1)}{3 - 1} = 22 + \frac{3}{2}(3^{11} - 1)$

LEVEL – III

21. A, B, C

22. Key : B

Given G.P is 3,6,12, … … …

$a = 3, r = 2$

$a_{10} = \text{ar}^9 = (3)(2)^9$

23. Key : C

Given 0.1, 0.11, 0.111

$\text{a} = 0.1\ \text{r} = \dfrac{0.11}{0.1} = 1.1$

$a_n = a.r^{n-1} = (0.1)(1.1)^{n-1}$

24. Key : D

Given G.P is $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$

$a = \dfrac{1}{2}, r = \dfrac{1}{2}$

$a_7 = a \cdot r^6 = \left(\dfrac{1}{2}\right)\left(\dfrac{1}{2}\right)^6 = \dfrac{1}{2^7} = \dfrac{1}{128}$

25. (a) $a = \frac{1}{3}\ ar^3 = 9$

$r = 3$

$t_3 = ar^2 \rightarrow x_2 = \dfrac{1}{3}(3)^r = 3$

(b) $ar^2 = 4$

a.ar. $ar^2 \cdot ar^3 \cdot ar^4 = a^5r^{10} = (ar^2)^5 = (4)^5$

(c) $a = \frac{1}{3}, r = \frac{1}{3} \Rightarrow s_\alpha = \frac{a}{1-r} = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$

(d) $a = 1, r = 3, \frac{a(r^n-1)}{r-1} = 364$

$1.\dfrac{(3^n - 1)}{3 - 1} = 364 \Rightarrow 3^n = 3^6$

$n = 6$

26. Key : D

Given G.P is $2, 2\sqrt{2}, 4$ … …

$a = 2; r = \sqrt{2}$

$a_x = 64 \Rightarrow a \cdot r^{x-1} = 64$

$2(\sqrt{2})^{x-1} = 64$

$2^{\frac{x-1}{2}} = 32$

$2^{\frac{x-1}{2}} = 2^5 \Rightarrow \dfrac{x-1}{2} = 5$

$\text{x} = 11$

$\therefore x - 7 = 11 - 7 = 4$

27. Key: B, C

Given in G.P $a = 1$

$a_3 + a_5 = 90 \Rightarrow a \cdot r^2 + a \cdot r^4 = 90$

$ar^2(1 + r^2) = 90$

$r^2(1 + r^2) = 10 \Rightarrow (r^2)^2 + r^2 - 10 = 0 \Rightarrow (r^2 + 10) + (r^2 - 9) = 0$

$\Rightarrow r^2 = 9 \Rightarrow r = \pm 3$

28

Key: $(a) \rightarrow r$

(b) $\rightarrow$ p

(c) $\rightarrow$ s

$(d) \rightarrow q$

(a) nth term in H.P $a_n = \frac{1}{a+(n-1)d}$

(b) given H.P IS $\frac{4}{3}, \frac{3}{2}, \frac{12}{7} \ldots\ldots$

$a = \frac{3}{4}, d = \frac{2}{3} - \frac{3}{4} = \frac{-1}{12}$

(c) G.M OF 5,125 $= \sqrt{5 \times 125} = \sqrt{625} = 25$

(d) A.M OF $(a + 2)^2$ and $(a - 2)^2$

$= \frac{(a+2)^2 + (a-2)^2}{2} = 2\frac{[a^2 + 4]}{2} = a^2 + 4$

29. Ans: 6

Given in G.P $a_1 = 3$

$S_n = 189$

$a_n = 96$

a. $r^{n-1} = 96$

$\frac{a(r^n - 1)}{r - 1} = 189$

$3 \cdot r^{n-1} = 96$

$r^{n-1} = 32$

$\frac{3(r^n - 1)}{r - 1} = 189$

$r^n = 32r$

(1) $\frac{r^n - 1}{r - 1} = 63$

From (1) & (2)

$32r - 1 = 63r - 63$

$62 = 31r \Rightarrow r = 2$

$(1) \Rightarrow 2^{n-1} = 2^5 \Rightarrow n - 1 = 5$

30. Key : C, D

Given a, b, c are in A.P

$\therefore$ 2 b = a + c

$a+1; b; c$ are in G.P $\Rightarrow b^2 = (a+1)c$

(or)

$a, b, c+2$ are in G.P $\Rightarrow b^2 = a(c+2)$

By solving (1), (2), & (3) we get $a = 8, b = 12, c = 16$

31. ans = 5

given a,b.c are in H.P

$\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

32. a, b are any two positive real numbers

(a) $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is A.M. of a and b $\Rightarrow n = 0$

(b) $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is G,M of a and b $\Rightarrow n = \frac{-1}{2}$

(c) $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is H.M of a and b $\Rightarrow n = -1$

(d) $\frac{1}{2}, H_1, H_2, \frac{H}{17}$ are in Hp

$2, \frac{1}{H_1}, \frac{1}{H_2}, \frac{17}{4}$ are in AP

$a + 3d = \frac{17}{4}$

$d = \frac{3}{4}$

$a + d = 2 + \frac{3}{4} = \frac{11}{4}, a + 2d = 2 + 2 \times \frac{3}{4} = \frac{7}{2}$

$H_1 = \frac{1}{a+d} = \frac{4}{11}, H_2 = \frac{1}{a+2d} = \frac{2}{7}$

10. Quadratic Equations

WORK SHEET – 1

LEVEL – I

1.B	2. A	3. D	4. C	5. D	6. D	
7. B	8. C	9. B				

LEVEL – II

1.A	2.C	3.A	4.A	5.B	6.B	7.C
8.C	9.A	10.B	11.B	12. A	13. A	14.B
15.9	16.C	17.B	18.C	19.A	20.A	21.C

LEVEL – III

22. A, C 23. B 24.A 25.C 26. a-s; b-q,t; c-r,t; d-p,q,r,t

27.A, B 28. a-p,q; b-r,t; c-q,s; d-p,s 29. A, D 30.A

31. a-p,r; b-p,q; c-s; d-p,r

LEVEL – II

1.(A) $b^2 - 4ac$

$= (2)^2 - 4(1)(-1)$

$= 4 + 4 = 8$

2.(C) $ax^2 + (2b + 6)x + (c - 4) = 0$

to make a pure quadratic equation

$2b + 6 = 0 \Rightarrow 2b = -6, b = -3$

3.(A) $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$

$\sqrt{3}x^2 + 12x - 2x - 8\sqrt{3} = 0$

$\sqrt{3}x(x + 4\sqrt{3}) - 2(x + 4\sqrt{3}) = 0$

$(x + 4\sqrt{3})(\sqrt{3}x - 2) = 0$

$x = -4\sqrt{3}$ (or) $x = \frac{2}{\sqrt{3}}$

4.(A) $\alpha^2 + \beta^2 = 10$

$(\alpha + \beta)^2 - 2\alpha\beta = 10$

$p^2 - 2(-3) = 10$

$p^2 = 10 - 6$

$p = \pm 2$

5.(B) $\alpha + \beta = \frac{-6}{9} = \frac{-2}{3}, \alpha\beta = \frac{1}{9}$

$x^2 - x\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + \frac{1}{\alpha\beta} = 0$

$$x^2 - x\left(\frac{-2/3}{1/9}\right) + \frac{1}{\frac{1}{9}} = 0$$

$$x^2 + 6x + 9 = 0$$

6. (B) $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

$$= \frac{(-1)^2 - 2(1)}{1} = 1 - 2 = -1$$

7.(C)

$x^2 - px - p - c = 0$

$\alpha + \beta = p,\ \alpha\beta = -p - c$

Now $1 + \alpha\beta + \alpha + \beta + c$

$= 1 - p - c + p + c$

8. (C)

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 3\ \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{(4)^2 - 2(2)}{(2)^2} = \frac{16-4}{4} = \frac{12}{4} = 3$$

9. (A)

$x^2 - (4)x + 4 - 3 = 0$
$x^2 - 4x + 1 = 0$

10. (B)

$x^2(a+b) + x \quad (2a - 3b + 1) + a - 2b + 1 = 0$
$a + b = 0 \quad 2a - 3b + 1 = 0$
$-2b - 3b + 1 = 0$
$-5b = 1$
$b = \frac{1}{5}, a = \frac{-1}{5}$

11. (B)

$\Delta = 0\ a = 1,\ b = 2(k+1), c = k^2$
$b^2 - 4ac = 0$
$[2)k + 1)]^2 - 4(1)(k^2) = 0$
$4(k^2 + 2k + 1) - 4k^2 = 0$

$8k + 4 = 0$

$k = \frac{-1}{2}$

12.(A)

13.(A)

$\alpha = 5 + \sqrt{3}, \beta = 5 - \sqrt{3}$

$\alpha + \beta = 10\ \alpha\beta = 22$

$QE = x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$x^2 - 10x + 22 = 0$

$x^2 - 10x + p = 0 \Rightarrow$ put $x = 2$

$(2)^2 - 10(2) + \mathrm{p} = 0 \Rightarrow \therefore \mathrm{p} = 16$

Now $x^2 - 16x + 60 = 0$

$x^2 - 10x - 6x + 60 = 0$

$x = 10,6\ \therefore x = 10$

14.(B) α, β are roots of the equation $x^2 + 2x + 1 = 0$ then

The quadratic equation whose roots are $4\alpha, 4\beta$ is

$x^2 - (4\alpha + 4\beta)x + (4\alpha.4\beta) = 0$

$x^2 + 8x + 16 = 0$

15(9) $.16 + 4a + 8 = 0$

$4a = -24$

$a = -6$

$x^2 - 6x + b = 0$

$36 - 4b = 0$

$\therefore \mathrm{b} = 9$

16. (C) Given $(x - a)(x - b) - c = 0$

$(x - \alpha)(x - \beta) + c = 0$

$x^2 - (\alpha + \beta)x + \alpha\beta + c = 0$

$x^2 - (a + b)x + ab - c + c = 0$

$x^2 - (a + b)x + ab = 0$

$x^2 - (a + b)x + ab - c = 0$

$\alpha + \beta = a + b\ \alpha\beta = ab - c$

root are a, b

17. (B) $x^2 + x - 6 = 0,\ x^2 - x - 6 = 0$

$x = 2,3$

$x = -2,3$

for $x = 2$ or -2 the given equation is satisfied

$\therefore$ sum of real root is $2 + (-2) = 0$

18. (C) $(x - b)(x - c) + (x - c)(x - a) + (x - a)(x - b) = 0$

$\Rightarrow 3x^2 - 2(a + b + c)x + (bc + ac + ab) = 0$

$\Delta = \mathrm{b}^2 - 4\mathrm{ac}$

$= [-2(a + b + c)]^2 - 4 \cdot 3(ab + bc + ba)$

$= 4(a^2 + b^2 + c^2 + 2ab + 2bc + 2ca) - 12ab - 12bc - 12ca$

$= 4a^2 + 4b^2 + 4c^2 - 4ab - 4bc - 4ca = 2[2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca]$

$= 2[a^2 - 2ab + b^2 + b^2 - 2bc + c^2 + c^2 - 2(a + a^2]$

$= 2[(a-b)^2 + (b-c)^2 + (c-a)^2] > 0$

$\therefore \Delta > 0 \Rightarrow$ roots are real and distinct

19. (A) Given α, β are root of $x^2 = 5x - 3$

$x^2 - 5x + 3 = 0$

$\alpha + \beta = -5,\ \alpha\beta = 3$

Sum roots of now Q.E

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{25 - 6}{3} = \frac{19}{3}$$

Products of roots = 1 Q.E $\Rightarrow x^2 - \frac{19}{3}x + 1 = 0 \Rightarrow 3x^2 - 19x + 3 = 0$

20.(A) Let $x = \sqrt{42 + \sqrt{42 + \sqrt{42 + \cdots \ldots}}}$

s.b.s

$$\Rightarrow x^2 = 42 + \sqrt{42 + \sqrt{42 + \sqrt{42 + \cdots}}}$$

$\Rightarrow x^2 = 42 + x$

$\Rightarrow x^2 - x - 42 = 0\ x$ cannot be negative

$\Rightarrow (x-7)(x+6) = 0$

$\therefore X = 7$

$x = 7$ or -6

21. (C) $a + b + c = 0$

$\Delta = (2b)^2 - 4(3a)(c)$
$= 4b^2 - 12ac = 4[-(a+c)]^2 - 12ac$
$= 4a^2 + 4c^2 + 8ac - 12ac = 2[a^2 + c^2 - 2ac]$
$= 2(a-c)^2 \ \because (a-c)^2 > 0$

$b = -(a + c)$

$\Delta > 0$

roots are Real and distinct

LEVEL – III

22. $(A, C) p + q = p$

$pq = q$

$2p = q$

p = 1 and q = −2

23. (B) $x^2 - x(\alpha + \beta) + \alpha\beta = 0$

$x^2 - x(-5) + (-6) = 0$

24. (A) $x^2 - x(3 + \sqrt{7} + 3 - \sqrt{7}) + (9 - 7) = 0$

$x^2 - 6x + 2 = 0$

25. (C) $6 + i + 6 - i = -p$

p = −12

26. a-s; b-q,t; c-r,t; d-p,q,r,t

27. (A, B)$\sin\alpha, \cos\alpha$ are roots

$\sin\alpha + \cos\alpha = \frac{-b}{a}$, $\sin\alpha \cdot \cos\alpha = \frac{c}{a}$

s.b.s

$\sin^2\alpha + \cos^2\alpha + 2\sin\alpha \cdot \cos\alpha = \frac{b^2}{a^2}$

$1 + 2\frac{c}{a} = \frac{b^2}{a^2}$ $a^2 - b^2 + 2ac + c^2 = 0 + c^2$

$\frac{a + 2c}{a} = \frac{b^2}{a^2}$ $(a^2 + c)^2 = b^2 + c^2$

$a(a + 2c) = b^2$

$a^2 + 2ac = b^2 \Rightarrow a^2 + 2ac - b^2 = 0$

28. a-p,q; b-r,t; c-q,s; d-p,s

29. (A, D)

$b^2 - 4ac < 0$
$1 - m^2 \cdot 4 < 0$
$(2m + 1)(2m - 1) > 0$
$m < \frac{-1}{2}$ (or) $m > \frac{1}{2}$

30. A

31. a-p,r; b-p,q; c-s; d-p,r

WORKSHEET – 2

LEVEL – I

1. A	2. A	3. B	4. C	5. D	6. C	7. C
8. B	9. A	10. B				

LEVEL – II

1.A	2.B	3.D	4.A	5.A	6.B	7.A
8.D	9.D	10.D	11.B	12.C	13.C	14.D
15. A	16. D	17. A	18. D	19. B	20. A	21. A

22. B 23. B 24. C 25. D

LEVEL – III

26. A, C 27.A 28. a-r; b-p; c-s; d-q 29. C, D 30. a-t; b-q; c-s; d-p
31. A, C 32. a-q,s; b-p,q; c-r; d-r

LEVEL – II

1. (A) Sum of coefficients of x^2, k and constant is

$I^2 - m^2 + m^2 - n^2 + n^2 - I^2 = 0$

$\therefore$ roots are $1, \frac{c}{a} \Rightarrow 1, \frac{n^2 - I^2}{I^2 - m^2}$

2. (B) $(bc - a^2)^2 = (ab - c^2)(ac - b^2)$

$a^4 + ab^3 + ac^3 = 3a^2bc$

$a(a^3 + b^3 + c^3) = 3abc \times a$

$a^3 + b^3 + c^3 = 3abc$

$\frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3$

3. (D) $x^2 - cx + d = 0$ let α, β areroots &$x^2 - ax + b = 0$ Let $\alpha_1 \alpha$ areroots

$\alpha + \beta = c, \alpha\beta = d$ $2\alpha = a$ $\alpha^2 = b$

$\beta = c - \alpha = c - \frac{a}{2} = \frac{2c - a}{2}$ $\alpha\beta = d$

$\left(\frac{2c - a}{2}\right)\left(\frac{a}{2}\right) = d$

$2ac - a^2 = 4\ d$

$2ac - 4a^2 = 4\ d$

$2ac = 4\ b + 4\ d \Rightarrow 2(\ b + d) = ac$

4. (A) eq (1) & (2) will have a common roots which is imaginary hence both the roots will be common

$\frac{a}{1} = \frac{b}{2} = \frac{c}{3} \Rightarrow a : b : c = 1 : 2 : 3$

5

(A) $x^2 + ax + b = 0$

$a + b + 1 = 0$

$x^2 \pm bx \pm a = 0$

$a + b + 1 + 1 = 0 + 1$

$(a - b)x = a - b$

$a + b + 2 = 1$

$x = 1$

6. (B) $\alpha:\beta = 1:3$

   $(1+3)^2 ac = (1\times 3)b^2$

   $16ac = 3\ b^2$

7. (A) coefficient of $x^2 = ab - ac$

   coefficient of $x = bc - ab$

   constant $= ac - bc$

   $ab - ac + bc - ab + ac - bc = 0$

   $\therefore$ one of the root is " 1 "

   Other root is $\frac{c(a-b)}{a(b-c)}$

8. key 4 $\frac{a}{2} = \frac{b}{6} = \frac{4}{8}$

   $a + b = 4$

9. (D) Given ' 2 ' is common factor

   $$\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = 2 \Rightarrow \frac{b-d}{c-a} = 2$$

10.

   (D) $\frac{a}{2} = \frac{b}{8} = \frac{3}{6} = \frac{1}{2}$

   $\Rightarrow a = 1, b = 4, \frac{b}{a} = 4$

11. (B) Let α be the other root, then

   $4a + 2b + c = 0$ and $2b = a + c$

   $5a + 2c = 0$

   $$\frac{c}{a} = \frac{-5}{2}$$

   Now, $2 \times \alpha = \frac{c}{a} = \frac{-5}{2}$

   $$\alpha = \frac{-5}{4}$$

12. (C) Given $\alpha + \beta = \alpha^2 + \beta^2$

   $\alpha + \beta = (\alpha+\beta)^2 - 2\alpha\beta$

   $-p = (-p)^2 - 2q$

   $p^2 + p = 2q$

13. (C) $ax^2 + 2cx + b = 0\ a\left(\frac{1}{2}\right)^2 + 2c\left(\frac{1}{2}\right) + b = 0$

   $ax^2 \pm 2bx \pm c = 0\ \frac{a}{4} + 2c + b = 0$

   $2(c-b)x = c - b\ a + 4c + 4b = 0$

$x = \frac{1}{2}$

14. (D) $x^2 + 2px + q = 0\ x^2 + 2qx + p = 0$

$\alpha - \beta = \frac{|b^2 - 4ac|}{|a|}$

$|\alpha - \beta| = \frac{4p^2 - 4 \cdot 1.q}{1}, |\alpha - \beta| = \frac{4q^2 - 4 \cdot 1 \cdot p}{1}$

$\therefore 4p^2 - 4q = 4q^2 - 4p$

$4[(p^2 - q^2) + (p - q)] = 0$

$(p + q)(p - q) + (p - q) = 0$

$(p - q)(p + q + 1) = 0$

$p - q = 0$ or $p + q + 1 = 0$

since $p \neq q$

$\therefore p + q + 1 = 0$

15. (A) Given $\alpha = \beta^2$, so $\alpha + \beta = \frac{-b}{a}$ and $\alpha\beta = \frac{c}{a}$

$\beta^2 + \beta = \frac{-b}{a}\ \beta^3 = \frac{c}{a}$

$\beta = \left(\frac{c}{a}\right)^{\frac{1}{3}}$

$\beta^2 + \beta = \frac{-b}{a}$

$\left(\frac{c}{a}\right)^{\frac{2}{3}} + \left(\frac{c}{a}\right)^{\frac{1}{3}} = \frac{-b}{a}$

c.b.s

$\left(\frac{c}{a}\right)^2 + \left(\frac{c}{a}\right) + 3\left(\frac{c}{a}\right)\left(\frac{-b}{a}\right) = \frac{-b^3}{a^3}$

$\frac{c^2 + ac - 3bc}{a^2} = \frac{-b^3}{a^3}$

$a(c^2 + ac - 3bc) = -b^3$

$b^3 + ac^2 + a^2c = 3abc$

16. (D) $x^2 + ax + 12 = 0$

$x^2 + bx + 15 = 0$

Adding (1) and (2)

$2x^2 + (a + b)x + 27 = 0$

equation (3) subtracting from $x^2 + (a + b)x + 36 = 0$

$x^2 - 9 = 0 \Rightarrow x = 3, -3$

The common positive root is 3 . Hence

$9 + 3a + 12 = 0 \Rightarrow 3a = -21 \Rightarrow a = -7$

and $a + 3b + 15 = 0$

$b = -8$

17. (A) Let roots $m:n$ ratio

roots are mx and nx

from eq (1) $mx:nx \Rightarrow mnb_1^2 = a_1c_1(m+n)^2$

from eq (2) $mx:nx \Rightarrow mnb_2^2 = a_2c_2(m+n)^2$

from (1) & (2)

$$\frac{mnb_1^2}{mnb_2^2} = \frac{a_1c_1(m+n)^2}{a_2c_2(m+n)^2}$$

$$\left[\frac{b_1}{b_2}\right]^2 = \frac{a_1c_1}{a_2c_2}$$

18. (D) $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$

$(p.6 - 12.3)^2 = (3 - (-17) - 6(-2))(-2.12 - (-17) \cdot p)$

$(6p - 36)^2 = (-39)(-24 + 17p)$

$12\mathrm{p}^2 - 144\mathrm{p} + 432 = (-13)(17\mathrm{p} - 24)$

$12p^2 - 144p + 432 + 221p - 312 = 0$

$12p^2 + 77p + 120 = 0$

$$\mathrm{p} = \frac{90}{24}, \mathrm{p} = \frac{-64}{24}$$

$\mathrm{p} = \frac{15}{4}$ or $\mathrm{p} = \frac{-8}{3}$

19. (B) The no. of roots in common for the equation

$3x^2 - 21x + 18 = 0$ and $5x^2 - 35x + 30 = 0$ [2]

Here $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ ∴ They will be both common roots

20. (A) (i) $a + b + c$ is +ve (ii) $a^2b + b^2c + c^2a$ is +ve

21. (A) $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$

$(2\mathrm{k} \times 1 + \mathrm{k} \times 4)^2 = ((4 \times -3) - 1 \times -11)(-11\mathrm{x} - \mathrm{k} + 3 \times 2\mathrm{k})$

$(6k)^2 = -17k$

$36k^2 + 17k = 0$

$\mathrm{k}[36\mathrm{k} + 17] = 0$

$$\therefore \mathrm{k} = \frac{-17}{36}$$

22. (B) α_1, β_1 are roots $a_1x^2 + b_1x + c_1 = 0$

$\alpha_1 + \beta_1 = \frac{-b_1}{a_1}$, $\alpha_1\beta_1 = \frac{c_1}{a_1}$

α_2, β_2 areroots of $a_2x^2 + b_2x + c_2 = 0$

$\alpha_2 + \beta_2 = \frac{-b_2}{a_2}$, $\alpha_2\beta_2 = \frac{c_2}{a_2}$

$\alpha_1\alpha_2 = \beta_1\beta_2 = 1$

$\alpha_1 = \frac{1}{\alpha_2}, \beta_1 = \frac{1}{\beta_2}$

$\alpha_1\beta_1 = \frac{c_1}{a_1} \Rightarrow \frac{1}{\alpha_2\beta_2} = \frac{c_1}{a_1} \Rightarrow \alpha_2\beta_2 = \frac{a_1}{c_1} \Rightarrow \frac{c_2}{a_2} = \frac{a_1}{c_1}$

$\therefore a_1a_2 = c_1c_2$

$\alpha_1 + \beta_1 = \frac{-b_1}{a_1}$

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{-b_1}{a_1} \Rightarrow \frac{\alpha_2 + \beta_2}{\alpha_2\beta_2} = \frac{-b_1}{a_1}$

$\frac{-b_2}{a_2} \times \frac{a_2}{c_2} = \frac{-b_1}{a_1} \Rightarrow \frac{b_2}{c_2} = \frac{b_1}{a_1} = \frac{b_1}{b_2} = \frac{a_1}{c_2} \Rightarrow \frac{b_1}{b_2} = \frac{a_1}{c_2} = \frac{c_1}{a_2}$

23. (B) (i) b-a-c is -ve (ii) $\frac{ab}{c}$ is -ve

24. C

25. key D:'a' is a common root of $x^2 + ax + bc = 0$

$x^2 + bx + ca = 0$

Similarly b, c are common roots

Sum = $a + b + c$

LEVEL – III

26. key A, C: Let a is a common root of equations

$\Rightarrow a^2 - 5a + 6 = a^2 - 7a + k$

2a = k – 6

$a = \frac{k-6}{2}$ is sub in $a^2 - 7a + k = 0$

then, k = 10,12

27. key A

28. $a - r, b - p, c - s, d - q$

29. key C, D $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\frac{a}{2} = \frac{b}{-7} = \frac{2}{1}$

$a = 4, b = -14$

30. $a-t; b-q; c-s; d-p$

31. (A &C)

$2x^2 + 8ax + 6 = 0$
$2x^2 \pm 3ax \mp 9 = 0$

Subtracting,

$5ax + 15 = 0$

$x = \frac{-3}{9}$

$\left(\frac{-3}{a}\right)^2 + 4a\left(\frac{-3}{a}\right) + 3 = 0 \Rightarrow \frac{9}{a^2} = 9 \Rightarrow a^2 = 1 \Rightarrow a = \pm 1$

32. a-q,s; b-p,q; c-r; d-r

11. MATRICES

WORKSHEET – 1

LEVEL – I

1. B	2. A	3. C	4. D	5. D	6. C	7. A
8. A	9. C	10. D	11. A	12. D		

LEVEL – II

1. B	2. A	3. C	4. D	5. A	6. D	7. C
8. A	9. A	10. D	11. A	12. A	13. A	14. D
15. B	16. A	17. D	18. A	19. B	20. B	21. B

LEVEL – III

22. A, B, D	23. B	24. A	25. D	26. a-q, b-p, c-t, d-p	27. B
28.A, D	29. a-r; b-p; c-t; d-q	30. B	31. A, B, C	32. A	33. C
34. A	35. A				

LEVEL – II

1. $A+B=\begin{bmatrix}-1+2 & 2+1 & 0+0\\ 0+0 & 1+0 & -1+1\\ 3+1 & 1+(-2) & 2+1\end{bmatrix}$

2. $A-B=\begin{bmatrix}2-1 & 8-1\\ 4-2 & 5-5\end{bmatrix}$

3. $x+3=0, 2y+x=-7$

   $4a-z=2a$

   $\Rightarrow x=-3, y=-2, z=4, a=2$

4. Additive inverse of ' A ' is ' $-A$ '

5. $A^2=\begin{bmatrix}0 & -i\\ -i & 0\end{bmatrix}\begin{bmatrix}0 & -i\\ -i & 0\end{bmatrix}=\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$

6. Order of ' B ' is 3×2

7. $\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

   $\Rightarrow x^2+1=1, x=0$

8. $[ax^2+2hxy+by^2]$

9. 1×13 or 13×1

10. $\begin{bmatrix}0 & 2k\\ 3k & -4k\end{bmatrix}=\begin{bmatrix}0 & 3a\\ 2b & 24\end{bmatrix}$

    $k=6, a=4, b=9$

11. Square matrices of same order

12. $PQ = \begin{bmatrix} 2 & -1 & 5 \\ 6 & -3 & 15 \\ 8 & -4 & 20 \end{bmatrix}$

13. $\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} 1+1 & 1+2 & 1+3 & 1+4 \\ 2+1 & 2+2 & 2+3 & 2+4 \\ 3+1 & 3+2 & 3+3 & 3+4 \end{bmatrix}$

14. $AB = \begin{bmatrix} 0+abc-abc & 0+b^2c-b^2c & bc^2-bc^2 \\ -a^2c+a^2c & -abc+abc & -ac^2+ac^2 \\ a^2b-a^2b & ab^2-ab^2 & abc-abc \end{bmatrix}$

$AB = 0$

15 $AB = A \Rightarrow (AB)A = A^2 \Rightarrow A(BA) = A^2 \Rightarrow AB = A^2 \Rightarrow A = A^2$ similarly $B = B^2$

16. $X = B - A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

17. $al + bE = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$

$(al + bE)^3 = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \times \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \times \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} = \begin{bmatrix} a^3 & 3a^2b \\ 0 & a^3 \end{bmatrix} = a^3I + 3a^2bE$

18. $A^2 = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}\begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$n = 2$

19. $2x = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix} \Rightarrow x = \begin{bmatrix} 1 & 3/2 \\ 1 & 5/2 \end{bmatrix}$

20. $p = 2,\ A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ +2 & -3 \end{bmatrix}$

Verify options with p = 2

21. $A^2 = \begin{bmatrix} x^2 & 0 & 0 \\ 0 & x^2 & 0 \\ 0 & 0 & x^2 \end{bmatrix} \Rightarrow A^3 \begin{bmatrix} 0 & 0 & x^3 \\ 0 & x^3 & 0 \\ x^2 & 0 & 0 \end{bmatrix}$

Similarly $A^{100} = \begin{bmatrix} x^{100} & 0 & 0 \\ 0 & x^{100} & 0 \\ 0 & 0 & x^{100} \end{bmatrix}$

LEVEL – III

22. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2-7 & 1+4 \\ 5+2 & 0+6 \end{bmatrix};\ a = -5, b = 5, c = 7, d = 6$

23. Key :2; $2\begin{bmatrix} -7 & 4 \\ 2 & 6 \end{bmatrix} - 3\begin{bmatrix} 7 & 6 \\ 5 & 4 \end{bmatrix} + 4\begin{bmatrix} 2 & 1 \\ 5 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -14 & 8 \\ 4 & 12 \end{bmatrix} - \begin{bmatrix} 21 & 18 \\ 15 & 12 \end{bmatrix} + \begin{bmatrix} 8 & 4 \\ 20 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -27 & -6 \\ 9 & 0 \end{bmatrix}$

24. Key : A; $\begin{bmatrix} 2 & 1 \\ 5 & 0 \end{bmatrix} - 3X = \begin{bmatrix} -7 & 4 \\ 2 & 6 \end{bmatrix}\ ;\ 3X = \begin{bmatrix} 9 & -3 \\ 3 & -6 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$

25. $X = \begin{bmatrix} -5 & 5 \\ 7 & 6 \end{bmatrix}, Y = \begin{bmatrix} -5 & -5 \\ 0 & -4 \end{bmatrix};\ 2X + 3Y = \begin{bmatrix} -10-15 & 10-15 \\ 14+0 & 12-12 \end{bmatrix} = \begin{bmatrix} -25 & -5 \\ 14 & 0 \end{bmatrix}$

26. a) $\begin{bmatrix} 12-4 & -3-10 \\ 4+12 & -1+30 \end{bmatrix}$

b) $\begin{bmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{bmatrix}$

c) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 9-8 & -4+4 \\ 18-18 & -8+9 \end{bmatrix}$

27. $3x-2y=11, -x+2y=-5 \Rightarrow x=3, y=-1 \Rightarrow 3x+7y=9-7=2$

28.; Given , $(A+B)^2=A^2+B^2+2AB \Rightarrow (A+B)(A+B)=A^2+B^2+2AB$

$\Rightarrow A^2+AB+BA+B^2=A^2+B^2+2AB \Rightarrow BA=AB$

$\Rightarrow \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} a+2 & -a+1 \\ b-2 & -b-1 \end{bmatrix}=\begin{bmatrix} a-b & 1+1 \\ 2a+b & 2-1 \end{bmatrix}$

The corresponding elements of equal matrices are equal.

$a+2=a-b, -a+1=2 \Rightarrow a=-1$

$b-2=2a+b, -b-1=1 \Rightarrow b=-2 \Rightarrow a=-1, b=-2$

29. $x-1=3 \Rightarrow x=4$

$z-1=4 \Rightarrow z=5 \Rightarrow x+2y+z=1 \Rightarrow y=-4$

$a+y=2 \Rightarrow a=6$

30. a) Ans: $2; x-y=1, x+y=1 \Rightarrow x=1, y=0$

$\therefore x^2+y^2+a^2=5 \Rightarrow a^2=4 \Rightarrow a=2, a>0$

31. $A^2=\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \Rightarrow A^3=\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$

$A^{2012}=\begin{bmatrix} 1 & 0 \\ 4024 & 1 \end{bmatrix} \Rightarrow a=d, a+b+c+d=4026$

32. $-3m+4n=10, 4m-3n=-11 \Rightarrow m=-2, n=1 \Rightarrow 2m+5n=1$

33. $(A-2I)(A-3I)=\begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}=\begin{bmatrix} 2-2 & 4-4 \\ -1+1 & -2+2 \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}=0$

34. $(A-3I)(A-2I)=\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}=\begin{bmatrix} 2-2 & 2-2 \\ -2+2 & -2+2 \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

35. $(A-2I)A(A-3I)=\begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 6 & 6 \\ -3 & -3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} 6-6 & 12-12 \\ -3+3 & -6+6 \end{bmatrix}=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

WORKSHEET – 2

LEVEL – I

1. B	2. A	3. B	4. B	5. C	6. A	7. B
8. B	9. C	10. C				

LEVEL – II

1. C	2. C	3. A	4. A	5. B	6. A	7. D
8. C	9. A	10. C	11. C	12. A	13. C	14. A

15. A 16. A 17. B 18. B 19. B 20. C 21. A

LEVEL – III

22. A, B, D 23. A 24. D 25. C 26. a-s,b-p,c-q,d-r 27. C

28. a-p,r, b-q, c-r, d-s 29. B 30. C 31. D

LEVEL – II

1. Order of A^{T} is 3×2

2. $A^{\mathrm{T}} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

4. $\mathrm{A}^2 = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

   $= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1$

5. $(A^{\mathrm{T}})^{\mathrm{T}} = A = \begin{bmatrix} -1 & -3 \\ -2 & -4 \end{bmatrix}$

6. $\mathrm{A}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$

7. $AA^{\mathrm{T}} = \begin{bmatrix} 0 & \mathrm{i} \\ -\mathrm{i} & 0 \end{bmatrix}\begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

8. [C]; $A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$ and $A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$

   $\therefore A$ is the least positive integer such that $A^3 = 0$ so A is nilpotent with index 3

   .

9. $r + 4 = 5 \Rightarrow r = 1$

10. $A \cdot A^T = I_2$

11. $3\ \mathrm{A}^{\mathrm{T}} + 4\ \mathrm{B} = \begin{bmatrix} 7 & 0 \\ -10 & 6 \\ 17 & 31 \end{bmatrix} \rightarrow (1)$

    $-3A^{\mathrm{T}} + 2B = \begin{bmatrix} -1 & 18 \\ 4 & -6 \\ -5 & -7 \end{bmatrix} \rightarrow (2)$

    $\mathrm{Eq}(1) + (2) \Rightarrow 6\ \mathrm{B} = \begin{bmatrix} 6 & 18 \\ -6 & 0 \\ 12 & 24 \end{bmatrix}$

12. $\mathrm{A} + \mathrm{A}^{\mathrm{T}} = \begin{bmatrix} 2 & 6 \\ 7 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 7 \\ 6 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 13 \\ 13 & 4 \end{bmatrix} \Rightarrow \mathrm{P} = \frac{\mathrm{A}+\mathrm{A}^{\mathrm{T}}}{2} = \begin{bmatrix} 2 & \frac{13}{2} \\ \frac{13}{2} & 2 \end{bmatrix}$

13. $(A+B)^{\top}=A^{\top}+B^{\top}$

$(A-B)^{\top}=A^{\top}-B^{\top},(AB)^{\top}=B^{\top}A^{\top},(ABC)^{\top}=C^{\top}B^{\top}A^{\top}$

14. $\mathrm{AA}^{\top}=\frac{1}{5}\begin{bmatrix}3 & -4\\ 4 & \mathrm{x}\end{bmatrix}\frac{1}{5}\begin{bmatrix}3 & 4\\ -4 & \mathrm{x}\end{bmatrix}=\frac{1}{25}\begin{bmatrix}25 & 12-4\mathrm{x}\\ 12-4\mathrm{x} & 16+\mathrm{x}^2\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$12-4\mathrm{x}=0\Rightarrow \mathrm{x}=3$

15. $(A^2)^{\top}=(A^{\top})^2=A^2$

16. Let the order of B be $m\times n$. Since A is 3×4 matrix, A' is 4×3 matrix Since A ' B is defined, number of columns of A' must be equal to number of rows of B $\therefore$ $m=3$

BA' is defined, number of columns of B must be equal to number of rows of A'.

$\therefore \mathrm{n}=4$ ' B ' is matrix type 3×4.

17. $\begin{bmatrix}2 & 4\\ -1 & K\end{bmatrix}\begin{bmatrix}2 & 4\\ -1 & K\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

$\Rightarrow\begin{bmatrix}0 & 8+4\,\mathrm{K}\\ -2-\mathrm{K} & -4+\mathrm{K}^2\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

$\Rightarrow \mathrm{K}=-2$

18. $(AB-BA)^{\top}=(AB)^{\top}-(BA)^{\top}=B^{\top}A^{\top}-A^{\top}B^{\top}$

$=\mathrm{BA}-\mathrm{AB}=-(\mathrm{AB}-\mathrm{BA})$

19. $\mathrm{A}^2=\mathrm{A},\mathrm{B}^2=\mathrm{B}\Rightarrow(\mathrm{A}+\mathrm{B})^2=\mathrm{A}^2+\mathrm{B}^2+\mathrm{AB}+\mathrm{BA}=\mathrm{A}+\mathrm{B}$

20. In a skew-symmetric matrix, and diagonal elements are zero.

$\begin{bmatrix}0 & x & x\\ x & 0 & x\\ x & x & 0\end{bmatrix}$ also $a_{ij}+a_{ji}=0$ Hence, number of matrices $=2\times 2\times 2=8$

21. $(A+B)^n=\sum_{r=0}^{n}n_{C_r}A^{n-r}B^r$ Since $A^2=A$; $B^2=B=2^n(A+B)$

LEVEL – III

22. $(A+B)^{\top}=A^{\top}+B^{\top},(A-B)^{\top}=A^{\top}-B^{\top}$

$(AB)^{\top}=B^{\top}A^{\top}$, $(ABC)^{\top}=C^{\top}B^{\top}A^{\top}$

23. $X=3$

24. $\mathrm{P}=\frac{\mathrm{A}+\mathrm{A}^{\top}}{2},\mathrm{Q}=\frac{\mathrm{A}-\mathrm{A}^{\top}}{2}$

25. $(x,y)=(0,0)$

26. a-s, b-p, c-q, d-r

27. $\mathrm{AA}^{\top}=\mathrm{I}$

$$\Rightarrow\frac{1}{9}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ x & 2 & y\end{bmatrix}\begin{bmatrix}1 & 2 & x\\ 2 & 1 & 2\\ 2 & -2 & y\end{bmatrix}=I\Rightarrow\frac{1}{9}\begin{bmatrix}9 & 0 & x+4+2y\\ 0 & 9 & 2x+2-2y\\ x+4+2y & 2x+2-2y & x^2+4+y^2\end{bmatrix}=I$$

$\Rightarrow x + 2y = -4$

$x - y = -1$

$\Rightarrow x = -2, y = -1$

28. $a^2 + b^2 + c^2 = 1,\ ab + bc + ca = 0 \Rightarrow (a + b + c)^2 = 1 \Rightarrow a + b + c = \pm 1$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= \pm 1 \Rightarrow a^3 + b^3 + c^3 = \pm 1 + 3 = 4$ or 2

29,30,31.

we observe that $A^2 = A, B^2 = 0, C^2 = I$ and $|A| = 0, |B| = 0$ and $|C| = -1$ and hence results

WORKSHEET – 3

LEVEL – I

1. A 2. B 3. A 4. C 5. A 6. A
7. A 8. C

LEVEL – II

1. B 2. D 3. A 4. B 5. B 6. D 7. A
8. B 9. A 10. C 11. D 12. D 13. B 14. B
15. A 16. C 17. B 18. D 19. A 20. C 21. C

LEVEL – III

22. A, B, C 23. 2 24. D 25. C 26. a -s,r b -s,r c - p d -q 27. D 28. C, D 29. a - s b - r c - q d - p 30. A, B 31. 8 32. B 33. D 34. D

LEVEL – II

1. $|A| = 30 - 35 = -5$
2. $x^2 - 36 = 0 \Rightarrow x = \pm 6$
3. $5x - 30 = 0 \Rightarrow x = 6$
4. $\det A = -\sin^2\alpha - \cos^2\alpha = -1$
5. [B] $(-1)\begin{vmatrix} b & c \\ a & b \end{vmatrix}$
6. [D] Use $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$ 15. [2] put $x = 1, y = 1, z = 2$
7. [A] Applying $C_1 \to C_1 + C_2 + C_3$
8. [B] Applying $C_1 \to C_1 - C_2$; $C_2 \to C_2 - C_3$
9. $(2 + i)(1 - i) - (2 - i)(1 + 1)$

   $= (2 - 2i + i + 1) - (2 + 2i - i + 1) = (3 - i) - (3 + i) = -2i$

   = Complex quantity

10. [C] applying $C_3 \to C_3 - C_2$ after that $C_2 \to C_2 - 3C_3$ and expand
11. [D] $R_1 \to R_1 + R_2 + R_3$ then $C_1 \to C_2 - C_1, C_3 \to C_3 - C_1$
12. [D] Take out ' 3 ' from C_1 and ' 2 ' from $R_1 \Rightarrow$ Det = 6k
13. [B] put $x = y = z = 1$
14. [B] put $x = 1, y = 1, z = 2$
15. $\Rightarrow (1 + xyz)\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0 \Rightarrow xyz = -1$
16. $R_1; R_1 - 2R_2$

    $\begin{vmatrix} a-6 & 0 & 0 \\ 3 & b & c \\ 4 & a & b \end{vmatrix} = 0$

    $(a-6)(b^2 - ac) = 0$

    $a \neq 6, b^2 = ac$

    $\Rightarrow b^3 = abc$
17. [B] Keep least of given values in principal diagonal, highest of given values in other places. $\Rightarrow \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2.$
18. [D] taking out common a, b, c from C_1, C_2, C_3 respectively and

    $$C_1 \to C_1 + C_2 + C_3$$
19. [A] Substitute options and verify
20. [C] $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$ we get

    $$\Delta = \begin{vmatrix} x & -y & 0 \\ 0 & y & -z \\ p-x & q-y & r \end{vmatrix} = 0 \Rightarrow pyz + qzx + rxy = 2xyz \Rightarrow \frac{p}{x} + \frac{q}{y} + \frac{r}{z} = 2$$
21. [C] $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}\begin{vmatrix} -a & -b & -c \\ c & a & b \\ b & c & a \end{vmatrix}$

LEVEL – III

22. expanding determinant and comparing co-efficients
23. (B); $(a-b)(b-c)(c-a)$ does not change if a is replaced by b, b by c and c by a. But the degree of expression is 5 , therefore it must have a homogeneous symmetric expression of degree 2 as factor. The two degree symmetric expressions are only

    $$a^2 + b^2 + c^2, ab + bc + ca, (a+b+c)^2\ a^2 + b^2 + c^2 - ab - bc - ac$$

    Thus one of the following must be an identity.

$$\begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix} = k(a-b)(b-c)(c-a)(a^2+b^2+c^2) \ldots (i)$$

$$\Rightarrow \begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix} = k(a-b)(b-c)(c-a)(a+b+c)^2 \ldots (\text{ ii })$$

$$\Rightarrow \begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix} = k(a-b)(b-c)(c-a)\,(a^2+b^2+c^2-ab-bc-ac) \ldots$$

$$\Rightarrow \begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix} = k(a-b)(b-c)(c-a)(ab+bc+ac)\ldots (\text{ (iv) Where } k \text{ is a}$$

constant.

In the first assumption putting $a = 0, b = 1, c = -1$, we get

$$\begin{vmatrix}1 & 0 & 0\\ 1 & 1 & 1\\ 1 & 1 & -1\end{vmatrix} = \mathrm{k}(0-1)(1+1)(-1-0)(2) \Rightarrow \ -2 = 4\mathrm{k} \Rightarrow \mathrm{k} = -\frac{1}{2}$$

Also putting $\mathrm{a} = 0, \mathrm{b} = 1, \mathrm{c} = 2$, we get $\mathrm{k} = \frac{5}{2}$

Since the two values of k are different; (i) cannot be an identity.

We can similarly show that second and third are not identities either. Thus the only possibility left is fourth.

Thus $\begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix} = k(a-b)(b-c)(c-a)(ab+bc+ac)$

On putting $\mathrm{a} = 0, \mathrm{b} = 1, \mathrm{c} = -1$, we get $\mathrm{k} = 1$

(any other choice of values for a, b, c will yield same k).

24. [D) ; Since the degree of D_2 is 4 , the fourth factor should be of degree 1 . This factor should be symmetric i.e., if $a \to b, b \to c$ and $c \to a$, the value of the factor must not change (since in D_2 if $a \to b, b \to c, c \to a$, then no change occurs.)

The only 1 dimensional factor which is symmetric in a, b, c is $a + b + c$.

25. (C) ; On putting $a = -b$, the determinant

$$= \begin{vmatrix}2b & 0 & c-b\\ 0 & 2b & b+c\\ c-b & b+c & -2c\end{vmatrix} = 2b[4bc-(b+c)^2] + (c-b)[2b(c-b)]$$

$$= 2b(b-c)^2 - 2b(b-c)^2 = 0$$

$\Rightarrow a + b$ is a factor. Since the determinant is symmetric, $b + c, c + a$ should also be the factors since the determinant is of degree 3. No other factor is possible.

$$\Rightarrow \begin{vmatrix}-2\mathrm{a} & \mathrm{a}+\mathrm{b} & \mathrm{c}+\mathrm{a}\\ \mathrm{a}+\mathrm{b} & -2\ \mathrm{b} & \mathrm{b}+\mathrm{c}\\ \mathrm{c}+\mathrm{a} & \mathrm{b}+\mathrm{c} & 2\mathrm{c}\end{vmatrix} = \mathrm{k}(\mathrm{a}+\mathrm{b})(\mathrm{b}+\mathrm{c})(\mathrm{c}+\mathrm{a})$$

Where k is a constant. Since this is an identity, we put a = b = c = 1

To get k, we get $\begin{vmatrix} -2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{vmatrix} = 8k \Rightarrow 32 = 8k \Rightarrow k = 4$

26. A-S,R; B-S-R, C-P, D-Q

A) 6 places with -1 or 1 in 2^6 ways = 64

B) $\Delta = (a_1b_2c_3 + b_1c_2a_3 + c_1a_2b_3)$

$-(a_3b_2c_1 + b_3c_2a_1 + c_3a_2b_1)$

All can not be +ve. Maximum value = 4

C) For the maximum trace principal diagonal = (1,1,1)

D) No. of skew symmetric matrices = 0

27. $x + 2 - 6 = 0$

$x = 4$

28. C, D

29. (A) The given determinant is $\Delta = \begin{vmatrix} x+2 & x+3 & x+5 \\ x+4 & x+6 & x+9 \\ x+8 & x+11 & x+15 \end{vmatrix}$

Applying $R_3 \to R_3 - R_2$ and $R_2 \to R_2 - R_1$, we have

$\Delta = \begin{vmatrix} x+2 & x+3 & x+5 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{vmatrix} = 2\begin{vmatrix} x & x & x+1 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix}$ [Applying $R_1 \to R_1 - R_2$ and $R_3 \to R_3 - R_2$]

$= 2\begin{vmatrix} x & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$ [Applying $C_3 \to C_3 - C_2$ and $C_2 \to C_2 - C_1$] =-2 [Expanding along R_3]

(B) $\begin{vmatrix} 7 & 6 & x^2-13 \\ 2 & x^2-13 & 2 \\ x^2-13 & 3 & 7 \end{vmatrix}$

Let $x^2 - 13 = t$. Then $t^3 - 67t + 126 = 0 \Rightarrow t = -9,2,7 \Rightarrow x = \pm 2, \pm\sqrt{20}, \pm\sqrt{15}$

Hence sum of other five roots is 2 .

(C) $\Delta\begin{vmatrix} \sqrt{6} & 2i & 3+\sqrt{6} \\ \sqrt{12} & \sqrt{3}+\sqrt{8}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{18} & \sqrt{2}+\sqrt{12}i & \sqrt{27}+2i \end{vmatrix}$

Taking $\sqrt{6}$ common from C_1, we get $\Delta = \sqrt{6}\begin{vmatrix} 1 & 2i & 3+\sqrt{6} \\ \sqrt{2} & \sqrt{3}+2\sqrt{2}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{3} & \sqrt{2}+2\sqrt{3}i & 3\sqrt{3}+2i \end{vmatrix}$

Applying $R_2 \to R_2 - \sqrt{2}R_1$ and $R_3 \to R_3 - \sqrt{3}R_{-1}$, we get

$$\Delta = \sqrt{6}\begin{vmatrix} 1 & 2i & 3+\sqrt{6} \\ 0 & \sqrt{3} & \sqrt{6}i - 2\sqrt{3} \\ 0 & \sqrt{2} & 2i - 3\sqrt{2} \end{vmatrix} = \sqrt{6}(-3\sqrt{6} + 2\sqrt{6}) = -6, \text{ which is an integer}$$

(D) $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$

Applying $R_1 \to R_1 + (\sin\theta)R_3$ and $R_2 \to R_2 - (\cos\theta)R_3$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin\theta \\ 0 & 1 & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix} = \sin^2\theta + \cos^2\theta = 1 \Rightarrow f(\pi/3) = 1$$

30. [A, B]; $A^2 = -kI$; $\begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix}\begin{bmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{bmatrix} = \begin{bmatrix} -K & 0 \\ 0 & -K \end{bmatrix}$

31. Let $m = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad \therefore a - b = -1, c - d = 2$ $\quad -a + 2b = 1, -c + 2d = 0$ $\Rightarrow m = \begin{bmatrix} -1 & 0 \\ 4 & 2 \end{bmatrix}$

$\therefore |m - xI| = 0 \Rightarrow (x-2)(x+1) = 0 \Rightarrow x_1 = +2, x_2 = -1$

$5x_1 + 2x_2 = 10 - 2 = 8$

32,33,34.

$A^{50} = A^{48} + A^2 - I$

$=25A^2 - 24I = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix} \Rightarrow \det A^{50} = 1$

and Trace $A^{50} = 3$

WORKSHEET – 4

LEVEL – I

1. B	2. A	3. B	4. B	5. A	6. C	7. C
8. B	9. A	10. A				

LEVEL – II

1. C	2. B	3. A	4. A	5. C	6. B	7. B
8. C	9. B	10. C	11. B	12. C	13. D	14. B
15. B	16. A	17. C	18. C	19. A	20. D	21. A
22. B						

LEVEL – III

23. A, B, C	24. A	25. B	26. C	27. *a -q,s b -q,s c - p d -q*
28. C	29. A	30. a-r; b-s; c-p; d-q		31. A, B, C, D
32. D	33. A	34. 2		

LEVEL – II

1. Co-factor matrix $= \begin{bmatrix} +(-7) & -(-14) & +(7) \\ -(4) & +6 & -(2) \\ +(-5) & -(-4) & +(1) \end{bmatrix}$

   $\text{adj}\, A = (\text{ co } - \text{ factor matrix })^{\top}$

   $= \begin{bmatrix} -7 & -4 & -5 \\ 14 & 6 & 4 \\ 7 & -2 & 1 \end{bmatrix}$

2. [B] $A\text{adj}\, A = |A|.I = kI$; $|A| = k$; $|\text{adj}\, A| = |A|^{n-1} = |A|^2 = K^2$

3. [A]; Hint: $\text{adj}\, A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & -3 & 2 \\ -1 & 2 & -1 \end{bmatrix}^{\top} = \begin{bmatrix} 2 & 0 & -1 \\ 0 & -3 & 2 \\ -1 & 2 & -1 \end{bmatrix}$, $|A| = -1$

   $\therefore A^{-1} = \dfrac{\text{adj}\, A}{|A|} = \dfrac{1}{-1}\begin{bmatrix} 2 & 0 & -1 \\ 0 & -3 & 2 \\ -1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 0 & 3 & -2 \\ 1 & -2 & 1 \end{bmatrix}$

4. [A] $A^2 - 5A - 14I = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} - 5\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} - 14\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

   $= \begin{bmatrix} 9+20 & -15-10 \\ -12-8 & 20+4 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$

   $= \begin{bmatrix} 29-15-14 & -25+25-0 \\ -20+20-0 & 24-10-14 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \;\; \therefore A^2 - 5A - 141 = 0$

   $|A| = \begin{vmatrix} 3 & -5 \\ -4 & 2 \end{vmatrix} = 6 - 20 = -14 \neq 0$

   $\therefore \text{A}^{-1}$ exists.

   Multiplying (1) by A^{-1}, we get $A^{-1}(A^2 - 5A - 14I) = A^{-1}O$

   $\Rightarrow (A^{-1}A)A - 5(A^{-1}A) - 14(A^{-1}I) = 0 \Rightarrow A - 5I - 14A^{-1} = 0 \Rightarrow 14A^{-1} = A - 5I$

   $\therefore\; A^{-1} = \frac{A-5I}{14} = \frac{1}{14}\left(\begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \frac{1}{14}\begin{bmatrix} -2 & -5 \\ -4 & -3 \end{bmatrix}$.

5. [C] $A(\text{Adj}.\, A) = |A|I_3 = |A| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix}$ (or)

   $|\text{A}|\,|\,\text{Adj}.\, A| = |\text{A}^3 = (\text{a}^3)^3 = \text{a}^9 \Rightarrow$ (C)

   $|A|\,|\,\text{Adj.A}\,| = |A|^{n-1}|A| = |A|^2|A| = a^6 \cdot a^3 = a^9 \Rightarrow (b)$

6. [B] $(B^{-1}A^{-1})^{-1} = AB$

7. [B] $|\text{adj}\, A| = |A|^{n-1} = 5^2 = 25$

8. $|A| = 0, (\text{adj}\, A)B \neq 0$

   it has no solution

9. [B] $A(\text{adj}\, A) = |A|.I$ 10. [3] $\det A^{-1} = \frac{1}{\det A}$

10. [B] $|\text{Adj}\, A| = |A^{n-1}|$

11. $|A| = 0, (\text{adj}\, A)B \neq 0$

It is inconsistent

13. [D] $4^3|\ \mathrm{A}|^2 = 64 \times \mathrm{K} = 64\ \mathrm{K}$

14. [B]; $\det A = -1(2-9) - 1(4-3) = 7 - 1 = 6$

$[A(\mathrm{adj}\, A)A^{-1}]A = A(\mathrm{adj}\, A)A^{-1}A = A(\mathrm{adj}\, A) = \det AI = 6I$

15. [B] Multiply we get $3A^{-1} - A + 2I = 0$

16. $\begin{vmatrix} 2 & -3 & 4 \\ 5 & -2 & -1 \\ 21 & -8 & a \end{vmatrix} = 0$

17. $|\mathrm{Adj\ A}| = (\det \mathrm{A})^2$

18. a = cofactor of ' -1 ' $= (-4) = 4$

b = cofactor of ' 1 ' $(\mathrm{a}_{33}) = 1$

19. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & \lambda \\ 1 & 3 & 3 \end{vmatrix} = 0$

20. [D] $B = A^{-1}$

21. [A] $A^2 = I \Rightarrow A = A^{-1}$ also $(KA)^{-1} = \frac{1}{K}A^{-1}$ Hence $\left(\frac{1}{2}A\right)^{-1} = 2A^{-1} = 2A$

22. $\begin{vmatrix} 4 & k & 2 \\ k & 4 & 1 \\ 2 & 2 & 1 \end{vmatrix} = 0$

LEVEL – III

23. [A, B, C]; $\because |A| = \begin{vmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{vmatrix} = 3(-3+4) + 3(2-0) + 4(-2-0) = 1$

$\therefore \mathrm{adj}(\mathrm{adj}\, A) = |A|^{3-2}A = A$ and $|\mathrm{adj}(\mathrm{adj}\, A)| = |A| = 1$

Also, $|\mathrm{adj}\, A| = |A|^{3-1} = |A|^2 = 1^2 = 1$

24. [A] $\mathrm{adj}(\mathrm{adj}\, A) = |A|^{n-2} \cdot A = |A|A = -A$

25. [B] $|AB| = -24$, then $|\mathrm{adj}\, AB| = |AB|^{n-1} = 24^2$

26. [C] $|\mathrm{adj}(\mathrm{adj}(2\ \mathrm{A}))| = 1$

27. $a \to q,s\ b \to q,s\ c \to p\ d \to q$

28. $\mathrm{Adj}\, A = \begin{bmatrix} -3 & +6 & +6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix} = 3A^{\mathrm{T}}$

29. $A^{-1} = \frac{\mathrm{Adj}\, A}{\det A}$

30. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}\ \therefore \mathrm{adj}(A) = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}\ \therefore A^{-1} = \frac{\mathrm{adj}(A)}{|A|}$

$= \frac{1}{1+\tan^2 x}\begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} = \begin{bmatrix} \cos^2 x & -\sin x\cos x \\ \sin x\cos x & \cos^2 x \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1+\cos 2x & -\sin 2x \\ \sin 2x & 1+\cos 2x \end{bmatrix}$

$\therefore (A) \to (R)\ \mathrm{Adj}(\mathrm{Adj}\, A) = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} = A$

$\therefore (\text{Adj } A)^{-1} = \frac{\text{Adj}(\text{Adj } A)}{|\text{Adj } A|} = \frac{1}{2}\begin{bmatrix} 1+\cos 2x & \sin 2x \\ -\sin 2x & 1+\cos 2x \end{bmatrix}$

$\therefore (B) \to (S)$; and $(C) \to (P)$

$2\text{ A} = 2\begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \therefore (\text{adj } 2\text{ A}) = 2^{2-1}\text{ adj (A)} = 2\begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \therefore \text{D} - \text{Q}$

31. $AB = \begin{bmatrix} \cos(\alpha+2\beta) & \sin(\alpha+2\beta) \\ \sin(\alpha+2\beta) & -\cos(\alpha+2\beta) \end{bmatrix} \Rightarrow (AB)^2 = I \Rightarrow (AB)(AB) = I \Rightarrow A^{-1} = BAB$

$(AB)^{-1} = AB \Rightarrow B^{-1}A^{-1} = AB \Rightarrow BB^{-1}A^{-1} = BAB \Rightarrow A^{-1} = BAB$

$BA^4B = A^{-1} \Rightarrow B^{-1}BA^4B = B^{-1}A^{-1} \Rightarrow A^4B = (AB)^{-1} = AB$

$\Rightarrow A^4 = A \Rightarrow \cos 4\alpha = \cos\alpha, \sin 4\alpha = \sin\alpha\ 4\alpha = 2\pi + \alpha \Rightarrow \alpha = \frac{2\pi}{3}$

32,,33&34;

$\Delta = -14a - 42, \Delta = -8ab - 9b + 6a + 13 = -14(a+3) = 15b - 5ata = -3$

$\therefore a \neq -3, b \in R$, unique solution

$a = -3, b \neq \frac{1}{3}$, no solution

$a = -3, b = \frac{1}{3}$, infinite solutions.

12. ANALYTICAL GEOMETRY

WORKSHEET – 1

LEVEL – I

1. A	2. C	3. D	4. C	5. B	6. A	7. A
8. D	9. C	10. C	11. B			

LEVEL – II

1. A	2. D	3. B	4. D	5. A	6. B	7. D
8. D	9. C	10. D	11. C	12. B	13. B	14. C
15. A	16. A	17. C	18. B	19. D	20. B	21. B
22. C	23. B	24. D	25. D			

LEVEL – III

26. A, C	27. A	28. C	29. D	30. a-s, b-s, c-q, d-(p,r)
31. a-s, b-r, c-p, d-q		32. D	33. C	34. 648

LEVEL – II

1. $\sqrt{a^2\cos^2\theta + a^2\sin^2\theta} = |a|$
2. D
3. B
4. D
5. Given vertices are $(0,0), (1,4), (2,5)$

   Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right) = \left(\frac{0+1+2}{3}, \frac{0+4+5}{3}\right) = (1,3)$.
6. Distance from origin to (x_1, y_1) is $\sqrt{x_1^2 + y_1^2}$.

   $\therefore$ Distance from origin to $(\tan\alpha, 1)$ is $\sqrt{\tan^2\alpha + 1} = |\sec\alpha|$.
7. Distance between 2 points is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

   $= \sqrt{(\cos\theta + \sin\theta)^2 + (\sin\theta - \cos\theta)^2} = \sqrt{2}$.
8. D
9. $2x + 3 = 1 \& 3y - 4 = 5 \Rightarrow x = -1 \& y = 3 \therefore (x, y) = (-1,3)$.
10. Given $A(7,9)$; $B(3,-7)$; $C(-3,3)$

    $AB = \sqrt{(7-3)^2 + (9+7)^2} = \sqrt{16 + 256} = \sqrt{272}$

    $BC = \sqrt{(3+3)^2 + (-7-3)^2} = \sqrt{36 + 100} = \sqrt{136}$

    $CA = \sqrt{(7+3)^2 + (9-3)^2} = \sqrt{100 + 36} = \sqrt{136}$

    Here $BC = CA$ and $BC^2 + CA^2 = 136 + 136 = 272 = AB^2$

    $\therefore ABC$ is right angled isosceles triangle

11. C

12. A point P divides the straight line joining two points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m:n$ is $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$.

    Since P divides the line segment joining points $A(0,0)$ and $B(9,12)$ internally in $1:2$ ratio $\because x = \frac{9\times1+0\times2}{3} = \frac{9}{3} = 3, y = \frac{12\times1+2\times0}{3} = \frac{12}{3} = 4$

    The coordinates of point P are (3,4).

13. Given $A(7,1), B(4,4), C(-2,-2), D(1,-5)$

    $AB = \sqrt{9+9} = 3\sqrt{2}, BC = \sqrt{36+36} = 6\sqrt{2}, CD = \sqrt{9+9} = 3\sqrt{2}, AD = \sqrt{36+36} = 6\sqrt{2}$
    $AC = \sqrt{81+9} = 3\sqrt{10}, BD = \sqrt{9+81} = 3\sqrt{10}$
    $AB = CD, BC = AD$ and $AC = BD$ $\therefore ABCD$ is a rectangle

14. Here$(2-7+2, 1-4+5) = (-3,2)$.

15. Given vertices are $(K, 2), (2,3), (3,4)$ of points are collinear then area of triangle in zero. $\Rightarrow \frac{1}{2}\begin{vmatrix} x_1-x_2 & x_1-x_3 \\ y_1-y_2 & y_1-y_3 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} k-2 & k-3 \\ 2-3 & 2-4 \end{vmatrix} = 0$

    $\Rightarrow (k-2)(-2) - (k-3)(-1) = 0 \Rightarrow -2k+4+k-3 = 0 \Rightarrow k = 1$

16. Third Vertex $\left(\frac{2+4\pm(4-2)}{2}, \frac{(4+2)+(2-4)}{2}\right) = (3 \pm 1, 3 \pm 1) = (4,4)$ or $(2,2)$

17. Here $S = P + R - Q$

18. By verification the ordinate is 3

19. Points are collinear = Area of triangle is zero $\Rightarrow \begin{vmatrix} x_1-x_2 & x_1-x_3 \\ y_1-y_2 & y_1-y_3 \end{vmatrix} = 0$

    $$\Rightarrow \begin{vmatrix} (k-1)-(k) & (k-1)-(k+1) \\ (k+2)-(k+1) & (k+2)-k \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -1 & -2 \\ 1 & 2 \end{vmatrix} = 0$$

    For all values of ' K ' points are collinear.

20. $SA = \sqrt{a(t^2-1)+4a^2t^2} = \sqrt{a^2[(t^2-1)+4t^2]} = \sqrt{a^2(t^2+1)^2} = \sqrt{a^2\left(\frac{1}{t^2}+1\right)^2} = a\left(\frac{1}{t^2}+1\right)$

$$SB = \sqrt{\left[a\left(\frac{1}{t^2}-1\right)+\left(\frac{2a}{t}\right)^2\right.} = \sqrt{a^2\left[\left(\frac{1}{t^2}-1\right)^2+\frac{4}{t^2}\right]} = \sqrt{a^2(t^2+1)^2} = \sqrt{a^2\left(\frac{1}{t^2}+1\right)^2} = a\left(\frac{1}{f^2}+1\right)$$

$$\frac{1}{SA}+\frac{1}{SB} = \frac{1}{a(t^2+1)}+\frac{1}{a\left(\frac{1}{t^2}+1\right)} = \frac{1}{a(t^2+1)} = \frac{1+t^2}{a(1+t^2)} = \frac{1}{a}$$

21. If P divides $\overline{QR}$ externally in the ratio $1:2$, then Q is the midpoint of PR. Let $R(x,y)$, then $(-2,-3) = \left(\frac{-5+x}{2}, \frac{4+y}{2}\right)$. $\therefore x = 1$ and $y = -10$ $\therefore R(x,y) = (1,-10)$.

22. Given points $(a,0), (0,b)$&$(1,1)$ are collinear = Area of triangle is zero.

    $$\Rightarrow \begin{vmatrix} x_1-x_2 & x_1-x_3 \\ y_1-y_2 & y_1-y_3 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a-0 & a-1 \\ 0-b & 0-1 \end{vmatrix} = 0 \Rightarrow -a+b(a-1) = 0$$

$$\Rightarrow a + b = ab \Rightarrow \frac{a+b}{ab} = 1 \Rightarrow \frac{1}{a} + \frac{1}{b} = 1 \text{ Option-3}$$

23. $x = \frac{(1-t)x_1+tx_2}{1-t+t}$; $y = \frac{(1-t)y_1+ty_2}{1-t+t}$; $1 - t > 0, t > 0$

24. Given vertices of the Quadrilateral are $A(2,-1), B(4,3), C(-1,2), D(-3,2)$

Area of quadrilateral $= \frac{1}{2}\begin{vmatrix} x_1 - x_3 & x_2 - x_4 \\ y_1 - y_3 & y_2 - y_4 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} 2+1 & 4+3 \\ -1-2 & 3-2 \end{vmatrix} = \frac{1}{2}(3(1) - 7(-3))$

$$= \frac{24}{2} \text{ sq.units } = 12 \text{ sq.units}$$

25. $AB = \sqrt{16+9} = 5, C = (-5,-2)$ then $AC = \sqrt{64+36} = 10; BC = \sqrt{144+81} = 15 \therefore AC + AB = BC \Rightarrow C = (-5,-2)$.

LEVEL – III

26. A, C

27. A

28. C

29. D

30. a-s,b-s,c-q,d-(p,r)

31. Given vertices are $(P-1,1), (2P+1,3) \& (2P+2,2P)$

Area of $\Delta le = \frac{1}{2}\begin{vmatrix} x_1 - x_2 & x_1 - x_3 \\ y_1 - y_2 & y_1 - y_3 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} (p-1)-(2p+1) & (p+1)-(2P+2) \\ 1-3 & 1-2p \end{vmatrix}$

$$= \frac{1}{2}\begin{vmatrix} -p & -p-1 \\ -2 & 1-2p \end{vmatrix} = \frac{1}{2}(-p + 2p^2 - 2p - 2) = \frac{1}{2}|(2p^2 - 3p - 2)|.$$

32. 1) Is incorrect since A, B, C are collinear

2) Is false since $AB^2 \neq AC^2$

3) Is incorrect since $AB^2 \neq AC^2$

4) Is correct since

$AB^2 = (1-a-b)^2 + \left(\frac{b}{a} - b + a\right)^2$ and $AC^2 = (1-a+b)^2 + \left(\frac{b}{a} - a - b\right)^2$

Note that $AB^2 = AC^2$.

33. $G\left(\frac{p+q+1}{3}, \frac{q+1+p}{3}\right) = (0,0) \therefore p + q + 1 = 0 \Rightarrow p^3 + q^3 + 1^2 = 3pq$

$\therefore a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc \ \therefore k = 3$.

34. Here $\frac{a+1+a^2+1}{2} = 11, \frac{a-1+a^2-1}{2} = 9 \Rightarrow a + a^2 - 20 = 0 \Rightarrow a = -5$ or 4 and $\frac{a^2+1+a^3+1}{2} = 41, \frac{a^2-1+a^3-1}{2} = 39 \Rightarrow a^2 + a^3 = 80$ which holds for $a = 4$

so, the given points are $A(5,3), B(17,15), C(65,63)$ and coordinates of F are and $(BF)^2 = (35-17)^2 + (33-15)^2 = (18)^2 + (18)^2 = 648$.

WORKSHEET – 2

LEVEL – I

1. A	2. B	3. A	4. C	5. B	6. A	7. B
8. C	9. A	10. B	11. A	12. B		

LEVEL – II

1. B	2. C	3. A	4. B	5. A	6. B	7. D
8. A	9. B	10. C	11. B	12. B	13. B	14. D
15. B	16. D	17. C	18. C	19. C	20. C	21. A
22. B	23. A	24. A	25. C			

LEVEL – III

26. B, C	27. C	28. A	29. B	30. C	31. a-r; b-q; c-t; d-p
32. B	33. A, B	34. a-q; b-p; c-s; d-r			

LEVEL – II

1. Slope of the line is $\tan 45° = 1$. Equation of the line passing through (1,0) and having slope 1 , is $(y - 0) = 1(x - 1) \Rightarrow x - y - 1 = 0$

2. $\frac{x}{1} + \frac{y}{2} = 1;\ 2x + y = 2$

3. Slope $(m) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2+4}{-3-5} = -\frac{3}{4}$

4. Given equation $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow bx + ay = ab \Rightarrow ay = -bx + ab \Rightarrow y = -\frac{b}{a}x + b$

   It is in the form of $y = mx + c$ $\therefore$ Slope $= \frac{-b}{a}$

5. Equation of the line is $\frac{y - y_1}{x - x_1} = \frac{y_1 - y_2}{x_1 - x_2} \Rightarrow \frac{y-3}{x-3} = \frac{3-6}{3-7} \Rightarrow 3x - 4y + 3 = 0.$

6. $\tan\theta = \frac{-1}{\sqrt{3}}.$

7. Slopes of the parallel lines are equal.

8. $y = m(x - 8), m = \tan 30.$

9. $m = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-2\sqrt{3}-1}{0-2} = \frac{-2\sqrt{3}}{-2} \tan\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}.$

10. Given points are $A(\mathrm{am}_1^2, 2\mathrm{am}_1), \mathrm{B}(\mathrm{am}_2^2, 2\mathrm{am}_2)$

    Slope of $AB(m) = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow m = \frac{2am_2 - 2am_1}{am_2^2 - am_1^2} = \frac{2(m_2 - m_1)}{m_2^2 - m_1^2} = \frac{2}{m_1 + m_2}$

    $\therefore$ The required equation is $\mathrm{y} - \mathrm{y}_1 = \mathrm{m}(\mathrm{x} - \mathrm{x}_1) \Rightarrow \mathrm{y} - 2\mathrm{am}_1 = \frac{2}{\mathrm{m}_1 + \mathrm{m}_2}(\mathrm{x} - \mathrm{am}_1^2)$

    $\Rightarrow \mathrm{y}(\mathrm{m}_1 + \mathrm{m}_2) - 2\mathrm{amm}_1(\mathrm{m}_1 + \mathrm{m}_2) = 2\mathrm{x} - 2\mathrm{amm}_1^2$

    $\therefore 2\mathrm{x} - \mathrm{y}(\mathrm{m}_1 + \mathrm{m}_2) - 2\mathrm{am}_1^2 - 2\mathrm{am}_1\ \mathrm{m}_2 + 2\mathrm{am}_1^2 = 0$

    $\Rightarrow 2\mathrm{x} - \mathrm{y}(\mathrm{m}_1 + \mathrm{m}_2) - 2\mathrm{am}_1\ \mathrm{m}_2 = 0$, is the required equation.

11. $\mathrm{m} = \tan 60^\circ = \sqrt{3}$ and the y-intercept is 3

    $\therefore$ The required equation is $y = \sqrt{3}x + 3$.

12. If the lines are perpendicular then the product of slopes $= -1$

    $m_1 \times m_2 = -1 \Rightarrow \frac{-2}{3} \times \frac{-3}{-a} = -1 \Rightarrow a = 2$

13. If the lines are Parallel, then Slopes are equal $\Rightarrow m_1 = m_2 \Rightarrow \frac{-1}{-3} = \frac{-k}{+6} \Rightarrow k = -2.$

14. $8x + 6y - 15 = 0, 6y = -8c + 15 \Rightarrow y = \frac{-8}{6}x + \frac{15}{6}, y = \frac{-4}{3}x + \frac{5}{2}.$

15. Midpoint of (1,2) and (3,4) is (2,3). Equation of line passing through (2,3) whose slope ' -1 ' is $y - 3 = -1(x - 2) \Rightarrow x + y = 5.$

16. Slope of line passing through $(1,2), (-1,5)$ is $m = \frac{5-2}{-1-2} = \frac{-3}{2}$

    Equation of line passing through (2,3) and whose slope is $\frac{-3}{2}$ is

    $y + 3 = \frac{-3}{2}(x - 2) \Rightarrow 2y + 6 = -3x + 6 \Rightarrow 3x + 2y = 0$

17. Given vertices of $\triangle ABC$ are $A(1,1), B(-3,4), C(2,-5)$

    Slope of Altitude perpendicular to BC $= \frac{5}{9}$

    Equation of line passing through (1,1) whose slope is 5/9 is $y - 1 = \frac{5}{9}(x -$

    1) $9y - 9 = 5x - 5 \Rightarrow 5x - 9y + 4 = 0.$

18. Given line is $\left(\frac{1}{a} + \frac{k}{b}\right)x + \left(\frac{1}{b} + \frac{k}{a}\right)y - (1 + k) = 0$

    $\therefore$ Slope $= -\frac{\left(\frac{1}{a}+\frac{k}{b}\right)}{\frac{1}{b}+\frac{k}{a}}$. But given that the slope is -1

    $\therefore \frac{\left(\frac{1}{a} + \frac{k}{b}\right)}{\left(\frac{1}{b} + \frac{k}{a}\right)} = 1 \Rightarrow \frac{1}{a} + \frac{k}{b} = \frac{1}{b} + \frac{k}{a} \Rightarrow \frac{1}{a} - \frac{1}{b} = \frac{k}{a} - \frac{k}{b} \Rightarrow \left(\frac{1}{a} - \frac{1}{b}\right) = k\left(\frac{1}{a} - \frac{1}{b}\right) \Rightarrow k = 1$

19. Given $P(\alpha, \beta)$ lies on $y = 6x - 1 \Rightarrow \beta = 6\alpha - 1 \Rightarrow 6\alpha - \beta - 1 = 0$

    Also $Q(\beta, \alpha)$ lies on $2x - 5y = 5 \Rightarrow 2\beta - 5\alpha = 5 \Rightarrow 5\alpha - 2\beta + 5 = 0$

    Solving (1) and (2), we have $\alpha = 1, \beta = 5 \therefore P = (1,5), Q = (5,1)$ Slope of

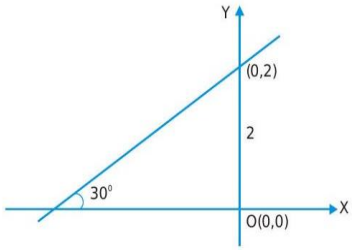
    $PQ = \frac{5-1}{1-5} = -1$

    Equation of PQ is $y - 5 = -1(x - 1) \Rightarrow y - 5 = -x + 1 \Rightarrow x + y = 6$

20. Intercepts between the axes made by the given lines are $a\sqrt{2}, ar\sqrt{2}, ar^2\sqrt{2}.$

21. Here $\mathrm{A} = (3,0), \mathrm{B} = (4,0); c = \left(2 - \frac{2}{m}, 0\right).$

22. [B] Given y - intercept (c) = 2 and $\theta = 30^\circ \Rightarrow \mathrm{m} = \tan\theta = \tan 30^\circ = \frac{1}{\sqrt{3}}$



∴ Required equation is $y = mx + c \Rightarrow y = \frac{1}{\sqrt{3}}x + 2 \Rightarrow x - \sqrt{3}y + 2\sqrt{3} = 0$.

23. $\sin\theta = \cos\theta \Rightarrow \tan\theta = 1$.

24. Here $(0, -4)$ lies on perpendicular bisector $\overline{PQ}$.

25. Equation of line in Intercept form is $\frac{x}{a} + \frac{y}{b} = 1$, where $a + b = 10$ and line passing through (3,2) is $\frac{3}{a} + \frac{2}{10-a} = 1 \Rightarrow 30 - 3a + 2a = 10a - a^2 \Rightarrow a^2 - 11a + 30 = 0 \Rightarrow (a-5)(a-6) = 0 \Rightarrow$ a = 5 or 6 $\Rightarrow$ b = 5 or 4 .

if $a = 5, b = 5$ the line is $x + y = 5$ and if $a = 6, b = 4$ the line is $\frac{x}{6} + \frac{y}{4} = 1$.

LEVEL – III

26. [B, C]

27. The equation of line parallel to $ax + by + c = 0$ and passing through (x_1, y_1) is $a(x - x_1) + b(y - y_1) = 0$

28. [A]

29. [B]

30. [C]

31. a-r; b-q; c-t; d-p ;

32. Slope $= \frac{-8}{-4} = 2$

33. A, B

34. a) $y - 2 = -1(x - 1) \Rightarrow x + y - 3 = 0$

b) $y - 5 = \frac{-3}{2}(x + 1) \Rightarrow 2y - 10 = -3x - 3 \Rightarrow 3x + 2y - 7 = 0$

c) $2x + 3y = 12 \Rightarrow \frac{2x}{12} + \frac{3y}{12} = 1 \Rightarrow \frac{x}{6} + \frac{y}{4} = 1 \Rightarrow \frac{x}{6} + \frac{y}{4} = 1$

d) $y - 4 = 1(x - 3) \Rightarrow x - y + 1 = 0$

WORKSHEET – 3

LEVEL – I

1. A	2. C	3. B	4. D	5. C	6. A	7. B
8. C	9. B	10. C				

LEVEL – II

1. A	2. D	3. B	4. A	5. B	6. B	7. B

8. D 9. C 10. B 11. D 12. C 13. C 14. D
15. A 16. C 17. D 18. C 19. C 20. C 21. C
22. D 23. C

LEVEL – III

24. B, D 25. A, C 26. D 27. A 28. B 29. C 30. C
31. B

1. Given equation is $4x - 3y + 12 = 0$

   $\Rightarrow \frac{-4}{\sqrt{4^2+3^2}}x + \frac{3}{\sqrt{4^2+3^2}}y = \frac{12}{\sqrt{4^2+3^2}} \Rightarrow \frac{-4}{5}x + \frac{3}{5}y = \frac{12}{5}$

2. Slope $= -\frac{1}{\sqrt{3}} = \text{Tan}\,\theta \Rightarrow \theta = 150^\circ \therefore$ Symmetric form is $\frac{x+2}{-\frac{\sqrt{3}}{2}} = \frac{y-0}{\frac{1}{2}}$

3. [B] $(x_1, y_1) = (3,2);\ \tan\theta = \frac{3}{4}, \sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}.\ (x, y) = (x_1 \pm r\cos\theta, y_1 \pm r\sin\theta).$

4. A

5. $x + y = 1 \Rightarrow \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow x + y = 1 \Rightarrow \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = \frac{1}{\sqrt{2}}$ Option:-B

6. $m = -1 \Rightarrow \tan\theta = -1 \Rightarrow \cos\theta = \frac{-1}{\sqrt{2}}; \sin\theta\,\frac{1}{\sqrt{2}} \therefore \frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} \Rightarrow \frac{x-1}{\frac{-1}{\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}}.$

7. Area of Δ formed by $ax + by + c = 0$ with axes is $\frac{c^2}{2|ab|} = \frac{(-1)^2}{2|(1)(-1)|} = \frac{1}{2}$ sq units.

8. Area of Δ formed by $\frac{x}{a} + \frac{y}{b} = 1$ is $\Delta = \frac{1}{2}|ab| = \frac{1}{2}\left|\frac{1}{2}\cdot\frac{1}{3}\right| = \frac{1}{12}$ sq uints. Option:-D.

9. [3] Line equation is $\sqrt{3}x + y - 4 = 0 \Rightarrow \frac{\sqrt{3}}{2}x + \frac{1}{2}y = 2 \Rightarrow x\cos\frac{\pi}{6} + y\sin\frac{\pi}{6} = 2.$

10. B

11. $\frac{C^2}{2|ab|} = 6 \Rightarrow \frac{K^2}{2|3\times4|} = 6 \Rightarrow K^2 = 144 \Rightarrow K = 12$ Option-D

12. $x\cos\alpha + y\sin\alpha = P \Rightarrow x\cos 150 + y\sin 150 = 6$

    $\Rightarrow\ x\left(\frac{-\sqrt{3}}{2}\right) + y\left(\frac{1}{2}\right) = 6$ Option-C

13. $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} \Rightarrow \frac{x-3}{\cos\frac{\pi}{3}} = \frac{y-4}{\sin\frac{\pi}{3}} \Rightarrow \frac{x-3}{\frac{1}{2}} = \frac{y-4}{\frac{\sqrt{3}}{2}}$ Option-C.

14. [D] perpendicular distance

    $= \frac{|-1|}{\sqrt{\frac{1}{p^2\sec^2\alpha} + \frac{1}{p^2\text{cosec}^2\alpha}}} = \frac{1}{\sqrt{\frac{1}{p^2}(\cos^2\alpha + \sin^2\alpha)}} = |p|$

15. $(x_1, y_1) = (2,3), \tan\theta = \frac{3}{4}; r = 5 \Rightarrow \sin\theta = \frac{3}{5}; \cos\theta = \frac{4}{5}$

    Req.points $= (x_1 \pm r\cos\theta, y_1 \pm r\sin\theta)$

$= \left(2 \pm 5 \cdot \left(\frac{4}{5}\right), 3 \pm 5 \cdot \left(\frac{3}{5}\right)\right) = ((2 \pm 4), 3 \pm 3) = (6,6)(\text{ or })(-2,0)$ Option-A

16. $x\cos 30° + y\sin 30° = 2$

$\Rightarrow \frac{\sqrt{3}x}{2} + \frac{y}{2} = 2 \Rightarrow \sqrt{3}x + y = 4$ Option-C

17. $\alpha = 60°; \mathrm{P} = 5\ \mathrm{x}\cos\alpha + \mathrm{y}\sin\alpha = \mathrm{P} \Rightarrow \mathrm{x}\cos 60° + \mathrm{y}\sin 60° = 5$

$\Rightarrow \frac{x}{2} + \frac{\sqrt{3}y}{2} = 5 \Rightarrow x + \sqrt{3}y = 10$ Option- D

18. $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow bx + ay = ab\ P = \frac{c}{\sqrt{a^2+b^2}} \Rightarrow P = \frac{ab}{\sqrt{b^2+a^2}}$

$P^2 = \frac{a^2b^2}{a^2+b^2} \Rightarrow \frac{1}{P^2} = \frac{a^2+b^2}{a^2b^2} \Rightarrow \frac{1}{P^2} = \frac{1}{a^2} + \frac{1}{b^2}$

19. $P(\sqrt{3}, 2), \theta = \frac{\pi}{6}\ PQ = 6$

$x = x_1 + r\cos\theta; y = y_1 + r\sin\theta \Rightarrow x = \sqrt{3} + r \cdot \cos 30°; y = 2 + r \cdot \sin 30°$

$x = \sqrt{3} + \frac{\sqrt{3}r}{2}; y = 2 + \frac{r}{2}.$ (x, y) lies on $\sqrt{3}x - 4y + 8 = 0$

$\sqrt{3}\left(\sqrt{3} + \frac{\sqrt{3}r}{2}\right) - 4\left(2 + \frac{r}{2}\right) + 8 = 0 \Rightarrow r = 6$

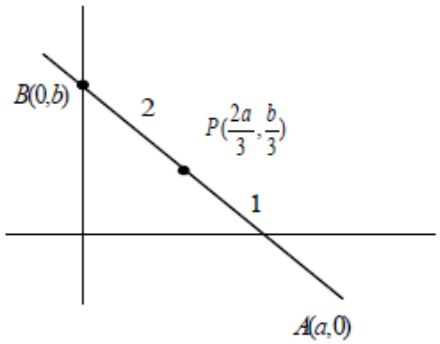
20. The line makes an angle ' θ ' with x-axis then its perpendicular makes an angle $\alpha = \theta - 90°$.

21. Equation of line passing through (3,4)&(5,6) is

$y - 4 = \frac{6-4}{5-3}(x-3)\ y - 4 = x - 3 \Rightarrow x - y + 1 = 0$

Area with the axes $= \frac{c^2}{2|ab|} = \frac{1}{2|1(-1)|} = \frac{1}{2}$ sq.units

22. Let $\frac{x}{a} + \frac{y}{b} = 1$ be line. When ' P ' divides $A(a,0)B(0,b)$ in the ratio then



$(-5,4) = P\left(\frac{1(0)+2(a)}{3}, \frac{1(b)+2(0)}{3}\right)$

$a = \frac{-15}{2}; b = 12\ \therefore$ Required line is $\frac{x}{\frac{-15}{2}} + \frac{y}{12} = 1.$

23. Here $\mathrm{p} = 2, \mathrm{x}\cos\alpha + \mathrm{y}\sin\alpha = 2$

LEVEL – III

24. The gradient of the perpendicular is $\frac{3}{4}$, so $\tan\alpha = \frac{3}{4}$ and $p = 6$. Now $\tan\alpha = \frac{3}{4}$ and $0 \leq \alpha < 2\pi$, then α either lies in the first quadrant or the third quadrant. $\therefore \cos\alpha = \frac{4}{5}$ and $\sin\alpha = \frac{3}{5}$ or $\cos\alpha = \frac{-4}{5}$ and $\sin\alpha = \frac{-3}{5}$. So the required equation is $x\left(\frac{4}{5}\right) + y\left(\frac{3}{5}\right) = 6 \Rightarrow 4x + 3y = 30$ or $x\left(\frac{-4}{5}\right) + y\left(\frac{-3}{5}\right) = 6 \Rightarrow 4x + 3y = -30$.

25. Here $\mathrm{p} = 2$ and as the line makes an angle $150°$ with positive direction of y axis so, clearly $\alpha = 150°$ or $330°$. $\therefore$ Required equation is

$$x(\cos 150°) + y(\sin 150°) = 2 \Rightarrow x\left(\frac{-\sqrt{3}}{2}\right) + y\left(\frac{1}{2}\right) = 2 \Rightarrow -\sqrt{3}x + y = 4 \text{ or}$$

$$x(\cos 330°) + y(\sin 330°) = 2 \Rightarrow x\left(\frac{\sqrt{3}}{2}\right) + y\left(\frac{-1}{2}\right) = 2 \Rightarrow \sqrt{3}x - y = 4.$$

26. Mid-point of the given vertices is $P(x_1, y_1) = (2,4)$ which lies on $2y = x + c$ then $\mathrm{c} = 6$. Now $\mathrm{r} = \mathrm{BP} = \mathrm{AP} = \sqrt{5}, \tan\theta = \frac{1}{2}$.

27. Hence third vertex $= (\mathrm{x}_1 + \mathrm{r}\cos\theta, \mathrm{y}_1 + \mathrm{r}\sin\theta) = (4,5)$.

28. Fourth vertex $= (x_1 - r\cos\theta, y_1 - r\sin\theta) = (0,3)$.

29. $\alpha = 135° - 90°, \mathrm{p} = 5$.

30. Here $r = \sqrt{\frac{2}{3}}$ and $(\mathrm{x}_1, \mathrm{y}_1) = (1,2)$.

31. Let $L = 0$ be the line passing through (1,5) and intersect the given lines at A and B. Let ' θ ' be the inclination of $L = 0$.

$\mathrm{A} = (1 - \mathrm{r}\cos\theta, 5 - \mathrm{r}\sin\theta), \mathrm{B} = (1 + \mathrm{r}\cos\theta, 5 + \mathrm{r}\sin\theta)$.

A lies on $3x + 4y - 4 = 0 \Rightarrow r = \frac{19}{3\cos\theta + 4\sin\theta}$,

B lies on $5x - y - 4 = 0 \Rightarrow r = \frac{4}{5\cos\theta - \sin\theta}$.

$\frac{19}{3\cos\theta + 4\sin\theta} = \frac{4}{5\cos\theta - \sin\theta} \Rightarrow \tan\theta = \frac{83}{35}$.

$\therefore$ Equation of the line is $83x - 35y + 92 = 0$.

13. BINOMIAL THEOREM

WORKSHEET – 1

LEVEL – I

1. B	2. B	3. B	4. D	5. C	6. A	7. D
8. D	9. B	10. A				

LEVEL – II

1. D	2. B	3. D	4. C	5. C	6. B	7. D
8. A	9. B	10. A	11. A	12. D	13. B	14. B
15. C	16. B	17. C	18. D	19. C	20. C	21. A

LEVEL – III

22. A, B, C, D 23. C 24. B 25. A 26. D

27. a-r,b-p,c-s,d-q 28. B, C, D 29. *a - q, b -t, c -s,r , d - p,q* 30. 5

31. D 32. A 33. A 34. 0

LEVEL – II

1. $n! = 720$

   $= 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6\ !$

   n = 6

2. If ${}^{n}C_{n-3} = 10 = \frac{n!}{(n-3)!(n-n+3)!} = \frac{n \times n-1 \times n-2}{3!} = 10$

   $5 \times 4 \times 3 = 60$

   n = 5

3. ${}^{n+1}C_3 = 4.\ {}^{n}C_2$

   $$\frac{(n+1)!}{(n-2)!\,(3!)} = 4 \cdot \frac{n!}{(n-2)! \cdot 2!}$$

   $$\frac{(n+1)(n!)}{6} = \frac{4 \cdot n!}{2}$$

   n + 1 = 12

   n = 11

4. ${}^{n}C_{12} = {}^{n}C_8$ then n =

   $n = r + s = 12 + 8 = 20$

5. Expand $(x+6)^5$

   $\Rightarrow x^5 + 30x^4 + 360x^3 + 2160x^2 + 6480x + 7776$

6. $(2+x)^{10}$

   6th term $\Rightarrow T_{r+1} = T_{5+1} \Rightarrow {}^{10}C_5(2)^{10-5}(x)^5$

$\Rightarrow {}^{10}C_5 2^5 x^5$

7. $\left(x-\frac{1}{2x}\right)^{10}$; middle term is $T_{\frac{10}{2}+1} = T_{5+1} = T_6$

$\therefore T_6 = T_{5+1} = {}^{10}C_5 \cdot (x)^{10-5} \cdot \left(\frac{-1}{2x}\right)^5 = 10_{C_5} \cdot x^5 \cdot \frac{-1}{2^5 \cdot x^5} = \frac{-63}{8}$

8. Independent term of x in $\left(x^2-\frac{1}{x}\right)^9$

$r = \frac{np}{p+q} = \frac{9(2)}{2+1} = \frac{18}{3} = 6 \Rightarrow {}^9C_6(x^2)^{9-6} \cdot \left(\frac{-1}{x}\right)^6 = {}^9C_6 \cdot x^6 \cdot \frac{1}{x^6} = {}^9C_6 = \frac{9!}{3!\,6!} = \frac{9\times8\times7}{6}$
$= 3 \times 4 \times 7 = 84$

9. ${}^{15}C_{r+3} = {}^{15}C_{2r-3}$ then $r =$

$15 = r+3+2r-r$
$3r = 15 = r = 5$

10. ${}^nP_r = 1320\ {}^nC_r = 220$

$r! = \frac{{}^nP_r}{{}^nC_r} = \frac{1320}{220} = 6$

$\frac{{}^nP_r}{{}^nC_r} = r! \Rightarrow r! = 6 = 3\times2\times1 = 3!$

$r = 3$

11. $(2x-1)^{13}$

12th term $\Rightarrow T_{11+1} = {}^{13}C_{11}(2x)^{13-11}(-1)^{11}$
$-{}^{13}C_2(2x)^2$

12. $\left(2x-\frac{3}{x}\right)^8$

$T_{r+1} = {}^nC_r(x)^{n-r}(y)^2$
$= {}^8C_r(2x)^{8-r}\left(\frac{-3}{x}\right)^r$
$= {}^8C_r(2x^{8-r})x^{8-2r}(-3)^r \cdot (x^{-r})$
$= {}^8C_r(2x^{8-r})(-3)^r x^{8-2r}$

r^{th} term contain x^{8-2r}.

$8-2r = -4$
$-2r = -12 \Rightarrow r = 6$
$\Rightarrow {}^8C_6(2)^{8-6}(-3)^6 x^{8-12}$
$2^2 = (-3)^6 = {}^8C_6(2)^2 \cdot 3^6$

13. ${}^{28}C_r \cdot {}^{24}C_{r-4} = 225:11$

$\frac{\frac{28!}{(28-r)!}r!}{24!}\frac{225}{(24-r+4)!\,(r-4)!}$

$\frac{28\times27\times26\times25\times24!}{r(r-1)(r-2)(r-3)(r-4)!} \times \frac{(r-4)!}{24!} = \frac{225}{11}$

$\therefore r = 14$

14. No. of non zero terms in $(x+a)^{100} + (x-a)^{100}$ is

No. of terms $\left[\frac{n}{2}+1\right] = \left[\frac{100}{2}+1\right]$

15. ${}^{10}C_5 + {}^{10}C_4 + {}^{10}C_4 + {}^{10}C_3$

$= {}^{11}C_5 + {}^{11}C_4$

$= {}^{12}C_5$

16. ${}^{n+1}P_5 : {}^{n}P_6 = 2:7$

$$\frac{(n+1)!}{n-4!} \times \frac{(n-6)!}{n!} = \frac{2}{7}$$

$$\frac{n+1}{(n-4)(n-5)} = \frac{2}{7}$$

$7(n+1) = 2(n^2 - 9n + 20)$

$(n-11)(2n-3) = 0$

$n = 11$

17. $\ln\left[x + \frac{1}{y}\right]^{23}$

$$T_{12} = {}^{23}C_{11}x^{12} \cdot \frac{1}{y^{11}}$$

$$T_{13} = {}^{23}C_{12}x^{11} \cdot \frac{1}{y^{12}}$$

$$\frac{T_{12}}{T_{13}} = \frac{{}^{23}C_{11} \cdot x^{120} \cdot y^{-11}}{{}^{23}C_{12}x^{11} \cdot y^{-12}}$$

$$\frac{T_{12}}{T_{13}} = \frac{{}^{23}C_{11}}{{}^{23}C_{12}}xy$$

$$\frac{T_{12}}{T_{13}} = \frac{11+1}{23-11} \times y$$

$$\frac{T_{12}}{T_{13}} = \frac{xy}{1}$$

$T_{12} = T_{13} \times y$

18. ${}^{50}C_4 + {}^{55}C_3 + {}^{54}C_3 + {}^{53}C_3 + {}^{52}C_3 + {}^{51}C_3 + {}^{50}C_3$

$= {}^{51}C_4 + {}^{51}C_3 + {}^{52}C_3 + {}^{53}C_3 + {}^{54}C_3 + {}^{55}C_3$

$= {}^{52}C_4 + {}^{52}C_3 + {}^{53}C_3 + {}^{54}C_4 + {}^{55}C_3$

$= {}^{53}C_4 + {}^{53}C_3 + {}^{54}C_3 + {}^{55}C_3 = {}^{56}C_4.$

19. $\left(\frac{q}{2x} - px\right)^8$

$t_5 = 1120,\ t_{4+1} = 1120$

$t_{4+1} = 8_{c_4} \cdot \left(\frac{q}{2x}\right)^4 \cdot (-px)^4 = 1120$

$= \frac{8 \times 7 \times 6 \times 5}{1 \times 2 \times 3 \times 4} \cdot \frac{q^4}{2^4} \cdot x^4 \cdot p^4 \cdot x^4 = 1120$

$= 70 \times p^4 \cdot q^4 = 1120 \times 2$

$p^4q^4 = 2^4 \cdot 2^4$

$(pq)^4 = (4)^4$

$p = 4 \quad q = 1$

$p = 4, \quad q = 2$

$p = 2, \quad q = 2\ p > q > 0,$

20. T_{r+1} is the term independent of x in $\left(ax + \frac{6}{x}\right)^{14}$

Then $r = \frac{np}{p+q} = \frac{14.1}{1+1} = \frac{14}{2} = 7$

21. $\left(\frac{3x}{50} - \frac{5}{2x}\right)^8 \Rightarrow p = 3, n = 8$

$T_{n-p+2} = T_{8-3+2} = T_7 = T_{6+1}$

$n = 8\ r = 6$

$T_{r+1} \Rightarrow T_{6+1} = 8_{c_{6^*}} \left(\frac{3x}{5}\right)^2 \cdot \left(\frac{-5}{2x}\right)^6$

$= 28 \cdot \frac{9 \cdot x^2}{25} \times \frac{5^6}{2^6} \cdot x^6 =$

$= \frac{39375}{16 \cdot x^4}$

LEVEL – III

22. Multi correct answers

$n = 5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$

$5 \times 4 \times 3! = 20 \times 3!$

23. ${}^nC_r = {}^nC_s \Rightarrow r = s$ (or) $n = r + s$

${}^{12}C_2 = \frac{12 \times 11}{2 \times 1} = 66$

24. ${}^nC_{24} = {}^nC_{35}$ then $n = 24 + 35 = 59 \Rightarrow {}^{60}C_{59} = 60$

25. ${}^{30}C_{2r+3} = {}^{30}C_{3r-2}$

$30 = 2r + 3 + 3r - 3 = 5r$

$5r = 30 \Rightarrow r = 6$

26. ${}^nC_4 = {}^nC_0$

$\Rightarrow n = 4$

27. a) Middle term $T_6 = -10_{c_5} \left(\frac{6}{7}\right)^5 x^5y^5$

b) Middle term $T_5 = 8_{c_4}(14)^4(pq)^4$

c) Middle terms $T_9,\ T_{10}$

$T_{10} = 17_{c_4} 4^8 5^9 \times 43$

d) Middle term $T_{11} = 20_{c_{10}}(15)^{10}a^{10}$

28. $(1-x)^2 = 1 - 2x + x^2$; co-efficient of x^2 is 1

a) $n_{c_r} = \frac{n}{r}\ {}^{n-1}C_{r-1}$

b) ${}^{n}P_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

c) If ' n ' is odd, middle term of $(1+x)^n$ is T_{r+1} where $r = \frac{n-1}{2}$ (or) $\frac{n+1}{2}$

d) $\frac{{}^{n}p_r}{r!} = {}^{n}c_r \Rightarrow {}^{n}c_r = {}^{n-1}c_r + {}^{n-1}c_{r-1}$

30. The number of terms are, if n is odd $\left[\frac{n+1}{2}\right] = \frac{9+1}{2} = 5$

31. Co-efficient of ' x ' in $\left(x^2 + \frac{k}{x}\right)^5$ is 270 ${}^{5}C_3 \cdot (1)^{5-3} \cdot (k)^3 = 270$

$\frac{5!}{(5-3)!\,31} \cdot k^3 = 270$ $\quad \frac{120}{2 \times 6} \cdot k^3 = 270$

$\frac{120}{12} k^3 = 270$ $\qquad k^3 = \frac{270}{10}$

$k^3 = 27$ $\qquad k = 3$

32. Independent term of x in $\left(x^2 - \frac{1}{x}\right)^9$

$$r = \frac{np}{p+q} = \frac{9(2)}{2+1} = \frac{18}{3} = 6 \Rightarrow {}^{9}C_6 (x^2)^{9-6} \cdot \left(\frac{-1}{x}\right)^6 = {}^{9}C_6 \cdot x^6 \cdot \frac{1}{x^6} = {}^{9}C_6 = \frac{9!}{3!\,6!} = \frac{9 \times 8 \times 7}{6}$$
$$= 3 \times 4 \times 7 = 84$$

33. $\ln(3+kx)^9$

coefficient x^2 = coefficient of x^3

$${}^{9}C_2 \cdot 3^7 \cdot k^2 = {}^{9}C_3 \cdot 3^6 \cdot k^3 \Rightarrow {}^{9}C_2 \cdot 3 = {}^{9}C_3 \cdot k \Rightarrow \frac{9!}{7!\,2!} \cdot 3 = \frac{9!}{6!\,3!} \cdot k \Rightarrow \frac{3}{7} = \frac{k}{3} \Rightarrow k = \frac{9}{7}$$

34. Co-efficient of x^{32} is 15_{c_r}

$r = \frac{60-32}{7} = 4$

$\Rightarrow 15_{c_4}$

Co-efficient of x^{-17} is $-15_{c_{11}}$

sum $= 15_{c_4} - 15_{c_{11}} = 0$

14. 2D & 3D SHAPES

WORKSHEET – I

LEVEL – I

1.D	2.C	3.D	4.C	5.B	6.B	7.D
8.C	9.C	10.C	11.C	12.D		

LEVEL – II

1. D	2. C	3. B	4. B	5. C	6. B	7. A
8. A	9. D	10. B	11. D	12. D	13.C	14. D
15. A	16. B	17. D	18. B	19. C	20. B	21. B
22. C	23. C					

LEVEL – III

24. A, B	25.A, B, C, D	26. C	27. C	28. B	
29.a-q; b-s; c-r; d-p	30. A, C, D	31. C	32. A, D	33. A	34. B
35. C					

LEVEL – II

1. $A = A = \frac{\sqrt{3}}{4}a^2$

   $\frac{\sqrt{3}}{4}(4)^2 = 4 \cdot \sqrt{3}\ cm^2$

2. $A = I \times b = 32\ cm^2$

3. $A = \frac{1}{2}bh = \frac{1}{2}8.6 = 24\ cm^2$

4. $A = (\text{ side })^2 = 100\ cm^2$

5. $A = \frac{1}{2} \cdot b \cdot h = \frac{1}{2} \cdot 15 \cdot 20 = 150\ cm^2$

6. Volume of two cylinders are equal, $d_1 : d_2 = 2:3 \Rightarrow r_1 : r_2 = 1:\frac{3}{2}$

   $\pi r_1^2 h_1 = \pi r_2^2 h_2 \Rightarrow \frac{h_1}{h_2} = \frac{r_2^2}{r_1^2} \Rightarrow \frac{h_1}{h_2} = \frac{\left(\frac{3}{2}\right)^2}{(1)^2} = 9:4 \ \therefore h_1 : h_2 = 9:4$

7. $A = \frac{h}{2}(a+b)$

   $= \frac{6}{2}(7+9)$

   $= 48\ cm^2$

8. $A = \frac{1}{2} \cdot d_1 \cdot d_2$

   $= 24\ cm^2$

9. $A = \pi r^2$

$= \frac{22}{7} \cdot (7)^2$

$= 154 \text{ cm}^2$

10. h = 11 cm, d = 7 cm ⇒ r = $\frac{7}{2}$ cm.

Volume of cylinder = πr^2 h = $\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 11 = \frac{11 \times 7 \times 11}{2} = \frac{847}{2}$ cm^3

∴ Volume of 50 rods = $\frac{847}{2} \times 50 = 847 \times 25 = 21175 \text{ cm}^3$

11. A = $\frac{\theta}{360} \cdot \pi r^2$

$= \frac{60°}{360°} \cdot \frac{22}{7} \cdot (21)^2 = 231 \text{ cm}^2$

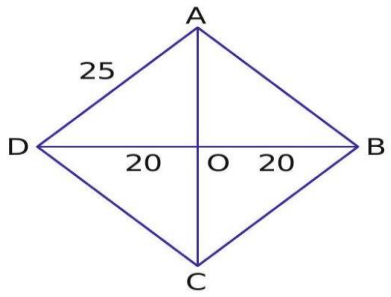
12. $A = \frac{\sqrt{3}}{4} \cdot a^2$

$64 \cdot \sqrt{3} = \frac{\sqrt{3}}{4} \cdot a^2$

⇒ a = 8.2

= 16 cm

13. 4a = 100 ⇒ a = 25



$OA = \sqrt{(25)^2 - (20)^2} = 15$

∴ $AC = 2(OA) = 30$ cm

∴ $A = \frac{1}{2} \cdot d_1 \cdot d_2$

$= \frac{1}{2} \cdot 30 \cdot 400 = 600 \text{ cm}^2$

14. $A = \frac{\sqrt{3}}{4} a^2 = 64\sqrt{3} \Rightarrow a^2 = 64 \times 4$

a = 16 ⇒ 3a = 48

15. $A_1 = A_2 + A_3$

$\pi r_1^2 = \pi(5)^2 + \pi(12)^2$

$\Rightarrow r_1 = 13$

16. Let $b = x$, $\ell = x + 1$

p = 2(ℓ + b)

$14 = 2(x + x + 1)$

x = 3

⇒ b = 3, $\ell = 4$

$A = \ell \times b = 12$ cm^2

17. $3x + 2x = 50$

x = 10

∴ Large piece length = 30, smaller piece length = 20

∴ 4a = 20

a = 5

∴ $A = a^2 = (5)^2 = 25$

18. $\ell^2 + b^2 = d^2$

$(4x)^2 + (3x)^2 = (200)^2$

$x^2 = \frac{200.200}{25}$

$x = \frac{200}{5} = 40$

∴ $\ell = 4(40) = 160$

$b = 3x = 3(40) = 120$

$A = \ell \times b = 19200$

19. $A = \pi r^2$

$4s^2\pi = \pi r^2$

⇒ r = 2 s

⇒ d = 4 s

∴ $A =$ (Side)^2 ∴ $A =$ (Side)$^2 = (d)^2 = 16s^2$

20. $d^2 = l^2 + b^2$

$d = \sqrt{3^2 + 4^2}$

d = 5

diameter of circle

∴ $C = \pi d = 5\pi$

21. A = bh

AB × DF = AD × BE

50 × 7 = 25 × BE

BE = 14 cm

(b)

22. $A_1 = I_1 \cdot b_1$

$I_2 = 120\% I_1$

$b_2 = 80\%\ b_1$

$A = \frac{120}{100} \times I_1 \times \frac{80}{100} \times b_1$

$= \frac{9600}{10{,}000} I_1 b_1$

$A_2 = 96\% \cdot A_1$

23. C.S.A of cone $(\pi r \ell) = 4070; d = 70$ cm $\Rightarrow r = 35$ cm

$\Rightarrow \frac{22}{7} \times 35 \times \ell = 4070 \Rightarrow \ell = \frac{4070}{22 \times 5} = \frac{4070}{110} = 37$ cm

LEVEL – III

24. $A = \frac{\sqrt{3}}{4} a^2 = 16\sqrt{3}$ cm^2

$h = \frac{\sqrt{3}}{2} \cdot a = \frac{\sqrt{3}}{2} \cdot 8 = 4 \cdot \sqrt{3}$

25. $A = \pi r^2 = \frac{22}{7} \cdot (14)^2 = 616$ cm^2

$p = 2\pi r = 2 \cdot \frac{22}{7} \cdot 14 = 88$ cm

$c = r(\pi + 2) = 14 \times \frac{36}{7} = 72$ cm

$I = \pi r = \frac{22}{7} \cdot 14 = 44$ cm

(a, b, c, d)

26. $A = I \times b$

$b = \frac{280}{20} = 14$ cm

(c)

27. $A = a^2$

$a = \sqrt{961} = 31$ cm

(c)

28. $B = \frac{A}{h} = \frac{38}{7.6} = 5$ cm

29. a) $A = bh = 4 \times 7 = 28$ cm^2

b) $A = \frac{1}{2} bh = \frac{1}{2} 4 \times 7 = 14$ cm^2

c) $A = \frac{1}{2} bh = \frac{1}{2} 4 \times 5 = 10$ cm^2

d) $A = (a)^2 = (5)^2 = 25 \text{ cm}^2$

30. A, C, D

31. Here, $h = 21 \text{ cm}, d = 18 \text{ cm} \Rightarrow r = 9 \text{ cm}$

Amount of water conical cup hold = Volume of cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 9 \times 9 \times 21 = 1782 \text{ cm}^3$

32. area it can graze $= \frac{\pi r^2}{4}$

$= \frac{1}{4} \cdot \frac{22}{7} \cdot 21 \cdot 21$

$= 346 \cdot 5 \text{ cm}^2$

remaining area = Area of rectangle = 346.5

$= (70 \times 52) - 346.5$

$= 3{,}293.5 \text{ cm}^2$

33. C.S.A of cone $= 1159\frac{5}{7}$; Area of base $= 254\frac{4}{7}$

$\pi r\ell = \frac{8118}{8}$; $\pi r^2 = \frac{1782}{7}$

$\Rightarrow \ell = \frac{8118}{8} \times \frac{7}{22} \times \frac{1}{9} = 41; \Rightarrow r^2 = \frac{1782}{7} \times \frac{7}{22}$

$\Rightarrow \ell = 41$; $\Rightarrow r = 9$

$\therefore \ell^2 = \text{r}^2 + \text{h}^2 \Rightarrow \text{h}^2 = \ell^2 - \text{r}^2 = 41^2 - 81$

$h^2 = 1600 \Rightarrow h = 40$

Volume of cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{1782}{7} \times 40 = 3394\frac{2}{7} \text{ cm}^3$

34. Here, $a^3 = 64 \text{ cm}^3 \Rightarrow a = 4 \text{ cm}$

So, $\ell = 8$ cm, b = 4 cm, h = 4 cm

T.S.A of cuboid $= 2(\ell b + bh + h\ell) = 160 \text{ cm}^2$

35. Volume of cuboid $= \ell \times b \times h = 50 \text{ m} \times 25 \text{ m} \times 3 \text{ m} = 3750 \text{ m}^3 = 3750\text{KL}$

15. COMPANING QUANTITIES

WORKSHEET – 1

LEVEL – I

1. A	2. B	3. C	4. C	5. C	6. D	7. B
8. C	9. A	10. C	11. C	12. C	13. D	14. A
15. B	16. C	17. D	18. B	19. C		

LEVEL – II

1.A	2. C	3. B	4. B	5. B	6. B	7. C
8. C	9. D	10. A	11. A	12. A	13. D	14. C
15. C	16. A	17. A	18. B	19.A	20. D	21. C
22. B	23. A	24. A				

LEVEL – III

25.A, B, C	26. A	27. B	28. C
29. a-s; b-r; c-t; d-q		30.A, B	31.A, B, C

LEVEL – I

1. $\frac{80}{112} = \frac{20}{28} = \frac{5}{7} = 5:7$

2. $\frac{4}{2.5} = \frac{40}{25} = \frac{8}{5} = 8:5$

3. $\frac{\frac{15}{4}}{\frac{41}{6}} = \frac{15}{44_2} \times \frac{6^6}{41} = \frac{45}{82} = 45:82$

4. $\frac{A}{B} = \frac{2}{3} : \frac{C}{B} = \frac{5}{9}$

   $\frac{A}{B} = \frac{2}{3} \times \frac{3}{3} : \frac{C}{B} = \frac{5}{9}$

   $\frac{A}{B} = \frac{6}{9} : \frac{C}{B} = \frac{5}{9}$

   $\therefore A:B:C = 6:9:5$

   $= 18:27:15$

5. $x\% = \frac{x}{100}$

6. $0.07 = \frac{7}{100} = 7\%$

7. $\frac{225}{135} = \frac{5}{x}$

   $\Rightarrow x = — = \frac{135}{45} = 3$

8. $\frac{27}{4} \times 100^{25}\%$

   $= 675\%$

9. $\left(\frac{28}{10\emptyset} \times 45\emptyset\right) + \left(\frac{45}{100} \times 280\right)$

$= 2 \times \frac{28}{10_5} \times 45^9$

$= 252$

10. $\frac{1}{2} = x\% \frac{1}{3}$

$\frac{1}{2} = \frac{x}{100} \times \frac{1}{3}$

$\Rightarrow x = \frac{100 \times 3}{2} = 150\%$

11. x% = 7 = 84

$\frac{x}{100} \times 7 = 84$

$\Rightarrow x = \frac{100 \times 84}{7} = 1200\%$

12. $\frac{5}{4} \times 100\% = 125\%$

13. $\frac{252}{27\emptyset} \times 10\emptyset\% = \frac{252^{28}}{2\pi_3} \times 10\% = \frac{280\%}{3} = 93\frac{1}{3}\%$

14. loss % $= \frac{(\text{cp} - \text{sp})}{\text{cp}} \times 100\%$

$= \frac{(490 - 465.5)}{490} \times 100\% = \frac{24.5}{490} \times 100\%$

$= \frac{245^{35}}{49_7} = 5\%$

15. Gain % $= \frac{\text{s.p} - \text{c.p}}{\text{c.p}} \times 100\%$

$= \frac{(28.6 - 27.5)}{27.5} \times 100\%$

$= \frac{1.1}{27.5} \times 100\%$

$= \frac{1100}{275}$

$= 4\%$

16. Ratio of their investments $= \frac{22500}{35000}$

$= \frac{225/^9}{350_{14}}$

$= 9:14$

$\therefore$ Deposit's share $= \frac{14}{23} \times 13{,}800$

$= 8400$

17. Ratio of their investments $= \frac{85,000}{1,15,000}$

$= \frac{85}{115} = \frac{17}{23}$

18. A's one day work $= \frac{1}{10}$

B's one day work $= \frac{1}{15}$

The taken by they to do the wise in 1 day is

$= \frac{1}{10} + \frac{1}{15}$

$= \frac{15+10}{150}$

$= \frac{25}{150}$

$= \frac{1}{6}$

∴ No of days taken by them to complete the work is 6 days(b)

19. Work done by A and B in one day $= \frac{1}{18}$

B and C in one day $= \frac{1}{24}$

C and A in one day $= \frac{1}{36}$

work done by they All to given in one day=

$2(A+B+C) = \frac{1}{18} + \frac{1}{24} + \frac{1}{36}$

$2(A+B+C) = \frac{4+3+2}{72} = \frac{1}{8}$

$A+B+C = \frac{1}{16}$

∴ time taken by them to complete the given work is 16 days

LEVEL – II

1. $\frac{A}{B} = \frac{3}{5} \times \frac{7}{5} = \frac{21}{25}$, $\frac{B}{C} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$

$\frac{A}{B} = \frac{21}{25}$: $\frac{C}{B} = \frac{8}{15}$

$\frac{A}{B} = \frac{21}{25} \times \frac{3}{3} = \frac{63}{75}; \frac{C}{B} = \frac{8}{15} \times \frac{5}{5} = \frac{40}{75}$

$\therefore A:B:C = 63:75:40$

2. $\frac{A}{B} = \frac{3}{4}; \frac{C}{B} = \frac{9}{8}$

$\frac{A}{B} = \frac{3}{4} \times \frac{2}{2} = \frac{6}{8}; \frac{C}{B} = \frac{9}{8}$

$\therefore A:B:C = 6:8:9$

$\therefore A:C = 6:9 = 2:8$

3. $2 = x\%$ of 50

$$2 = \frac{x}{100} \times 50$$
$$x = 4\%$$

4. $\frac{0.004 \times 100}{100}$

$$= \frac{0.4}{100} = 0.4\%$$

5. $\frac{5}{2250} \times 100\%$

$$\frac{55 \times 10}{225_{45}}$$
$$\frac{10}{45} = \frac{2}{9}$$

6. Gain $\% \frac{\text{Pr ofit}}{\text{C.P}} \times 100\%$

$$= \frac{7\emptyset}{7\emptyset\emptyset\emptyset} \times 1\emptyset\emptyset\%$$
$$= 1\%$$

7. Given SP = 100

Profit $= 15$

$\therefore$ Profit $=$ S.P.C.P

$\Rightarrow CP = S.P -$ Pr ofit $= 100 - 15 = 85$

$\therefore$ Gain $\% = \frac{15}{85_{17}} \times 100^{20}\%$

$$= \frac{300\%}{17}$$
$$= 17\frac{11}{17}\%$$

8. C. $p = 1400$; Loss $\% = 15\%$

C. $p = \frac{100}{100 - L\%} \times S.P$

$$\therefore S.p = \frac{C.P(100 - L\%)}{100}$$
$$= \frac{1400 \times 85}{100}$$
$$= 1190$$

9. ratio of their investments $= 120:135:150$

$= 8:9:10$

$\therefore$ B's Share $= \frac{}{2T_\beta} \times 56700^{18900}$

$= 18{,}900$

10. Work done by A on one day $= \frac{1}{18}$

work done by B in one day $= \frac{1}{9}$

work done by they together in one day $= \frac{1}{18} + \frac{1}{9} = \frac{3}{18} = \frac{1}{6}$

11. $x^2 + 4y^2 = 4xy$

$(x)^2 - 2 \cdot x \cdot 2y + (2y)^2 = 0$

$(x - 2y)^2 = 0$

$x = 2y$

$\frac{x}{y} = \frac{2}{1} = 2:1$

12. x% of 24 hours = 3 hours

$\frac{x}{100} \times 24 = 3$

$x = \frac{x \times 100}{24_8} = \frac{25}{2} = \frac{25}{2}\% = 12\frac{1}{2}\%$

13. Let $\frac{x}{5} = \frac{y}{8} = k$

$\Rightarrow x = 5k$ and $y = 8k$

$\therefore \frac{x+5}{y+8} = \frac{5(k+1)}{8(k+1)} = 5:8$

14. $\frac{2.5}{25} \times 100^4\%$

$= 10\%$

15. Work done by A and B in one day $= \frac{1}{8}$

Work done by B and C in one day $= \frac{1}{12}$

Work done by A, B and C in one day $= \frac{1}{6}$

Let work done by A and C in one day $= \frac{1}{\text{X}}$

$\therefore 2(A + B + C) = \frac{1}{6}$

$\Rightarrow 2\left(\frac{1}{8} + \frac{1}{12} + \frac{1}{x}\right) = \frac{1}{6}$

$= \frac{1}{x} = \frac{1}{3} - \frac{1}{8} - \frac{1}{12}$

$\frac{1}{x} = \frac{8 - 3 - 2}{24} = \frac{1}{8}$

$\Rightarrow x = 8$ days.

16. $\frac{(2^2)^{3.5}}{2^5} = \frac{2^{+7}}{2^5} = 2^2 = \frac{4}{1} = 4:1$

17. $\frac{x}{100} \times 25 = 2.125$

$\Rightarrow x = 8.5$

18. Let the distance travelled by train is $x$kms.

$\therefore 10 + x + 2x = 70 \Rightarrow x = 20$kms

19. Let the present ages are $7x$ and $8x$

$$\therefore \frac{7x + 10}{5x + 10} = \frac{9}{7}$$
$$49x + 70 = 45x + 90$$
$$4x = 20$$
$$x = 5$$

$\therefore$ Rafi's age $= 5x = 5(5) = 25$ years

20. Let the original population be x,

$(x + 1200) - 11\%(x + 1200) = x - 32$

$$(x + 1200)\left(1 - \frac{11}{100}\right) = x - 32$$
$$\frac{(x + 1200)(89)}{100} = x - 32$$
$$89x + 1{,}06{,}800 = 100x - 3200$$
$$11x = 1{,}06{,}800 + 3200$$
$$11x = 1{,}10{,}000$$
$$x = 10{,}000$$

21. Let the distance be $x$kms.

Given two-third of it was covered in 4 km/hr.

$\therefore$ time taken $= \frac{x}{4}$hr.

remaining distance: $x - \frac{2x}{3} = \frac{x}{3}$

time taken to cover $\frac{x}{3}$ distance at 5kmph speed $= \frac{\left(\frac{x}{3}\right)}{5} = \frac{x}{15}$hr.

Given total time taken = 76 mins.

$= \frac{76}{60}$hrs

$$\therefore \frac{x}{4} + \frac{x}{15} = \frac{76}{60}$$
$$\frac{15x + 4x}{60} = \frac{76}{60}$$
$$19x = 76$$
$$x = 4$$

22. Let the distance be $x$kms.

Difference between the delays is $\frac{x}{40} - \frac{x}{50} = (11 - 5)$ mins.

$\frac{50x - 40x}{40 \times 50} = \frac{6}{60}$ hrs

$\frac{10x}{40.50} = \frac{1}{10}$

$x = 20$kms

23. let the C.p. be x.

profit = 12%

Rate of sales tax = 10%

sale price of on article including sales tax = 616

$\therefore \frac{112x}{100} + 10\%$ of $\frac{112x}{100} = 616$

$\frac{112x}{100}(1 + 10\%) = 616$

$\frac{112x}{100} \times \frac{110}{100} = 616$

$x = \frac{56 \times 10 \times 100}{112}$

$x = 500$

24. let the speed be x km/hr.

$\therefore \frac{360}{x}\frac{360}{x+5} = 1$

$360\left(\frac{x+5-x}{x(x+5)}\right) = 1$

$x^2 + 5x - 1800 = 0$

$\therefore x = 40(\text{ or }) - 45$

$\therefore x = \frac{40 \text{ km}}{\text{hr}}$.

LEVEL – III

25. Let the notes one $2x, 3x, 5x$

$\therefore 2x \times 100 + 3x \times 50 + 5x \times 10 = 4{,}00{,}000$

$200x + 150x + 50x = 4{,}00{,}000$

$x = 1000$

26. Given $2A = 3B$ and $B = 4C$

$\frac{A}{B} = \frac{3}{2}; \frac{C}{B} = \frac{1}{4}$

$\frac{\text{A}}{\text{B}} = \frac{3}{2} \times \frac{2}{2}; \frac{\text{C}}{\text{B}} = \frac{1}{4}$

$\frac{\text{A}}{\text{B}} = \frac{6}{4}; \frac{\text{C}}{\text{B}} = \frac{1}{4}$

$\therefore$ A's share is $= \frac{6}{11} \times 16500$

$= 9000$

27. B's share $= \frac{4}{11} \times 16{,}500 = 6000$

28. C's share $= \frac{1}{11} \times 16500 = 1500$

29. Conceptual

30. No. of fifty paise coins coins $= x$

    $\therefore$ No. of twenty five paise coins $= 2x$

    $x\left(\frac{1}{2}\right) + 2x\left(\frac{1}{4}\right) = 9$

    $\frac{x}{2} + \frac{x}{2} = 9$

    x = 9

    $\therefore$ No.of fifty paise coins 19 and twenty five paise coins $= 9 \times 2 = 18$

31. $A : B : C = 35 : 45 : 55 = 7 : 9 : 11$

    $\therefore$ A's share $\frac{7}{27} \times 40{,}500 =$ Rs. 10,500

    $\therefore B'$ share $= \frac{0}{27} \times 40{,}500 =$ Rs. 13500

    $\therefore C'$ Share $= \frac{11}{27} \times 40{,}500 =$ Rs. 16,500